Olympiad Arena

Question Bank

An archive of problem set, mock test
and handout produced for the Olympiad Arena community

IOQM · RMO · INMO

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Three years of problem, 2024–2026

t.me/OlympiadArena

Preface

Olympiad Arena began as a Telegram channel and grew, over roughly three years, into a community for mathematics. What you are holding is the sediment of that work: most of the problems we set, adapted, or curated for the community, gathered for the first time into a single volume.

What is in here

The material comes from four kinds of source. There are the *IOQM mock tests* across the 2024, 2025 and 2026 seasons. There are the *RMO mock tests*, and *INMO mock*. And there are the *handouts*.

How this book is organised

Problems are sorted into four subjects — Algebra, Number Theory, Geometry, Combinatorics — and then into subtopics within each. A problem that could plausibly sit in two chapters has been placed where its *decisive* idea lives, not where its statement happens to point. For instance, a functional equation whose whole content is a divisibility argument is filed under Number Theory.

Within every subtopic the ordering is by difficulty, easiest first. This is a deliberate break from the original documents, several of which explicitly warned that their ordering meant nothing. The judgement is ours, and it is recorded only in the ordering itself: there are no difficulty labels.

A word on how to use it

Do not read this book. Pick a subtopic, start at its beginning, and stop when you are genuinely stuck rather than merely uncomfortable — those are different states and the difference is worth learning to feel.

The mock tests were written to be sat under time pressure and they still work that way; the index of source material at the back lets you reassemble any original paper if you want the full three hours. But most of the value here is in the second mode: slow, one problem at a time, with paper.

Devansh Kumar Tripathi, 2026

Credits

The mock tests and handouts collected in this volume were written and compiled by

Devansh Kumar Tripathi

and

Aarav Arora

over the course of the last three years, for the Olympiad Arena community.

Individual problems are, for the most part, not ours. They are drawn from national and regional olympiads, shortlists, team selection tests and invitational competitions from around the world — among them Vietnam, Iran, Japan, Korea, Turkey, Russia, Bulgaria, Romania, Greece, Austria, Australia, Canada, Argentina, Spain, Germany, the Balkan and Junior Balkan olympiads, the APMO, and the IMO and its shortlists. Where we know the origin of a problem it is recorded in the answer key at the back. Original problems, and problems adapted heavily enough that the original is no longer recognisable, carry no attribution.

We are grateful to the people who set those problems, and to the students of Olympiad Arena who sat the papers, argued about the answer keys, and found the errors.

Special thanks to Our Founder for his guidance.

This compilation is distributed free of charge for the practice and benefit of students preparing for the Indian olympiad programme. It is not for sale.

Contents

Preface **i**

Credits **ii**

1 Algebra **1**
1.1 Systems of Equations . . . 1
1.2 Polynomials . . . 4
1.3 Inequalities . . . 7
1.4 Functional Equations . . . 11
1.5 Sequences and Recurrences . . . 12
1.6 Miscellaneous Algebra . . . 14

2 Number Theory **16**
2.1 Divisibility, GCD and LCM . . . 16
2.2 Congruences and Orders . . . 19
2.3 Primes . . . 21
2.4 Diophantine Equations . . . 22
2.5 Arithmetic Functions . . . 26
2.6 Digits and Base Representation . . . 29
2.7 Advanced and Miscellaneous . . . 31

3 Geometry **34**
3.1 Angle Chasing and Cyclic Quadrilaterals . . . 34
3.2 Triangle Centres . . . 36
3.3 Circles, Tangency and Power of a Point . . . 38
3.4 Lengths, Areas and Computation . . . 41
3.5 Collinearity, Concurrency and Configurations . . . 46
3.6 Constructions . . . 51

4 Combinatorics **56**
4.1 Counting and Enumeration . . . 56
4.2 Probability and Expectation . . . 59
4.3 Graphs and Networks . . . 60
4.4 Games and Strategy . . . 61
4.5 Invariants, Colourings and Monovariants . . . 63
4.6 Extremal Combinatorics and Pigeonhole . . . 64
4.7 Grids, Tilings and Boards . . . 65

A Answer Key and Sources **67**
A.1 Algebra . . . 67
A.2 Number Theory . . . 71
A.3 Geometry . . . 77
A.4 Combinatorics . . . 83

B Index of Source Material **87**

C A Reference of Olympiad Mathematics **89**
C.1 Algebra . . . 89
C.1.1 Polynomials . . . 89
C.1.2 Complex Numbers and Roots of Unity . . . 90
C.1.3 Recurrences and Generating Functions . . . 91
C.1.4 Inequalities . . . 91
C.1.5 Functional Equations . . . 93
C.1.6 Algebraic Structures . . . 93
C.2 Number Theory . . . 93
C.2.1 Divisibility . . . 93
C.2.2 Congruences . . . 94
C.2.3 Arithmetic Functions . . . 94
C.2.4 Quadratic Residues . . . 95
C.2.5 Diophantine Equations . . . 95
C.2.6 Rational Approximation . . . 96
C.3 Geometry . . . 96
C.3.1 The Triangle and its Centres . . . 96
C.3.2 Concurrency and Collinearity . . . 97
C.3.3 Circles . . . 98
C.3.4 Transformations . . . 98
C.3.5 Metric Relations and Trigonometry . . . 99
C.3.6 Analytic Methods . . . 99
C.3.7 Geometric Inequalities . . . 100
C.4 Combinatorics . . . 100
C.4.1 Counting . . . 100
C.4.2 Special Numbers . . . 101
C.4.3 Graph Theory . . . 101
C.4.4 Extremal and Algebraic Methods . . . 102
C.4.5 Invariants and Games . . . 103
C.4.6 Combinatorial Geometry . . . 103
C.5 Methods of Proof . . . 104

Chapter 1

Algebra

The subtopics below follow the subtopics division as understood in general. Sequences whose content is really divisibility have been moved to Number Theory, and counting arguments dressed as algebra have been moved to Combinatorics.

1.1. Systems of Equations

The ordering below roughly tracks how well hidden the deciding move for the problem is.

1. Determine $x+y+z$ for all integer triples solving

$$xy+yz+zx=-4, \qquad x^2+y^2+z^2=24, \qquad x^3+y^3+z^3+3xyz=16.$$

2. Let $a,b,c,d,e,f\in\mathbb{R}$ satisfy $a^2+b^2+c^2=14$, $d^2+e^2+f^2=77$ and $ad+be+cf=32$. Find

$$(bf-ce)^2+(cd-af)^2+(ae-bd)^2.$$

3. Find $\dfrac{x^2y^2}{7}$ for integers x,y satisfying

$$y^2+3x^2y^2=30x^2+517.$$

4. Solve the system of equations

$$\frac{1}{xy}=\frac{x}{z}+1, \qquad \frac{1}{yz}=\frac{y}{x}+1, \qquad \frac{1}{zx}=\frac{z}{y}+1.$$

5. Solve the system

$$1-\frac{12}{y+3x}=\frac{2}{\sqrt{x}}, \qquad 1+\frac{12}{y+3x}=\frac{6}{\sqrt{y}}.$$

6. In the reals, solve

$$x+y+z=2,$$
$$(x+y)(y+z)+(y+z)(z+x)+(z+x)(x+y)=1,$$
$$x^2(y+z)+y^2(z+x)+z^2(x+y)=-6.$$

7. Find all real a, b, c, d, e, f satisfying

$$4a = (b+c+d+e)^4, \quad 4b = (c+d+e+f)^4, \quad 4c = (d+e+f+a)^4,$$
$$4d = (e+f+a+b)^4, \quad 4e = (f+a+b+c)^4, \quad 4f = (a+b+c+d)^4.$$

8. Find all real solutions (x, y) of

$$x^3 + 3xy^2 = 49, \qquad x^2 + 8xy + y^2 = 8y + 17x.$$

9. Find the number of real solutions (x, y, z) to

$$x + y + z = 3xy, \qquad x^2 + y^2 + z^2 = 3zx, \qquad x^3 + y^3 + z^3 = 3yz.$$

10. Solve the system

$$x + \frac{3x - y}{x^2 + y^2} = 3, \qquad y - \frac{x + 3y}{x^2 + y^2} = 0.$$

11. Given a positive integer n, find all non-negative reals $x_1, \ldots, x_n$ with

$$x_1 + x_2^2 + x_3^3 + \cdots + x_n^n = n, \qquad x_1 + 2x_2 + 3x_3^3 + \cdots + nx_n^n = \frac{n(n+1)}{2}.$$

12. Find all ordered pairs of reals (x, y) for which

$$(1+x)(1+x^2)(1+x^4) = 1 + y^7, \qquad (1+y)(1+y^2)(1+y^4) = 1 + x^7.$$

13. Let x, y, z be positive reals with

$$4x^2 - 2xy + y^2 = 64, \qquad y^2 - 3yz + 3z^2 = 36, \qquad 4x^2 + 3z^2 = 49.$$

Find the closest integer to the maximum possible value of $2xy + yz - 4zx$.

14. Let x, y, z be pairwise distinct complex numbers with

$$x^2 = y + 6, \qquad y^2 = z + 6, \qquad z^2 = x + 6.$$

If the largest possible value of $x + y + z$ is $\dfrac{a + \sqrt{b}}{c}$ with a, b, c integers, find the minimum value of $a + b + c$.

15. Let a, b, c, d, e be reals with

$$a \le b \le c \le d \le e, \quad a + e = 1, \quad b + c + d = 3, \quad a^2 + b^2 + c^2 + d^2 + e^2 = 14.$$

Determine the absolute value of the maximum possible value of ae.

16. Let $a, b \in \mathbb{R}$ with

$$\sqrt[3]{a} - \sqrt[3]{b} = 12, \qquad ab = \left(\frac{a+b+8}{6}\right)^3.$$

Find $\dfrac{a}{12} - b$.

17. Find all triples (x, y, z) of positive reals satisfying

$$2x^3 = 2y(x^2+1) - (z^2+1),$$
$$2y^4 = 3z(y^2+1) - 2(x^2+1),$$
$$2z^5 = 4x(z^2+1) - 3(y^2+1).$$

18. Given a positive integer a and an integer $n \geq 4$, find all n-tuples $(x_1, \ldots, x_n)$ of positive numbers with

$$x_i x_{i+1}(3a - 2x_{i+2}) = a^3 \qquad \text{for all } i \text{ (indices mod } n).$$

19. When t is a positive real, find the number of positive real solutions (a, b, c, d) to

$$a(1-b^2) = t, \quad b(1-c^2) = t, \quad c(1-d^2) = t, \quad d(1-a^2) = t.$$

20. Find all triples (x, y, z) with $x, y, z \in (0, 1)$ satisfying

$$\left(x + \frac{1}{2x} - 1\right)\left(y + \frac{1}{2y} - 1\right)\left(z + \frac{1}{2z} - 1\right) = \left(1 - \frac{xy}{z}\right)\left(1 - \frac{yz}{x}\right)\left(1 - \frac{zx}{y}\right).$$

21. Given $x, y, z > 3$, find all real solutions of

$$\frac{(x+2)^2}{y+z-2} + \frac{(y+4)^2}{z+x-4} + \frac{(z+6)^2}{x+y-6} = 36.$$

22. Solve the system

$$\frac{1}{\sqrt{1+2x^2}} + \frac{1}{\sqrt{1+2y^2}} = \frac{2}{\sqrt{1+2xy}}, \qquad \sqrt{x(1-2x)} + \sqrt{y(1-2y)} = \frac{2}{9}.$$

23. How many positive integer solutions does the following system have?

$$\sqrt{2020(a+\sqrt{b})} = \sqrt{(b+2020)(c+2020)},$$
$$\sqrt{2020(b+\sqrt{c})} = \sqrt{(c+2020)(d+2020)},$$
$$\sqrt{2020(c+\sqrt{d})} = \sqrt{(d+2020)(a+2020)},$$
$$\sqrt{2020(d+\sqrt{a})} = \sqrt{(a+2020)(b+2020)}.$$

24. Let a, b, c be positive reals satisfying

$$\sqrt{2a-ab}+\sqrt{2b-ab}=1,$$

$$\sqrt{2b-bc}+\sqrt{2c-bc}=\sqrt{2},$$

$$\sqrt{2c-ca}+\sqrt{2a-ca}=\sqrt{3}.$$

Then $\left[(1-a)(1-b)(1-c)\right]^2$ can be written as $\frac{m}{n}$ with $\gcd(m,n)=1$. Find $m+n$.

25. Pick reals x, y, z satisfying

$$(-x+y+z)^2-13=4(y-z)^2,$$

$$(x-y+z)^2-14=4(z-x)^2,$$

$$(x+y-z)^2-15=4(x-y)^2.$$

If $xy+yz+zx=\frac{p}{q}$ in lowest terms, find the product of the non-zero digits of $p+q$.

26. Let the number of ordered triples (a, b, c) of integers in $[-100, 100]$ satisfying

$$a^3-2a=abc-b-c, \quad b^3-2b=bca-c-a, \quad c^3-2c=cab-a-b$$

be x. Compute $\dfrac{x}{3}$.

27. Determine all n for which the following system is solvable in $\mathbb{R}$:

$$\sum_{k=1}^{n} x_k = 27, \qquad \prod_{k=1}^{n} x_k = \left(\frac{3}{2}\right)^{24}.$$

1.2. Polynomials

The recurring instruments are Vieta, the factor theorem, interpolation, and the observation that a polynomial identity holding on infinitely many points holds everywhere. Problems whose real subject is roots of unity or algebraic integers appear later, in Number Theory.

28. Let $P(x)=x^3+x^2-r^2x-2020$ have roots r, s, t. Compute the sum of the digits of $-P(1)$.

29. Let a, b, c be the roots of x^3-3x^2-4x+5. Compute

$$\frac{a^4+b^4}{a+b}+\frac{b^4+c^4}{b+c}+\frac{c^4+a^4}{c+a},$$

and find the sum of the digits of the numerator.

30. If $P(x)$ denotes a polynomial of degree n such that $P(k)=\dfrac{k}{k+1}$ for $k=0,1,\ldots,n$, determine $P(n+1)$.

31. Suppose P is a cubic polynomial with $P(0) = 3$ and

$$\left(x^3 - 2x + 1 - P(x)\right)\left(2x^3 - 5x^2 + 4 - P(x)\right) \leq 0 \quad \forall x \in \mathbb{R}.$$

Determine the value of $9\,|P(-1)|$.

32. Consider a quadratic $P(x) = ax^2 + bx + c$ with distinct positive roots r_1, r_2, and a second polynomial $Q(x) = cx^2 + bx + a$ with roots r_3, r_4. Four numbers r_1, r_2, r_3, r_4 are written on a board. What is the smallest possible integer value of their sum?

33. Suppose $f(x)$ is a monic quadratic such that there is an increasing arithmetic progression $x_1 < x_2 < x_3 < x_4$ with $|f(x_1)| = |f(x_2)| = |f(x_3)| = |f(x_4)| = 2020$. Compute the absolute difference of the two roots of f. If the answer is $a\sqrt{b}$ in simplest form, find $b - a$.

34. $P(x)$ is a polynomial of degree $3n$ such that

$$P(0) = P(3) = \cdots = P(3n) = 2,$$

$$P(1) = P(4) = \cdots = P(3n-2) = 1,$$

$$P(2) = P(5) = \cdots = P(3n-1) = 0,$$

and $P(3n+1) = 70$. Determine n.

35. The equation $x^5 - 2x^4 - 1 = 0$ has five complex roots $r_1, \ldots, r_5$. Find

$$\frac{1}{r_1^8} + \frac{1}{r_2^8} + \frac{1}{r_3^8} + \frac{1}{r_4^8} + \frac{1}{r_5^8}.$$

36. Let $\dfrac{3}{4} < a < 1$. Prove that the equation

$$x^3(x+1) = (x+a)(2x+a)$$

has four distinct real solutions, and find these solutions explicitly.

37. Determine all polynomials $P(x) \in \mathbb{R}[x]$ satisfying $P(2017) = 2016$ and $\left(P(x)+1\right)^2 = P(x^2+1)$ for all real x.

38. The polynomial $f(x) = x^3 - 6x^2 + 1$ has distinct real roots p, q, r. Let δ be the arithmetic mean of $a^2b + b^2c + c^2a$ over all permutations (a, b, c) of (p, q, r). Compute 4δ.

39. Let a and b be reals, and let r, s, t be the roots of $f(x) = x^3 + ax^2 + bx - 1$. Suppose r^2, s^2, t^2 are the roots of $g(x) = x^3 + mx^2 + nx + p$. If $g(-1) = -5$, find the maximum possible value of b. If the answer is $\sqrt{a} + b$, find $a^2 + b^2$.

40. Let $f(x) = ax^3 + bx^2 + cx + d$ have real coefficients, three positive real roots, and $f(0) < 0$. Prove that

$$2b^3 + 9a^2d - 7abc \leq 0.$$

41. Let $P(x) = x^3 + 3x^2 + 1$. There is a unique line $y = mx + b$ with $m > 0$ meeting the graph at three points A, B, C with $AB = BC = 2$. Find $\lfloor 100m \rfloor$.

42. How many cubics of the form $x^3 - ax^2 + (a + d)x + (a + 2d)$, with a, d integers, have all roots non-negative integers?

43. Given the polynomial $P(x) = x^4 + x^3 + 3x^2 - 6x + 1$, find the closest integer to $P(\alpha^2 + \alpha + 1)$, where

$$\alpha = \sqrt[3]{\frac{1 + \sqrt{5}}{2}} + \sqrt[3]{\frac{1 - \sqrt{5}}{2}}.$$

44. Let $P(x) = 8x^3 + ax + b + 1$ with $a, b \in \mathbb{Z}$. It is known that P has a root $x_0 = p + \sqrt{q} + \sqrt[3]{r}$ with $p, q, r \in \mathbb{Q}$, $q \geq 0$, but P has no rational root. Find the smallest possible value of $a + b$, and report its absolute value.

45. Find all polynomials $P(x)$ with real coefficients such that for all non-zero real x,

$$P(x) + P\left(\frac{1}{x}\right) = \frac{P\left(x + \frac{1}{x}\right) + P\left(x - \frac{1}{x}\right)}{2}.$$

46. Determine all polynomials $f(x)$ of degree 2024 such that

$$f(x^2) = f(x)\, f(x + 2) \qquad \forall x \in \mathbb{R}.$$

(Report the number of such polynomials.)

47. Given a polynomial $P(x)$ of degree $n > 1$ with real coefficients, the equation $P(P(P(x))) = P(x)$ has n^3 distinct real roots. Prove that these roots can be split into two groups with equal arithmetic mean.

48. There are 8 distinct quadratic trinomials on a board, no two of which sum to the zero polynomial. It turns out that for any two chosen trinomials g_1, g_2, the remaining six can be labelled $g_3, \ldots, g_8$ so that

$$g_1 + g_2, \quad g_3 + g_4, \quad g_5 + g_6, \quad g_7 + g_8$$

all share a common root. Must all eight trinomials share a common root? (The root need not be real.)

49. Let $p(x), q(x), r(x)$ be monic quadratics with only real roots, satisfying

$$p(q(x)) = (x-1)(x-3)(x-5)(x-7), \qquad q(r(x)) = (x-2)(x-4)(x-6)(x-8)$$

for all $x \in \mathbb{R}$. Find the sum of the digits of $p(0) + q(0) + r(0)$.

50. Denote by $F_0(x), F_1(x), \ldots$ the Fibonacci polynomials, defined by $F_0(x) = 1$, $F_1(x) = x$ and $F_n(x) = xF_{n-1}(x) + F_{n-2}(x)$ for $n \geq 2$. There exist unique integers $\lambda_0, \ldots, \lambda_{1000}$ with

$$x^{1000} = \sum_{i=0}^{1000} \lambda_i F_i(x)$$

for all real x. For which k is $|\lambda_k|$ maximised?

51. Let α be the unique real root of $x^3 - 2x^2 + x - 1$, so that $1 < \alpha < 2$. Define $p_0(x) = x$ and $p_{n+1}(x) = \big(p_n(x)\big)^2 - \alpha$. How many distinct real roots does $p_{10}(x)$ have?

52. Let $\alpha, \beta, \gamma \in \mathbb{C}$ be the roots of $x^3 - 3x^2 + 3x + 7$. For a complex number z set

$$f(z) = |z-\alpha| + |z-\beta| + |z-\gamma| - 2 \max_{w \in \{\alpha,\beta,\gamma\}} |z-w|.$$

Let A be the area of the region bounded by the locus of all $z \in \mathbb{C}$ at which f attains its global minimum. Find $\lfloor A \rfloor$.

53. Let f be a monic polynomial with integer coefficients of degree m, having a root that is not an integer. Let $a_1, \ldots, a_m$ be integers. Prove that

$$f(x) = (y + a_1)(y + a_2) \cdots (y + a_m)$$

is satisfied by only finitely many pairs of integers (x, y).

1.3. Inequalities

The standard — AM–GM, Cauchy–Schwarz, power mean, Chebyshev, Jensen, SOS, and the tangent line trick — covers most of what follows. The harder entries reward normalisation and a willingness to guess where equality sits before doing any work.

54. Let a, b, c be positive reals with $a + b + c = 3$. Prove that

$$\frac{1}{2 + a^2 + b^2} + \frac{1}{2 + b^2 + c^2} + \frac{1}{2 + c^2 + a^2} \leq \frac{3}{4}.$$

55. If a, b, c are positive reals, prove that

$$\left(1 + \frac{4a}{b+c}\right)\left(1 + \frac{4b}{c+a}\right)\left(1 + \frac{4c}{a+b}\right) > 25.$$

56. Let a, b, c be distinct positive reals. Prove that

$$\left(\frac{2a-b}{a-b}\right)^2 + \left(\frac{2b-c}{b-c}\right)^2 + \left(\frac{2c-a}{c-a}\right)^2 \geq 5.$$

57. Let x, y, z be positive reals. Prove that

$$\frac{xy}{z} + \frac{yz}{x} + \frac{zx}{y} > 2\sqrt[3]{x^3 + y^3 + z^3}.$$

58. Given $x, y, z \geq 0$ with $x + y + z = 2$, prove that

$$x^2y^2 + y^2z^2 + z^2x^2 + xyz \leq 1.$$

59. Let a, b, c be positive reals with $a^2 < bc$. Prove that

$$b^3 + ac^2 > ab(a + c).$$

60. Let a, b, c be positive reals with $abc = 1$. Prove that

$$\left(a - 1 + \frac{1}{b}\right)\left(b - 1 + \frac{1}{c}\right)\left(c - 1 + \frac{1}{a}\right) \leq 1.$$

61. If $x, y, z > 0$ and $k > 2$, prove that

$$\frac{x}{x + ky + kz} + \frac{y}{kx + y + kz} + \frac{z}{kx + ky + z} \geq \frac{3}{2k + 1}.$$

62. Let a, b, c be positive reals with $ab + bc + ca = 1$. Prove that

$$\sqrt[3]{\frac{1}{a} + 6b} + \sqrt[3]{\frac{1}{b} + 6c} + \sqrt[3]{\frac{1}{c} + 6a} \leq \frac{1}{abc}.$$

63. Let $a, b, c > 0$ with $ab + bc + ca = 1$. Prove that

$$\sqrt{a^3 + a} + \sqrt{b^3 + b} + \sqrt{c^3 + c} \geq 2\sqrt{a + b + c}.$$

64. Let a, b, c be positive reals with $a + b + c = 1$. Prove that

$$\frac{a - bc}{a + bc} + \frac{b - ca}{b + ca} + \frac{c - ab}{c + ab} \leq \frac{3}{2}.$$

65. Let $x, y, z \geq 0$ with $x^2 + y^2 + z^2 = 3$. Prove that

$$\frac{x}{\sqrt{x^2 + y + z}} + \frac{y}{\sqrt{x + y^2 + z}} + \frac{z}{\sqrt{z^2 + x + y}} \leq 3.$$

66. Let a, b, c be positive reals. Prove that

$$\frac{a^2b(b-c)}{a+b} + \frac{b^2c(c-a)}{b+c} + \frac{c^2a(a-b)}{c+a} \geq 0.$$

67. Show that for all positive reals a, b, c,

$$\frac{(b+c)(a^4-b^2c^2)}{ab+2bc+ca} + \frac{(c+a)(b^4-c^2a^2)}{bc+2ca+ab} + \frac{(a+b)(c^4-a^2b^2)}{ca+2ab+bc} \geq 0.$$

68. Let x_1, x_2, x_3 be reals with $x_k > -2k$ for $k = 1, 2, 3$ and $x_1 + 3x_2 + 5x_3 \leq 28$. The minimum value of

$$\frac{1}{x_1+2} + \frac{3}{x_2+4} + \frac{5}{x_3+6}$$

can be written $\frac{m}{n}$ in lowest terms. What is $m+n$?

69. Let a, b, c, d be positive reals with $abcd = 1$. Prove that

$$\sum_{\text{cyc}} \sqrt{\frac{a}{b+c+d^2+a^3}} \leq 2.$$

70. For $n \geq 2$, let $a_1, \ldots, a_n$ be positive reals with

$$(a_1 + \cdots + a_n)\left(\frac{1}{a_1} + \cdots + \frac{1}{a_n}\right) \leq \left(n + \frac{1}{2}\right)^2.$$

Prove that $\max(a_1, \ldots, a_n) \leq 4\min(a_1, \ldots, a_n)$.

71. Let $a, b, c > 0$ with $a + b + c = 3$. Prove that

$$\frac{a^2+3b^2}{ab^2(4-ab)} + \frac{b^2+3c^2}{bc^2(4-bc)} + \frac{c^2+3a^2}{ca^2(4-ca)} \geq 4.$$

72. Let $x, y, z > 0$ with $x + y + z = 3$. Prove that

$$\frac{x^3}{y^3+8} + \frac{y^3}{z^3+8} + \frac{z^3}{x^3+8} \geq \frac{1}{9} + \frac{2}{27}(xy + yz + zx).$$

73. Given positive reals $a_1, \ldots, a_n$ with $n \geq 2$ and $a_1 + \cdots + a_n = 1$, prove that

$$\sum_{i=1}^{n} \frac{\prod_{j \neq i} a_j}{a_i + n - 2} \leq \frac{1}{(n-1)^2}.$$

74. Let x, y, z be positive reals with $\sqrt{x} + \sqrt{y} + \sqrt{z} = 1$. Prove that

$$\frac{x^2+yz}{\sqrt{2x^2(y+z)}} + \frac{y^2+zx}{\sqrt{2y^2(z+x)}} + \frac{z^2+xy}{\sqrt{2z^2(x+y)}} \geq 1.$$

75. Let x, y, z be positive reals with $xyz \geq 1$. Prove that

$$\frac{x^5 - x^2}{x^5 + y^2 + z^2} + \frac{y^5 - y^2}{x^2 + y^5 + z^2} + \frac{z^5 - z^2}{x^2 + y^2 + z^5} \geq 0.$$

76. If a, b, c are the sides of a triangle, prove that

$$\frac{\sqrt{b+c-a}}{\sqrt{b}+\sqrt{c}-\sqrt{a}} + \frac{\sqrt{c+a-b}}{\sqrt{c}+\sqrt{a}-\sqrt{b}} + \frac{\sqrt{a+b-c}}{\sqrt{a}+\sqrt{b}-\sqrt{c}} \leq 3.$$

77. Let a, b, c be positive reals with $abc = 8$. Prove that

$$\frac{a^2}{\sqrt{(1+a^3)(1+b^3)}} + \frac{b^2}{\sqrt{(1+b^3)(1+c^3)}} + \frac{c^2}{\sqrt{(1+c^3)(1+a^3)}} \geq \frac{4}{3}.$$

78. Let x, y, z, m, n be positive reals with $m + n \geq 2$. Prove that

$$x\sqrt{yz(x+my)(x+nz)} + y\sqrt{xz(y+mx)(y+nz)} + z\sqrt{xy(z+mx)(z+ny)}$$
$$\leq \frac{3(m+n)}{8}(x+y)(y+z)(z+x).$$

79. Find the largest positive γ such that

$$|\gamma xy + yz| \leq \frac{\sqrt{5}}{2}$$

holds for all reals x, y, z with $x^2 + y^2 + z^2 = 1$.

80. Find the smallest positive real k such that for any integer $n \geq 2$ and positive reals $a_0, a_1, \ldots, a_n$,

$$\frac{1}{a_0 + a_1} + \frac{1}{a_0 + a_1 + a_2} + \cdots + \frac{1}{a_0 + a_1 + \cdots + a_n} < k\left(\frac{1}{a_0} + \frac{1}{a_1} + \cdots + \frac{1}{a_n}\right).$$

81. Let $x_1, \ldots, x_{2010} > 0$ with $\sum_{i=1}^{2009} x_i^{2009} = 1$. Find

$$\min \sum_{i=1}^{2010} \frac{x_i^{2008}}{1 - x_i^{2009}}$$

and prove your result.

82. If $0 < x_1 \leq x_2 \leq \cdots \leq x_s$ and $0 < y_1 \leq y_2 \leq \cdots \leq y_s$ satisfy

$$x_1^r + x_2^r + \cdots + x_s^r = y_1^r + y_2^r + \cdots + y_s^r$$

for all $r \in \mathbb{N}$, prove that $x_i = y_i$ for $i = 1, \ldots, s$.

83. Positive numbers $x_1, \ldots, x_n$ with $n \geq 2$ satisfy

$$\frac{1}{x_1 + 1998} + \cdots + \frac{1}{x_n + 1998} = \frac{1}{1998}.$$

Prove that $\dfrac{\sqrt[n]{x_1 \cdots x_n}}{n-1} \geq 1998$.

84. If a, b, c, d are positive reals with $(a+c)^2 = 4(ad+bc)$, prove that

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} + \frac{4bd}{ac} \geq 6.$$

85. For all positive reals a, b, c, prove that

$$\frac{(a^2+bc)^2}{b+c} + \frac{(b^2+ca)^2}{c+a} + \frac{(c^2+ab)^2}{a+b} \geq \frac{2abc(a+b+c)^2}{ab+bc+ca}.$$

86. For positive reals a, b, c, d with

$$\frac{a^2}{b+c+d} + \frac{b^2}{a+c+d} + \frac{c^2}{a+b+d} = \frac{3d^2}{a+b+c},$$

prove that

$$3a + 3b + 3c + 3d \geq \frac{16}{a+b+2d} + \frac{16}{b+c+2d} + \frac{16}{a+c+2d}.$$

87. Let n be a positive integer and let x, y be positive reals with $x^n + y^n = 1$. Prove that

$$\left(\sum_{k=1}^{n} \frac{1+x^{2k}}{1+x^{4k}}\right)\left(\sum_{k=1}^{n} \frac{1+y^{2k}}{1+y^{4k}}\right) < \frac{1}{(1-x)(1-y)}.$$

88. Call a triple (a, b, c) of positive numbers *mysterious* if

$$\sqrt{\frac{a^2+1}{a^2c^2+2ab}} + \sqrt{\frac{b^2+1}{b^2a^2+2bc}} + \sqrt{\frac{c^2+1}{c^2b^2+2ca}} = 2(a+b+c).$$

Prove that if (a, b, c) is mysterious then so is (c, b, a).

1.4. Functional Equations

89. Let $f(x) = 1 - \frac{1}{x}$. Suppose a, b, c are the roots of

$$x + f(x) + f(f(x)) = 26.$$

Find $a^2 + b^2 + c^2$.

90. Let f map non-negative integers to non-negative integers with $f(0) = 0$ and

$$f(m) = f\left(\left\lfloor \tfrac{m}{2} \right\rfloor\right) + \left\lceil \tfrac{m}{2} \right\rceil^2$$

for all positive integers m. Compute

$$\frac{f(1)}{1 \cdot 2} + \frac{f(2)}{2 \cdot 3} + \cdots + \frac{f(31)}{31 \cdot 32}.$$

91. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$(x + y^2) f(yf(x)) = xy\, f(y^2 + f(x)) \qquad \forall x, y \in \mathbb{R}.$$

92. Let $f(n)$ satisfy: $f(n)$ is a non-negative integer for each natural n; $f(n) = 2010$ whenever n ends in the digit 7; and $f(b/a) = |f(b) - f(a)|$ whenever $a \mid b$. Find $f(2009^{2009^{2009}})$.

93. Consider the function $f(m) = \sum_{n=0}^{\infty} \dfrac{(n-m)^2}{(2n)!}$. This can be written $f(m) = a_m e + \dfrac{b_m}{4e}$ for integer sequences $\{a_m\}, \{b_m\}$. Determine the sum of the digits of $\lim_{m\to\infty} \dfrac{2024\, b_m}{a_m}$.

1.5. Sequences and Recurrences

Recurrences whose point is a divisibility or a p-adic valuation live in Number Theory; what remains here is growth, closed forms, telescoping, and monotonicity.

94. Let $a_1, a_2, \ldots$ be arithmetic and $b_1, b_2, \ldots$ geometric, and set $c_n = a_n + b_n$. If $c_1 = 1$, $c_2 = 4$, $c_3 = 15$ and $c_4 = 2$, compute c_5.

95. The sequence $a_0, a_1, a_2, \ldots$ is determined by $a_0 = 0$ and

$$a_n = \begin{cases} 1 + a_{n-1} & n \text{ positive and odd,} \\ 3a_{n/2} & n \text{ positive and even.} \end{cases}$$

How many of these numbers are less than 2007?

96. Let $\{a_n\}$ be an infinite geometric sequence with common ratio satisfying $0 < |q| < 1$. Given $\sum_{n\geq 1} a_n = \sum_{n\geq 1} a_n^2$, find the sum of all integers in the largest possible range of a_2.

97. Let $\{a_n\}$ be defined by $a_1 = 0$ and

$$a_n = \frac{1}{n} + \frac{1}{\lceil n/2 \rceil} \sum_{k=1}^{\lceil n/2 \rceil} a_k$$

for every positive integer n. Find the maximal term of this sequence, and report your answer after multiplying it by 132.

98. Let $a_1, a_2, \ldots$ be a strictly increasing sequence of positive reals with $a_1 = 1$, $a_2 = 4$, such that for every positive integer k the subsequence $a_{4k-3}, a_{4k-2}, a_{4k-1}, a_{4k}$ is geometric and $a_{4k-1}, a_{4k}, a_{4k+1}, a_{4k+2}$ is arithmetic. Let r_k be the common ratio of the k-th geometric block. Compute

$$12 + \sum_{k=1}^{\infty} (r_k - 1)(r_{k+1} - 1).$$

99. Define $a_0 = 1$, $a_1 = 1$ and $a_{n+2} = a_{n+1} + xa_n$ for all $n \geq 0$, where $x \in \mathbb{R}$. Given that $\sum_{n=0}^{\infty} \frac{a_n}{10^n} = 1$, compute x.

100. The sequence a_n is defined by $a_0 = a_1 = 1$ and

$$a_n = \frac{a_{n-1}a_{n-2} + a_{n-1}^2}{a_{n-2}} \quad \forall n \geq 2,$$

and b_n by $b_0 = 1$, $b_1 = 3$ and the same recurrence. Consider $T_n = b_n/a_n$; writing $T_n = f(n)$, find $f(1)$.

101. A sequence $\{a_n\}$ of reals is defined by $a_1 = 1$ and, for all $n \geq 1$,

$$a_{n+1} = \frac{a_n\sqrt{n^2+n}}{\sqrt{n^2+n+2a_n^2}}.$$

Compute the product of the digits of the sum of all positive integers $n < 1000$ for which a_n is rational.

102. Determine all sequences of reals $a_1, \ldots, a_{1995}$ satisfying

$$2\sqrt{a_n - (n-1)} \geq a_{n+1} - (n-1) \quad (n = 1, \ldots, 1994),$$

and $2\sqrt{a_{1995} - 1994} \geq a_1 + 1$.

103. Find the largest constant M such that for every sequence $a_0, a_1, a_2, \ldots$ of reals with $a_0 = 1$, $a_1 = 3$ and

$$a_0 + a_1 + \cdots + a_{n-1} \geq 3a_n - a_{n+1} \quad \text{for all } n \geq 1,$$

one has $\frac{a_{n+1}}{a_n} > M$ for every $n \geq 0$.

104. The sequence of reals $a_0, a_1, a_2, \ldots$ is defined recursively by $a_0 = -1$ and

$$\sum_{k=0}^{n} \frac{a_{n-k}}{k+1} = 0 \quad \text{for } n \geq 1.$$

Show that $a_n > 0$ for all $n \geq 1$.

105. Let $(a_n)_{n\geq 1}$ be a sequence of positive reals with

$$a_{n+1}^2 + a_n a_{n+2} \leq a_n + a_{n+2}$$

for all positive integers n. Show that $a_{2022} \leq 1$.

106. Given a sequence of reals, a move consists of choosing two terms and replacing each with their arithmetic mean. Show that there exists a sequence of 2015 distinct reals such that after *any* single initial move there is always a way to continue with finitely many moves so as to obtain a constant sequence.

107. Let $n \geq 2$ and let $a_1, \ldots, a_n$ be positive reals with $a_1 + a_2 + \cdots + a_n = 2n - 1$. Find the sum of the digits of the minimum possible value of

$$\frac{a_1}{1} + \frac{a_2}{1+a_1} + \frac{a_3}{1+a_1+a_2} + \cdots + \frac{a_n}{1+a_1+\cdots+a_{n-1}}$$

when $n = 2025$.

108. Determine all sequences of real numbers $a_1, a_2, \ldots, a_{1995}$ which satisfy

$$2\sqrt{a_n - (n-1)} \geq a_{n+1} - (n-1) \;\; (n = 1, 2, \ldots, 1994), \qquad 2\sqrt{a_{1995} - 1994} \geq a_1 + 1.$$

1.6. Miscellaneous Algebra

109. If A, B, C are real angles with

$$\cos(B-C) + \cos(C-A) + \cos(A-B) = -\tfrac{3}{2},$$

find $\cos A + \cos B + \cos C$.

110. Let $(\log_2 x)(\log(yz)) = 2$, $(\log_2 y)(\log(xz)) = 4$ and $(\log_2 z)(\log(xy)) = 10$. If k is the product of all distinct values of $\log\left(x^{\log(y^{\log z})}\right)$, find $|k|$.

111. Let a, b, c be reals with $1 < a < b < c$ satisfying

$$\log_a b + \log_b c + \log_c a = 6.5, \qquad \log_b a + \log_c b + \log_a c = 5.$$

Then $\max\{\log_a b, \log_b c, \log_c a\}$ can be written $\sqrt{x} + \sqrt{y}$ with x, y positive integers. What is $x + y$?

112. For all real solutions of $\lfloor x \rfloor^3 - 7\lfloor x + 13 \rfloor = -13$, find the sum of all possible values of $\lfloor x \rfloor$.

113. For a real number x let $\lfloor x \rfloor$ be the greatest integer at most x and $\{x\} = x - \lfloor x \rfloor$. The sum of all real α satisfying

$$\alpha^2 + \{\alpha\} = 21$$

can be expressed as $\dfrac{\sqrt{a} - \sqrt{b}}{13} - d$ with a, b, c, d positive integers and a, b squarefree. Compute $\dfrac{a+b}{13} + c + d$.

114. If $|a| \leq 8$, find the number of positive divisors of the number of real solutions of $\{a\} + \{a^2\} = 1$.

115. Let $n \geq 1$ be an integer and consider

$$2\left\lfloor \tfrac{1}{2}x \right\rfloor - n + 1 = (n+1)(1-nx),$$

where x is the unknown real variable.

(a) Solve the equation for $n = 8$.

(b) Prove that there exists an integer n for which the equation has at least 2021 solutions.

116. Prove that for every irrational real a there exist irrationals b and b' such that $a+b$ and ab' are both rational while ab and $a+b'$ are both irrational.

117. Let λ be the largest real with the following property: for any positive reals p, q, r, s there is a complex number $z = a + bi$ $(a, b \in \mathbb{R})$ with

$$|b| \geq \lambda |a| \quad \text{and} \quad (pz^3 + 2qz^2 + 2rz + s)(qz^3 + 2pz^2 + 2sz + r) = 0.$$

Determine λ^8.

118. Let $0 < \frac{1}{q} \leq \frac{1}{p} < 1$ with $\frac{1}{p} + \frac{1}{q} = 1$. Let u_k, v_k, a_k, b_k be non-negative sequences with $u_k^2 > a_k^p$ and $v_k > b_k^q$ for $k = 1, \ldots, n$. If $0 < m_1 \leq u_k \leq M_1$ and $0 < m_2 \leq v_k \leq M_2$, find the minimum value of

$$\left(\sum_{k=1}^{n}\left(l\sqrt{(u_k+v_k)^2} - (a_k+b_k)\right)^p\right)^{1/p} - \left(\sum_{k=1}^{n}(u_k^2 - a_k^p)\right)^{1/p}\left(\sum_{k=1}^{n}(v_k^2 - b_k^p)\right)^{1/p},$$

where $l = \dfrac{M_1M_2 + m_1m_2}{2\sqrt{m_1M_1m_2M_2}}$.

119. Let $a_1, a_2, \ldots, a_n$ be reals and let m be a fixed positive integer less than n. Call an index k with $1 \leq k \leq n$ *good* if there is some ℓ with $1 \leq \ell \leq m$ such that $a_k + a_{k+1} + \cdots + a_{k+\ell-1} \geq 0$, indices taken modulo n. Let T be the set of good indices. Determine the minimum possible value of $\sum_{k \in T} a_k$.

Chapter 2

Number Theory

The subtopics below run roughly in following order — divisibility, then congruences, then primes, then equations, then the multiplicative functions — and each is internally ordered from easiest to hardest.

A note on placement. Many problems here could be filed elsewhere: sequences whose content is a divisibility, polynomials studied modulo p, counting arguments over residue classes. They are here because the decisive step is arithmetic. Conversely, a few problems that look arithmetic have been moved to Combinatorics because their real difficulty is a construction.

2.1. Divisibility, GCD and LCM

120. Determine the sum of the digits of the sum of all positive integers n between 1 and 100 inclusive such that $\gcd(n, 2^n - 1) = 3$.

121. Find the number of positive integers $n < 100$ such that

$$\gcd(n^2, 2023) = \gcd(n, 2023^2).$$

122. Find the number of positive integers $n < 9000$ such that $n^3 + 95$ is divisible by 96.

123. Find all positive integers m, n such that $\dfrac{2m-1}{n}$ and $\dfrac{2n-1}{m}$ are both integers.

124. Find the number of positive integer pairs (m, n) such that $\dfrac{2m-1}{n}$ and $\dfrac{2n-1}{m}$ are both integers.

125. Let m and n be natural numbers such that

$$A = \frac{(m+3)^n + 1}{3m}$$

is an integer. Prove that A is odd.

126. Show that there are infinitely many pairs of positive integers (m, n) such that

$$\frac{m+1}{n} + \frac{n+1}{m}$$

is a positive integer.

127. Find all ordered pairs (m, n) of positive integers such that $\dfrac{n^3+1}{mn-1}$ is an integer.

128. Let a and b be positive integers such that $ab+1$ divides a^2+b^2. Show that $\dfrac{a^2+b^2}{ab+1}$ is the square of an integer.

129. Find the least possible value of

$$\frac{\operatorname{lcm}(a,b)+\operatorname{lcm}(b,c)+\operatorname{lcm}(c,a)}{\gcd(a,b)+\gcd(b,c)+\gcd(c,a)}$$

over all distinct positive integers a, b, c.

130. Find all quadruples (a,b,c,d) of positive integers with $\gcd(a,b,c,d)=1$ and

$$a \mid b+c, \quad b \mid c+d, \quad c \mid d+a, \quad d \mid a+b.$$

131. Let n be a positive integer with $n \geq 3$. Show that

$$n^{n^{n^n}} - n^{n^n}$$

is divisible by 1989.

132. Let m and n be relatively prime positive integers. Determine all possible values of

$$\gcd\left(2^m - 2^n,\ 2^{m^2+mn+n^2}-1\right).$$

133. Let a, b, c be positive integers such that one of

$$\gcd(a,b)\cdot\operatorname{lcm}(b,c), \qquad \gcd(b,c)\cdot\operatorname{lcm}(c,a), \qquad \gcd(c,a)\cdot\operatorname{lcm}(a,b)$$

equals the product of the remaining two. Prove that one of a, b, c is a multiple of another.

134. Determine all integers $n > 1$ such that

$$\gcd\left(n,\ \frac{n-m}{\gcd(n,m)}\right) = 1$$

for all integers $1 \leq m < n$.

135. Let $\ell(n)$ denote the largest prime divisor of n, and let $a_{n+1} = a_n + \ell(a_n)$ with $a_1 = 2$. Determine all natural numbers m such that $a_i = m^2$ for some i.

136. Prove that there exist infinitely many positive integers n such that $\dfrac{4^n+2^n+1}{n^2+n+1}$ is a positive integer.

137. Let $n > 1$ be an integer. Prove that there exist $m > n^n$ such that $\dfrac{n^m - m^n}{m+n}$ is a positive integer.

138. Determine all positive integers n such that numbers from 1 to n can be sorted in some order $x_1, x_2, \ldots, x_n$ with the property that $x_1 + x_2 + \cdots + x_k$ is divisible by k for all $1 \leq k \leq n$.

139. Let a and b be positive integers such that $a^n + n$ divides $b^n + n$ for every positive integer n. Show that $a = b$.

140. Let a and b be given positive integers. Prove that there are infinitely many positive integers n such that $n^b + 1$ does not divide $a^n + 1$.

141. Find all triples (x, y, z) of positive integers with $z > 1$ satisfying simultaneously

$$x \mid y + 1, \qquad y \mid z - 1, \qquad z \mid x^2 + 1.$$

142. For each positive integer n, let $d(n)$ be the number of positive divisors of n. Prove that for all pairs of positive integers (a, b),

$$d(a) + d(b) \leq d(\gcd(a, b)) + d(\operatorname{lcm}(a, b)),$$

and determine all pairs for which equality holds.

143. Let $(a_n)_{n \geq 1}$ be an infinite sequence of positive integers such that for every positive integer n,

$$a_{n+1} = (n + 1)(a_n - n + 1).$$

In terms of a_1, determine the greatest positive integer k such that $\gcd(a_i, a_{i+1}) = k$ for some positive integer $i \geq 2$.

144. Let P be a polynomial with integer coefficients such that $P(0) = 0$ and $\gcd(P(0), P(1), P(2), \ldots) = 1$. Show there are infinitely many n such that

$$\gcd\big(P(n) - P(0),\, P(n + 1) - P(1),\, P(n + 2) - P(2),\, \ldots\big) = n.$$

145. Let $P(x)$ be a nonzero polynomial with integer coefficients. Let $a_0 = 0$ and $a_{i+1} = P(a_i)$ for $i \geq 0$. Show that $\gcd(a_m, a_n) = a_{\gcd(m,n)}$ for all $m, n \in \mathbb{N}$.

146. Let $a_1 < a_2 < a_3 < \cdots$ be positive integers such that a_{k+1} divides $2(a_1 + a_2 + \cdots + a_k)$ for every $k \geq 1$. Suppose that for infinitely many primes p there exists k with $p \mid a_k$. Prove that for every positive integer n there exists k with $n \mid a_k$.

2.2. Congruences and Orders

147. Let $k = 11^{23} + 23^{11}$. What is the units digit of k^{27}?

148. Consider the numbers

$$x_n = \underbrace{abaabaaab\cdots}_{n \text{ blocks}} b$$

where a and b are digits. Determine for how many integers n with $1 \leq n \leq 2020$ the number x_n is divisible by 11 regardless of the values of a and b. Find the digital sum of your answer.

149. For a positive integer n, prove that 3^{2023} divides

$$3^2 \cdot F_4 + 3^3 \cdot F_6 + 3^4 \cdot F_8 + \cdots + 3^{2023} F_{4046},$$

where F_n denotes the n-th Fibonacci number.

150. Let n be a positive integer. Prove that there exists a positive integer m such that $7^n \mid 3^m + 5^m - 1$.

151. Let k be an integer and a, b reals. Prove that $a - b$ is an integer divisible by k if and only if $\lfloor an \rfloor \equiv \lfloor bn \rfloor \pmod{k}$ for every positive integer n.

152. Prove that for any positive integer m there exist infinitely many pairs of integers (x, y) such that x and y are relatively prime, x divides $y^2 + m$, and y divides $x^2 + m$.

153. Let p be an odd prime.

(a)Show that p divides $n2^n + 1$ for infinitely many positive integers n.

(b)Find all such n when $p = 3$.

154. Let $x_1, x_2, \ldots, x_{2015}$ satisfy: for a positive integer a, the machine prints exactly one of $n, 2n, 3n$ for every positive integer n. Given that the machine prints 2, prove that it does not print 13824.

155. For any integer x, let

$$f(x) = 100! \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^{100}}{100!}\right).$$

A positive integer a is chosen such that $f(a) - 20$ is divisible by 101^2. Compute the remainder when $f(a + 101)$ is divided by 101^2.

156. Let $P(x) = x^{16} - x^{15} + \cdots - x + 1$, and let p be a prime such that $p - 1$ is divisible by 34 (for example $p = 103$). How many integers a with $1 \leq a \leq p - 1$ satisfy $p \mid P(a)$?

157. Call a positive integer x *n-cube-invariant* if the last n digits of x equal the last n digits of x^3. How many 2015-cube-invariant numbers x are there with $x < 10^{2015}$?

158. Let λ be the number of ordered triples (a, b, c) of integers with $1 \leq a, b, c \leq 31$ such that $ab + bc + ca \equiv 8 \pmod{31}$. Find 10λ.

159. What is the number of ordered pairs (p, a) with p prime, $3 \leq p < 100$, and $1 \leq a < p$, such that

$$a + a^2 + a^3 + \cdots + a^{(p-2)!}$$

is not divisible by p?

160. Determine the number of positive integers $2 \leq n \leq 50$ such that all coefficients of

$$x^{\varphi(n)} - 1 - \prod_{\substack{1 \leq k \leq n \\ \gcd(k,n)=1}} (x - k)$$

are divisible by n.

161. Let n be a positive integer. Prove that

$$3^{\frac{5^{2^n}-1}{2^{n+2}}} \equiv (-5)^{\frac{3^{2^n}-1}{2^{n+2}}} \pmod{2^{n+4}}.$$

162. Let p be a prime number. Prove that there exists a prime q such that for every integer n, the number $n^p - p$ is not divisible by q.

163. Given a prime p, show that

$$1 + p\left(\sum_{k=1}^{p-1} k^{-1}\right)^2 \equiv 1 - p^2 \sum_{k=1}^{p-1} k^{-2} \pmod{p^4}.$$

164. Let p be an odd prime and $a_1, a_2, \ldots, a_p$ integers. Prove that the following are equivalent.

(1)There exists a polynomial $P(x)$ of degree $\leq \frac{p-1}{2}$ with $P(i) \equiv a_i \pmod{p}$ for all $1 \leq i \leq p$.

(2)For every natural $d \leq \frac{p-1}{2}$, $\sum_{i=1}^{p}(a_{i+d} - a_i)^2 \equiv 0 \pmod{p}$, indices modulo p.

165. Let $\{m_1, m_2, \ldots\}$ be a (finite or infinite) set of positive integers. Consider the system of congruences

$$x \equiv 2^{m_i^2} \pmod{2^{m_i} - 1} \qquad (i = 1, 2, \ldots).$$

Give a necessary and sufficient condition for this system to be solvable.

166. Let $p = 4m-1$ be a prime, and let x, y be relatively prime integers with $x^2+y^2 = z^{2m}$ for some integer z. Prove that $p \mid xy$.

167. Let p be a prime and let n be the smallest integer strictly greater than 1 for which $p \mid n^6 - 1$. Prove that at least one of $(n+1)^6 - 1$ and $(n+2)^6 - 1$ is divisible by p.

168. An integer $n > 1$ is called *good* if there is a permutation $a_1, \ldots, a_n$ of $1, \ldots, n$ such that a_i and a_{i+1} have different parities for every $1 \leq i \leq n-1$, and $a_1 + a_2 + \cdots + a_k$ is a quadratic residue modulo n for every $1 \leq k \leq n$. Prove that there are infinitely many good numbers and infinitely many positive integers that are not good.

2.3. Primes

169. Let p, q, r, s be four distinct primes such that $p+q+r+s$ is prime, and the numbers p^2+qr and p^2+qs are both perfect squares. What is the value of $p+q+r+s$?

170. Suppose p is a prime which divides infinitely many numbers of the form $10^{n!} + 2023$, where n is a positive integer. What is the sum of all possible values of p?

171. Find all ordered pairs (p, q) of primes such that pq divides $5^p + 5^q$.

172. Find all ordered pairs (p, q) of primes such that pq divides $p^p + q^q + 1$.

173. Find all pairs of positive primes (p, q) such that $p^5 + p^3 + 2 = q^2 - q$.

174. Find all pairs of prime numbers (p, q) such that $p^5 + p^3 + 2 = q^2 - q$.

175. Find all pairs (p, q) of primes such that $p(p^2 - p - 1) = q(2q + 3)$.

176. Find all pairs of positive prime numbers (p, q) such that $p^5 + p^3 + 2 = q^2 - q$.

177. Find all triples (p, q, r) of primes such that all of

$$\frac{p^2+2q}{q+r}, \qquad \frac{q^2+9r}{r+p}, \qquad \frac{r^2+3p}{p+q}$$

are integers.

178. Find all pairs of primes p, q with $p \leq q$ satisfying: there exists a natural number n such that $2^n + 3^n + \cdots + (2pq-1)^n$ is a multiple of $2pq$.

179. Find all pairs of positive prime numbers (p, q) such that $p^5 + p^3 + 2 = q^2 - q$. Additionally, find all primes p and positive integers a, m with $a \leq 5p^2$ and $(p-1)! + a = p^m$.

180. Find all triples (m, p, q) such that $2^m p^2 + 1 = q^7$, where p, q are primes and m is a positive integer.

181. Find all prime numbers p for which there exist positive integers x, y, z such that $x^p + y^p + z^p - x - y - z$ is a product of exactly three distinct primes.

182. If m is a positive integer and $2^{m+1} + 1$ divides $3^{2^m} + 1$, prove that $2^{m+1} + 1$ is prime.

183. It is given that $p = 6k + 1$ where $k > 1$ is a natural number and p is prime. Let $m = 2^p - 1$. Prove that $\dfrac{2^{m-1} - 1}{127m}$ is an integer.

184. Let $p \neq 5$ be a prime such that $\frac{p+1}{2}$ is also prime. Suppose there exist positive integers $a < b$ such that $\dfrac{p^2 + a}{p + a^2}$ and $\dfrac{p^2 + b}{p + b^2}$ are integers. Show that $b = (a-1)^2 + 1$.

185. Find all positive integers a, b, c and primes p such that

$$73p^2 + 6 = 9a^2 + 17b^2 + 17c^2.$$

186. If p, q are primes such that p divides $2^{6q} - 1$ and p divides $2^{6p} + 1$, find the sum of the digits of the sum of all possible values of $p + q$.

187. Prove that there are infinitely many distinct pairs (a, b) of relatively prime integers $a > 1$ and $b > 1$ such that $a^b + b^a$ is divisible by $a + b$.

188. Prove that for primes p, q, r, if $p^n + q^n = r^2$, then $n = 1$.

189. Consider 2024 distinct primes $p_1, p_2, \ldots, p_{2024}$ such that

$$p_1 + p_2 + \cdots + p_{1012} = p_{1013} + p_{1014} + \cdots + p_{2024}.$$

Let $A = p_1 p_2 \cdots p_{1012}$ and $B = p_{1013} p_{1014} \cdots p_{2024}$. Prove that $|A - B| \geq 4$.

190. Suppose a and b are positive integers with the property that $(a^3 - 3ab + 12)^n + (b^3 + 12)^n$ is an integer for at least three, but at most finitely many, choices of positive integer n. What is the least possible value of $a + b$?

2.4. Diophantine Equations

The methods are factorisation, bounding, descent, Vieta jumping, and reduction modulo a well-chosen number. The handout from which many of these come opens with a table of quadratic and cubic residues; it is worth reconstructing that table yourself before starting.

191. How many ordered pairs of integers (x, y) satisfy

$$x^2 + y^2 = 2(x + y) + xy?$$

192. Find the number of integral solutions to

$$x^3 + 3y^3 + 9z^3 = 9xyz.$$

193. Determine all pairs (x, y) of integers such that $1 + 2^x + 2^{2x+1} = y^2$.

194. Find all pairs (x, y) of positive integers satisfying $3^x - 2^y = 7$.

195. Find all pairs (x, y) of positive integers for which $x^2 - y! = 2001$.

196. Determine the integers x such that $2^x + x^2 + 25$ is the cube of a prime.

197. Find all triples (x, y, z) of integers such that $x^2 + 2y^2 + 3z^2 = 23$.

198. Find all triples (x, y, z) of non-negative integers such that $5^x \cdot 7^y + 4 = 3^z$.

199. Let x and y be integers with $x^2 + y^2 + 1 = 2xy$. Prove that $x = y$.

200. Find all pairs (x, y) of rational numbers such that $xy^2 = x^2 + 2x - 3$.

201. Find the number of pairs of positive integers (a, b) such that

$$11ab \leq a^3 - b^3 \leq 12ab.$$

202. Find all positive integers $p, q, r, s > 1$ such that $p! + q! + r! = 2^s$.

203. Find all ordered pairs (a, b) of integers such that $3^a + 7^b$ is a perfect square.

204. Determine all ordered pairs (a, p) of positive integers, with p prime, such that $p^a + a^4$ is a perfect square.

205. Determine all ordered pairs (x, y) of nonnegative integers satisfying

$$3x^2 + 2 \cdot 9^y = x(4^{y+1} - 1).$$

206. Find all pairs of integers (x, y) such that $y^3 = 8x^6 + 2x^3y - y^2$.

207. Determine all pairs of integers (x, y) satisfying

$$y(x + y) = x^3 - 7x^2 + 11x - 3.$$

208. Find all pairs (m, n) of nonnegative integers for which

$$m^2 + 2 \cdot 3^n = m\left(2^{n+1} - 1\right).$$

209. Find all real x such that both $4x^5 - 7$ and $4x^{13} - 7$ are perfect squares.

210. Find all triples (a, b, c) of positive integers such that $a^2 + b + 3 = (b^2 - c^2)^2$.

211. Find all integers m and n such that the fifth power of m minus the fifth power of n equals $16mn$.

212. Find all prime numbers a, b, c and positive integers k satisfying

$$a^2 + b^2 + 16c^2 = 9k^2 + 1.$$

213. Determine all positive integers x, y, z such that $x^5 + 4^y = 2013^z$.

214. Find all x, y, z in positive integers with $z + y^2 + x^3 = xyz$ and $x = \gcd(y, z)$.

215. Find all natural numbers $x, y > 1$ and primes p satisfying

$$(x^2 - 1)(y^2 - 1) = (p + 1)^2.$$

216. Solve the equation $x^2 + 4 = y^n$, where n is an integer greater than 1.

217. Find all triples (a, b, c) of positive integers such that $a^{2a} = b^{2b} + 2c^c$.

218. Find the number of possible non-negative integer solutions (x, y) of $x! + 2^y = (x + 1)!$.

219. Find the total number of triples of integers (x, y, n) satisfying $\frac{1}{x} + \frac{1}{y} = \frac{1}{n^2}$, where n is either 2012 or 2013. Find the product of the digits of your answer.

220. Find the number of all triples (p, a, b) of positive integers such that

$$p = b\sqrt{\frac{a - 8b}{a + 8b}}$$

is prime.

221. Prove that there are unique positive integers a and n such that $a^{n+1} - (a+1)^n = 2001$.

222.

(a)Prove that if there is a triple (x, y, z) of positive integers with $x^2 + y^2 + 1 = xyz$, then $z = 3$.

(b)Find all such triples.

223. Let $x_1, \ldots, x_n$ be n positive integers. Prove that

$$x_1^2 + x_2^2 + \cdots + x_n^2 = kx_1x_2 \cdots x_n$$

has no solutions when $k > n$ is an integer.

224. Distinct positive integers a, b, c, d satisfy

$$a \mid b^2 + c^2 + d^2, \quad b \mid a^2 + c^2 + d^2, \quad c \mid a^2 + b^2 + d^2, \quad d \mid a^2 + b^2 + c^2,$$

and none of them is larger than the product of the other three. What is the largest possible number of primes among them?

225. Let $a > b > c > d$ be positive integers with

$$ac + bd = (b + d + a - c)(b + d - a + c).$$

Prove that $ab + cd$ is not prime.

226. Prove that for all integers $n \geq 2$ there are odd integers x, y with $|x^2 - 17y^2| = 4^n$.

227. Prove that every positive rational number can be represented in the form $\dfrac{a^3 + b^3}{c^3 + d^3}$ where a, b, c, d are positive integers.

228. Prove that there are infinitely many triples (a, b, c) of integers with $a^4 + b^3 = c^2$.

229. Find all pairs of positive integers $m, n \geq 3$ for which there exist infinitely many positive integers a such that

$$\frac{a^m + a - 1}{a^n + a^2 - 1}$$

is an integer.

230. Given that m, n are relatively prime integers, prove that the equation in x, t, y, s, v, r

$$x^m n^t + y^m n^s = v^m r^n$$

has infinitely many positive integer solutions.

231. Find all pairs of positive integers (x, p) such that p is prime, $x \leq 2p$, and x^{p-1} divides $(p - 1)^x + 1$.

232. Find all integers a, b, c with $1 < a < b < c$ such that $(a - 1)(b - 1)(c - 1)$ is a divisor of $abc - 1$.

233. In the range of integers, find the solutions of

$$x^{2010} - 2006 = 4y^{2009} + 4y^{2008} + 2007y.$$

234. Show that there do not exist natural numbers $a_1, \ldots, a_{2018}$ such that all the numbers

$$a_1^{2018} + a_2, \quad a_2^{2018} + a_3, \quad \ldots, \quad a_{2018}^{2018} + a_1$$

are powers of 5.

235. Does there exist a positive integer n with exactly 2000 prime divisors such that n divides $2^n + 1$?

236. Consider the system $x + y = z + u$, $2xy = zu$. Find the greatest real constant m such that $m \leq x/y$ for any positive integer solution (x, y, z, u) with $x \geq y$.

237. Let $K > 1$ be a positive integer. Prove that there always exist k positive integers $n_1, \ldots, n_k$ such that

$$n_1^2 + n_2^2 + \cdots + n_k^2 = 5^{m+k}$$

for each non-negative integer m.

238. Find all quadruples (r, u, v, s) satisfying the generalised Pythagorean equation $r^2 + u^2 + v^2 = s^2$.

239. Prove that if the difference of two consecutive cubes is n^2, then $2n - 1$ is a square.

240. Prove that there exist two strictly increasing sequences (a_n) and (b_n) such that $a_n(a_n + 1)$ divides $b_n^2 + 1$ for every natural n.

241. Show that the equation $y^2 = x^3 + 2a^3 - 3b^2$ has no integer solutions if $ab \neq 0$, $a \not\equiv 1 \pmod 3$, $3 \nmid b$, a is odd if b is even, and $p = t^2 + 27u^2$ is solvable in integers t, u whenever $p \mid a$ and $p \equiv 1 \pmod 3$.

2.5. Arithmetic Functions

Throughout, $d(n)$ or $\tau(n)$ is the number of divisors, $\sigma(n)$ the sum of divisors, φ Euler's totient, and μ the Möbius function.

242. Compute the sum of the digits of all positive integers whose positive divisors sum to 186.

243. Consider positive integers n having at least six positive divisors. Let the divisors of n be $1 = d_1 < d_2 < \cdots < d_k = n$ with $k \geq 6$. Find all n such that $n = d_5^2 + d_6^2$.

244. For a positive integer n write its divisors in increasing order $1 = d_1 < d_2 < \cdots < d_k = n$. Find all positive integers n divisible by 2019 such that $n = d_{19} \cdot d_{20}$.

245. Determine all positive integers n for which $d(n)^3 = 4n$.

246. Let $d(k)$ denote the number of positive divisors of k. Prove that for all positive integers n,

$$d(1) + d(3) + d(5) + \cdots + d(2n-1) \leq d(2) + d(4) + d(6) + \cdots + d(2n).$$

247. Define a function $f(n)$ from the positive integers to the positive integers such that $f(f(n))$ is the number of positive divisors of n. Prove that if p is prime then $f(p)$ is prime.

248. Let n be an odd positive integer. Prove that $\sigma(n)^3 < n^4$.

249. Show that for any positive integer n,

$$\frac{\sigma(n!)}{n!} \geq \sum_{k=1}^{n} \frac{1}{k}.$$

250. If n is composite, prove that $\varphi(n) \leq n - \sqrt{n}$.

251. Determine all positive integers k such that $\dfrac{d(n^2)}{d(n)} = k$ for some $n \in \mathbb{N}$.

252. Let m, n be positive integers. Prove that for some positive integer a, each of $\varphi(a), \varphi(a+1), \ldots, \varphi(a+n)$ is a multiple of m.

253. Prove that for infinitely many k one has $\varphi(2^k + 1) < 2^{k-1}$, and for infinitely many m one has $\varphi(2^m + 1) > 2^{m-1}$.

254. For all pairs (m, n) of positive integers having the same number k of divisors, define $m \circ n$ as follows: write the divisors in ascending order $1 = m_1 < \cdots < m_k = m$ and $1 = n_1 < \cdots < n_k = n$, and set $m \circ n = m_1 n_1 + \cdots + m_k n_k$. Find all pairs (m, n) with $m \neq n$ such that $m \circ n = 497$.

255. Define $m(n)$ to be the greatest proper natural divisor of n. Find all $n \in \mathbb{N}$ such that $n + m(n)$ is a power of 10.

256. Prove that for any $\delta > 1$ and any positive ϵ, there is an n with $\left| \dfrac{\sigma(n)}{n} - \delta \right| < \epsilon$.

257. Let $\tau(n)$ denote the number of positive divisors of n. Prove that there exist infinitely many positive integers a such that the equation $\tau(an) = n$ has no positive integer solution n.

258. Suppose that n and k are positive integers such that $1 = \underbrace{\varphi(\varphi(\cdots\varphi}_{k \text{ times}}(n)\cdots))$. Prove that $n \leq 3^k$.

259. For every positive integer N, let $\sigma(N)$ denote the sum of the positive divisors of N. Find all integers $m \geq n \geq 2$ satisfying

$$\frac{\sigma(m)-1}{m-1} = \frac{\sigma(n)-1}{n-1} = \frac{\sigma(mn)-1}{mn-1}.$$

260. Let φ and Ψ denote the Euler totient and Dedekind's totient respectively. Determine all n such that $\varphi(n)$ divides $n + \Psi(n)$.

261. For a positive integer n, let $\mu(n) = 0$ if n is not squarefree and $(-1)^k$ if n is a product of k distinct primes. Prove that for all n,

$$\sum_{d|n} \frac{\mu(d)\sigma(d)}{d} \geq \frac{1}{n},$$

and determine when equality holds.

262. Prove that if n is a perfect number, then

$$\sum_{p|n} \frac{1}{p+1} < \ln 2 < \sum_{p|n} \frac{1}{p-1},$$

the sums being over the prime divisors of n.

263. Let a, n be positive integers such that a^n is a perfect number. Prove that $a^n > (\frac{\mu}{2})^\mu$, where μ is the number of distinct prime divisors of a^n.

264. For all natural n, define $f(n) = \tau(n!) - \tau((n-1)!)$. Prove that there exist infinitely many composite n such that $f(m) < f(n)$ for all naturals $m < n$.

265. Let $(b_n)_{n\geq 0}$ be a sequence of positive integers with $b_n = d\Big(\sum_{i=0}^{n-1} b_i\Big)$ for all $n \geq 1$.

(a)Prove that (b_n) is unbounded.

(b)Prove that there are infinitely many n with $b_n > b_{n+1}$.

266. Let g be the multiplicative function given by $g(p^\alpha) = \alpha p^{\alpha-1}$ for all $\alpha \in \mathbb{Z}^+$ and primes p. Prove that there exist infinitely many integers n with $g(n+1) = g(n) + g(1)$.

267. For every positive integer $n = \prod_{i=1}^k p_i^{\alpha_i}$, define $\Phi(n) = \sum_{i:\, p_i > 10^{100}} \alpha_i$. Find all strictly increasing functions $f : \mathbb{Z} \to \mathbb{Z}$ such that $\Phi(f(a) - f(b)) \leq \Phi(a-b)$ for all integers $a > b$.

268. Let S be the set of positive integers such that $n \in S$ if and only if $\sum_{d|n,\ d<n,\ d\in S} d \leq n$. Find all positive integers $n = 2^k p$, where $k \geq 0$ and p is an odd prime, such that $\sum_{d|n,\ d<n,\ d\in S} d = n$.

2.6. Digits and Base Representation

269. Compute the base b for which $(253)_b \cdot (341)_b = (74XYZ)_b$ for some base b digits X, Y, Z.

270. Let $S(n)$ denote the sum of the digits of n. For some positive integer k it is known that $S(k) = 152$ and $S(k+1)$ is a multiple of 5. What is the difference between the largest and smallest possible values of $S(k+1)$?

271. Consider the first set of 38 consecutive positive integers all of whose digit sums are not divisible by 11. Find the smallest integer in this set; if the answer is X, report the sum of the digits of X.

272. Find the sum of the digits of the sum of all three-digit numbers whose digits, read left to right, form a strictly increasing sequence. (Numbers with a leading zero do not count as three-digit.)

273. Determine all n for which 11111 written in base n is a perfect square; give the sum of all such integers $n \geq 2$.

274. For any nonnegative integer n, let $S(n)$ be the digit sum of n. Let K be the number of nonnegative integers $n \leq 10^{10}$ satisfying $S(n) = \big(S(S(n))\big)^2$. Find K modulo 100.

275. Determine the value of $\log_4 x$ if

$$x = \frac{60!}{59!} + \frac{60!}{3!\,57!} + \frac{60!}{5!\,55!} + \cdots + \frac{60!}{27!\,33!} + \frac{60!}{29!\,31!}.$$

276. Given that $\log_2 5 \approx 2.321$, how many positive integers $k \leq 100$ are there such that there are exactly four powers of 2 with exactly k digits?

277. There exists a unique five-digit positive integer $\overline{abcde}$ with $a \neq 0$, $c \neq 0$ and $\overline{abcde} = (\overline{ab} + \overline{cde})^2$. Find $a^2 - b^2 + c^2 - d^2 + e^2$.

278. An integer N is worth one point for each pair of digits it contains that forms a prime in its original order. For example 6733 is worth three points and 20304 is worth two. Compute the sum of the squares of the digits of the smallest positive integer worth exactly eleven points. (Leading zeros are not allowed.)

279. A palindromic number reads the same when the order of its digits is reversed. Prove that the arithmetic progression $18, 37, \ldots$ contains infinitely many palindromic numbers.

280.

(a) Decide whether there exist two decimal digits a, b such that every integer with decimal representation $\overline{ab222\ldots231}$ is divisible by 73.

(b)Decide whether there exist two decimal digits c, d such that every integer with decimal representation $\overline{cd222\ldots231}$ is divisible by 79.

281. For a positive integer n let $S(n)$ be the digit sum. Any positive integer obtained by removing at least one digit from the right-hand end of the decimal representation of n is called a *stump* of n. Let $T(n)$ be the sum of all stumps of n. Prove that $n = S(n) + 9T(n)$.

282. Find an integral solution of

$$\left\lfloor \frac{x}{1!} \right\rfloor + \left\lfloor \frac{x}{2!} \right\rfloor + \left\lfloor \frac{x}{3!} \right\rfloor + \cdots + \left\lfloor \frac{x}{10!} \right\rfloor = 2019.$$

283. Evaluate $1 \oplus 2 \oplus \cdots \oplus 987654321$, where $\oplus$ denotes bitwise exclusive OR.

284. For a positive integer k, let $s(k)$ denote the number of 1s in the binary representation of k. Prove that for any positive integer n,

$$\sum_{i=1}^{n} (-1)^{s(3i)} > 0.$$

285. Let n be a positive integer. What is the smallest sum of digits that $5^n + 6^n + 2022^n$ can take?

286. Determine all the pairs (a, b) of positive integers such that: $b > a$ and $b - a$ is prime; the last digit of $a + b$ is 3; and ab is a perfect square.

287. Let $r = 0.d_0d_1d_2\cdots$ be a real number. Let e_n denote the number formed by the digits $d_n, d_{n-1}, \ldots, d_0$ written left to right (leading zeros permitted). Given that $d_0 = 6$ and, for each $n \geq 0$, e_n equals the number formed by the $n+1$ rightmost digits of e_n^2, show that r is irrational.

288. For a positive integer a, let a' be the integer obtained by reversing the decimal writing of a (the writing of a' may begin with zeros, but that of a may not); for instance if $a = 2370$ then $a' = 0732 = 732$. Let a_1 be a positive integer and define $a_{n+1} = a_n + a_n'$ for $n \geq 1$. Investigate the sequence $(a_n)_{n\geq 1}$.

289. Let a be a 2000-digit natural number of the form $a = \overline{2(A)99\ldots99(B)(C)}$ in base ten, not a multiple of 10, with $2(A) + (B)(C) = 99$ (for example $a = 2899\ldots9971$). Let b be the number obtained by reversing the digits of a. Find the sum of the digits of every positive integer a making ab a perfect square.

290. Prove that for each positive integer n there exists a positive integer with exactly n digits, none of which is 0, that is divisible by the sum of its digits.

2.7. Advanced and Miscellaneous

291. Let $x_1 = \frac{1}{2}$ and $x_{k+1} = \dfrac{x_k}{x_1^2 + \cdots + x_k^2}$. Prove that $\sqrt{x_k^4 + 4} \cdot \dfrac{x_{k-1}}{x_{k+1}}$ is rational.

292. Let $F_1 = F_2 = 1$ and $F_{n+1} = F_n + F_{n-1}$. Find all pairs of positive integers (x, y) such that $5F_x - 3F_y = 1$.

293. Let $\{F_n\}$ be the Fibonacci sequence. Show that $\gcd(F_m, F_n) = F_{\gcd(m,n)}$ for all $m, n \in \mathbb{N}$.

294. An integer sequence $\{a_n\}$ is defined by $a_0 = 0$, $a_1 = 1$ and $a_{n+2} = 2a_{n+1} + a_n$. Show that 2^k divides a_n if and only if 2^k divides n.

295. The sequence $\{x_n\}_{n\geq 1}$ is defined by $x_1 = x_2 = 1$ and $x_{n+2} = 14x_{n+1} - x_n - 4$. Prove that x_n is always a perfect square.

296. Let $\{u_n\}_{n\geq 0}$ be a sequence of positive integers with $u_0 = 1$ and $u_{n+1} = au_n + b$ for $a, b \in \mathbb{N}$. Prove that for any choice of a and b the sequence contains infinitely many composite numbers.

297. An integer sequence $\{a_n\}_{n\geq 1}$ is defined by $a_1 = 1$ and $a_{n+1} = a_n + \lfloor \sqrt{a_n} \rfloor$. Show that a_n is a square if and only if $n = 2^k + k - 2$ for some $k \in \mathbb{N}$.

298. Let $a_1, \ldots, a_{2023}$ be positive integers with $a_1 = 5$, $a_2 = 25$ and $a_{n+2} = 7a_{n+1} - a_n - 6$ for $n = 1, \ldots, 2021$. Prove that there exist integers x, y with $a_{2023} = x^2 + y^2$.

299. An integer sequence $\{a_n\}$ satisfies $2^n = \sum_{d|n} a_d$ for all $n \in \mathbb{N}$. Show that $n \mid a_n$ for all n.

300. Suppose $a_1, a_2, \ldots$ is a sequence of integers and d is some integer such that for all natural n: $|a_n|$ is prime, and $a_{n+2} = a_{n+1} + a_n + d$. Show that the sequence is constant.

301. Do there exist a sequence $a_1, a_2, \ldots$ of positive integers such that for any positive integers i, j we have $d(a_i + a_j) = i + j$, where $d(n)$ is the number of positive divisors of n?

302. Let a and b be positive integers such that $2a^2 + a = 3b^2 + b$. Prove that $a - b$ is a perfect square.

303. Let n be a positive integer such that $2 + 2\sqrt{28n^2 + 1}$ is an integer. Show that it is the square of an integer.

304. A natural number n has property P if, for all a, n^2 divides $a^n - 1$ whenever n divides $a^n - 1$.

(a)Show that every prime n has property P.

(b)Show that there are infinitely many composite n with property P.

305. Let k be a positive integer. Show that there are infinitely many perfect squares of the form $n \cdot 2^k - 7$, where n is a positive integer.

306. Determine all composite integers $n > 1$ with the following property: if $d_1, \ldots, d_k$ are all the positive divisors of n with $1 = d_1 < d_2 < \cdots < d_k = n$, then d_i divides $d_{i+1} + d_{i+2}$ for every $1 \leq i \leq k-2$.

307. Determine all integers $m \geq 2$ for which there exists an integer $n \geq 1$ with $\gcd(m, n) = d$ and $\gcd(m, 4n + 1) = 1$.

308. Determine all the sets of six consecutive positive integers such that the product of some two of them, added to the product of some other two, equals the product of the remaining two.

309. Find the largest positive integer k such that there is a set $A \subseteq \{1, 2, \ldots, 100\}$ with k elements such that for any $a, b \in A$, a divides b if and only if $s(a)$ divides $s(b)$, where $s(k)$ is the digit sum of k.

310. Let m be a positive integer. Define $\{a_n\}_{n\geq 0}$ by $a_0 = 0$, $a_1 = m$, $a_{n+1} = m^2 a_n - a_{n-1}$. Prove that an ordered pair (a, b) of non-negative integers with $a \leq b$ solves $\dfrac{a^2 + b^2}{ab + 1} = m^2$ if and only if $(a, b) = (a_n, a_{n+1})$ for some $n \geq 0$.

311. Let $\{K_n\}_{n\geq 1}$ be defined by $K_1 = 2$, $K_2 = 8$ and $K_{n+2} = 3K_{n+1} - K_n + 5(-1)^n$. Prove that if K_n is prime then n is a power of 3.

312. The sequence $\{a_n\}_{n\geq 1}$ is defined by $a_1 = 1$, $a_2 = 2$, $a_3 = 24$ and

$$a_n = \frac{6a_{n-1}^2 a_{n-3} - 8a_{n-1}a_{n-2}^2}{a_{n-2}a_{n-3}} \quad (n \geq 4).$$

Show that a_n is an integer for all n, and that $n \mid a_n$ for every $n \in \mathbb{N}$.

313. Let k be a positive integer. Prove that there exists an infinite monotone increasing sequence of integers $\{a_n\}_{n\geq 1}$ such that a_n divides $a_{n+1}^2 + k$ and a_{n+1} divides $a_n^2 + k$ for all $n \in \mathbb{N}$.

314. An integer $n \geq 2$ with exactly s divisors $1 = d_1 < \cdots < d_s = n$ is *good* if there is k with $2 \leq k \leq s$ and $d_k > 1 + d_1 + \cdots + d_{k-1}$; otherwise it is *bad*.

(a)Show that there are infinitely many bad integers.

(b)Prove that among any seven consecutive integers greater than 2 there are at least four good integers.

(c)Show that there are infinitely many runs of seven consecutive good integers.

315. Let $n > 1$ and let $a_1, \ldots, a_{n+1}$ be distinct positive integers. Does there exist a polynomial $p(x) \in \mathbb{Z}[x]$ of degree $\leq n$ such that $\gcd(p(a_i), p(a_j)) > 1$ for all $i < j$, while $\gcd(p(a_i), p(a_j), p(a_k)) = 1$ for all $i < j < k$?

316. Let $N \geq 1$ and let ℓ be a positive integer. Call a positive integer k *nice* if $k! + \ell$ is a perfect square. Prove that for every positive integer $n \geq \ell$, the set $\{1, 2, \ldots, n^2\}$ contains at most $n^2 - n + \ell$ nice integers.

317. Let S be a set of 100 positive integers with the property: among every four numbers of S there is a number dividing each of the other three, or a number equal to the sum of the other three. Prove that S contains a number dividing all the other 99 numbers of S.

318. Let $\alpha_1, \ldots, \alpha_d, \beta_1, \ldots, \beta_e \in \mathbb{C}$ be such that $f_1(x) = \prod_{i=1}^{d}(x - \alpha_i)$ and $f_2(x) = \prod_{i=1}^{e}(x - \beta_i)$ have integer coefficients. Suppose there exist $g_1, g_2 \in \mathbb{Z}[x]$ with $f_1 g_1 + f_2 g_2 = 1$. Prove that

$$\left| \prod_{i=1}^{d} \prod_{j=1}^{e} (\alpha_i - \beta_j) \right| = 1.$$

319. For a non-empty finite set A of positive integers let $\operatorname{lcm}(A)$ be the least common multiple of its elements and $d(A)$ the number of prime factors of $\operatorname{lcm}(A)$ counted with multiplicity. For a finite set S of positive integers put

$$f_S(x) = \sum_{\emptyset \neq A \subseteq S} \frac{(-1)^{|A|} x^{d(A)}}{\operatorname{lcm}(A)}.$$

Prove that $-1 \leq f_S(x) \leq 0$ whenever $0 \leq x \leq 2$.

320. Let n be an odd positive integer and let $f_n(x, y, z) = x^n + y^n + z^n + (x + y + z)^n$. Prove that there exist infinitely many values of n such that

$$f_n(x, y, z) \equiv (x + y)(y + z)(z + x) g_n(x, y, z) h_n(x, y, z) \pmod 2$$

for some integer polynomials g_n, h_n, neither constant modulo 2.

Chapter 3

Geometry

Geometry is very diverge subject. The IOQM papers ask for a length, an area or an angle and expect a number; the RMO and INMO papers ask you to prove three points are collinear. Both are here, sorted by what you actually have to do rather than by what the answer looks like.

The first section is angle chasing, which is where the Olympiad Arena geometry handout began and where most students should begin too. After that the sections run through triangle centres, circle configurations, computation, and finally collinearity and concurrency. Ruler-and-compass constructions have a section of their own at the end.

3.1. Angle Chasing and Cyclic Quadrilaterals

321. Quadrilateral $ABCD$ has $AB = BC = CD$, $\angle ABC = 70^\circ$ and $\angle BCD = 170^\circ$. What is the degree measure of $\angle BAD$?

322. If $ABCDE$ is a regular pentagon and $MNCD$ is a square with M and N inside $ABCDE$, compute $m\angle AMN - m\angle EAM$ in degrees.

323. Triangle ABC is isosceles with $AB = AC$. The bisectors of $\angle ABC$ and $\angle ACB$ meet at I. If $\angle CIA = 130^\circ$, compute $\angle CAB$.

324. In $\triangle PQR$, $\angle Q + 10^\circ = \angle R$. Let M be the midpoint of QR. If $m\angle PMQ = 100^\circ$, find $\angle Q$ in degrees.

325. AMC is an isosceles triangle with $AC = BC$, and CBD is an isosceles triangle with $CB = DB$. The line BD meets AC at a right angle. If $\angle A = 57^\circ$, what is $\angle D$?

326. A triangle $\triangle ABC$ lies in the plane and a point E is given on segment AC. Let D be a point in the plane such that lines AD and BE are parallel. Given $\angle EBC = 25^\circ$, $\angle BCA = 32^\circ$ and $\angle CAB = 60^\circ$, find the smallest possible value of $\angle DAB$ in degrees.

327. Consider quadrilateral $ABCD$ with $\angle DAC = 70^\circ$, $\angle BAC = 40^\circ$, $\angle BDC = 20^\circ$ and $\angle CBD = 35^\circ$. Let P be the intersection of AC and BD. Find $\angle BPC$.

328. An acute isosceles triangle ABC is inscribed in a circle. Tangents to the circle at B and C meet at D. If $\angle ABC = \angle ACB = 2\angle D$, find the radian measure of $\angle A$.

329. Cyclic quadrilateral $ABCD$ satisfies $\angle ABD = 70^\circ$, $\angle ADB = 50^\circ$ and $BC = CD$. Suppose AB meets CD at P and AD meets BC at Q. Compute $\angle APQ - \angle AQP$.

330. Triangle ABC has a right angle at B. The perpendicular bisector of AC meets segment BC at D, and the perpendicular bisector of AD meets AB at E. Suppose CE bisects the acute angle $\angle ACB$. What is $\angle ACB$?

331. In $\triangle ABC$, $m\angle A = 90°$, $m\angle B = 45°$ and $m\angle C = 45°$. A point P inside $\triangle ABC$ satisfies $m\angle BPC = 135°$. Given that $\triangle PAC$ is isosceles, the largest possible value of $\tan\angle PAC$ can be written $s + t\sqrt{u}$ with s, t integers and u squarefree. Compute $100s + 10t + u$.

332. Let $ABCD$ be a cyclic quadrilateral. Let I_A, I_B, I_C, I_D be the incentres of triangles BCD, ACD, ABD, ABC respectively. Prove that $I_AI_BI_CI_D$ is also cyclic.

333. Let ABC be an acute-angled triangle with incentre I. Let D, E, F be the feet of the perpendiculars from I to BC, CA, AB. Prove that AI, BI, CI are the angle bisectors of triangle DEF.

334. Consider a trapezium $AB\Gamma\Delta$ with $A\Delta \parallel B\Gamma$ and $\angle A = 120°$. Let E be the midpoint of AB and let O_1, O_2 be the circumcentres of triangles $AE\Delta$ and $BE\Gamma$. Find the ratio of the area of the trapezium to the area of triangle O_1EO_2.

335. $ABCD$ is a convex quadrilateral with $\angle ABC = \angle ADC$, and E, F, G, H are the midpoints of AC, BD, AD, CD respectively. Prove that

(a)$EFGH$ is a cyclic quadrilateral;

(b)$\angle AEF = \angle ACB - \angle ACD$.

336. In quadrilateral $ABCD$, $\angle CBA = 90°$, $\angle BAD = 45°$ and $\angle ADC = 105°$. Suppose $BC = 1 + \sqrt{2}$ and $AD = 2 + \sqrt{6}$. What is the closest integer to the length of AB?

337. In quadrilateral $ABCD$, CD is perpendicular to BC and to DA. A point E is chosen on segment CD so that $\angle AED = \angle BEC$. If $AB = 6$, $AD = 7$ and $\angle ABC = 120°$, compute $\dfrac{(AE + EB)^2}{2}$.

338. A circle is centred at O. Distinct pairs of points A, B and C, D are diametrically opposite on this circle. A point P is chosen on segment AD so that line BP meets the circle again at M and meets AC at X, with M the midpoint of PX. A point $Y \neq X$ is taken with $BX = BY$ and $CD \parallel XY$. If $\angle PYB = 10°$, find $\angle XCM$.

339. Let ABC be a triangle with $\angle ABC = 90°$. A circle Γ with centre on segment AC is tangent to AB at D and to BC at E. Let F and G be the points where Γ meets line AC, with A, F, G, C in that order. Let P be on segment AD with $PD = AF$, and Q on segment CE with $QE = CG$. Find the angle between lines PQ and AC.

340. In a convex quadrilateral $ABCD$, $\angle ABC = \angle ADC = 90°$. A point P is chosen on the diagonal BD such that $\angle APB = 2\angle CPD$; points X, Y are chosen on segment AP such that $\angle AXB = 2\angle ADB$ and $\angle AYD = 2\angle ABD$. Prove that $BD = 2XY$.

341. In triangle ABC the angle at B is $120°$. Let A_1, B_1, C_1 be points on sides BC, CA, AB such that AA_1, BB_1, CC_1 are the angle bisectors of the triangle. Determine $\angle A_1B_1C_1$.

342. Let ABC be an acute triangle with $\angle BAC = 60°$ and $AB > AC$. Let I be the incentre and H the orthocentre. Prove that $2\angle AHI = 3\angle ABC$.

343. O is the circumcentre of convex quadrilateral $ABCD$. Given that $AC \neq BD$ and AC, BD meet at E, and P is an interior point with $\angle PAB + \angle PCB = \angle PBC + \angle PDC = 90°$, prove that O, P, E are collinear.

344. In an acute-angled triangle ABC, H is the orthocentre and D is any point on side BC. Points E, F lie on segments AB, AC so that A, B, D, F and A, C, D, E are concyclic. Segments BF and CE meet at P. Let L be a point on HA such that LC is tangent to the circumcircle of $\triangle PBC$ at C, and let BH and CP meet at X. Prove that D, X and L are collinear.

345. Let $ABCD$ be a convex quadrilateral with $\angle BAC = \angle ADC$. Let M be the midpoint of diagonal AC. The side AD contains a point E such that $ABME$ is a parallelogram. Let N be the midpoint of AE. Prove that line AC touches the circumcircle of triangle DMN at M.

3.2. Triangle Centres

346. In triangle ABC with orthocentre H, the internal bisector of $\angle BAC$ meets BC at Y. Given $AH = 4$, $AY = 6$ and the distance from Y to AC equal to $\sqrt{15}$, write BC in simplest form $a\sqrt{b}$ and find $a + b$.

347. Rectangle $HOMF$ has $HO = 11$ and $OM = 5$. Triangle ABC has orthocentre H and circumcentre O; M is the midpoint of BC and the altitude AF meets BC at F. Find the length of BC.

348. In an acute triangle ABC, D and E are the feet of the altitudes from B and C. Lines BD and CE meet at H. The circle with diameter DE meets sides AB and AC again at F and G. Lines FG and AH meet at K. Given $BC = 25$, $BD = 20$ and $BE = 7$, write $AK = a/b$ in lowest terms and find the sum of the digits of $a - b$.

349. Let ABC be an acute triangle with $BC = 48$. Let M be the midpoint of BC, and let D and E be the feet of the altitudes from B and C to AC and AB. Let P be the intersection of the line through A parallel to BC with line DE. If $AP = 10$, compute PM.

350. ABC is a triangle, AD and CE are the bisectors of $\angle A$ and $\angle C$, and BM, BN are the perpendiculars from B to CE and AD respectively. Given $AB = 2023$, $BC = 2024$ and $AC = 2025$, find the sum of the squares of the digits of MN.

351. Let $\triangle ABC$ have A-excircle, B-excircle and C-excircle of radii R_A, R_B, R_C. If $R_A R_B R_C = 384$ and the perimeter of $\triangle ABC$ is 32, what is the area of $\triangle ABC$?

352. In $\triangle ABC$, D is on AC so that BD bisects $\angle B$. A point E is on AB and CE meets BD at P. Quadrilateral $BCDE$ is cyclic, $BP = 12$ and $PE = 4$. Compute the ratio $\dfrac{AC}{\overline{AE}}$.

353. Given that the incentre of $\triangle ABC$ is I, and the circle with centre I touches AC, AB at E, F respectively. M is a point on segment EF. Prove that the areas of triangles MAB and MAC are equal if and only if $MI \perp BC$.

354. Given that CH is the altitude of acute triangle ABC, O is the circumcentre and T is the projection of C on line AO. Prove that TH passes through the midpoint of BC.

355. In triangle ABC with $AB > AC$, AE is tangent to the circumcircle at A, and D lies on AB with $AD = AC = AE$. Prove that segment DE passes through the incentre of $\triangle ABC$.

356. Let ABC be an acute-angled isosceles triangle with $AB = AC$ and circumcentre O. Lines BO and CO meet AC, AB at B', C'. A line ℓ is drawn through C' parallel to AC. Prove that ℓ is tangent to the circumcircle of $\triangle B'OC$.

357. Let ABC be an acute triangle with orthocentre H. Let D be the foot of the altitude from A to BC. Let T be a point on the circle with diameter AH such that this circle is internally tangent to the circumcircle of triangle BDT. Let N be the midpoint of segment AH. Prove that $BT \perp CN$.

358. Given an acute-angled triangle ABC with circumcentre O and orthocentre H, let A_1, B_1, C_1 be the midpoints of BC, AC, AB. Rays HA_1, HB_1, HC_1 cut the circumcircle at A_0, B_0, C_0. Prove that O, H and H_0 are collinear, where H_0 is the orthocentre of $\triangle A_0 B_0 C_0$.

359. In an acute triangle ABC, D, E and F are the points of tangency of the incircle (centre I) with BC, CA, AB. Let M be the foot of the perpendicular from D to EF, and let P be on DM with $DP = MP$. If H is the orthocentre of $\triangle BIC$, prove that PH bisects EF.

360. In $\triangle ABC$ let G be the centroid and let the circumcentres of $\triangle BCG$, $\triangle CAG$ and $\triangle ABG$ be I, J, K. The line through I and the midpoint of BC meets KJ at Y. If the radius of circle K is 5, the radius of circle J is 8, and $AG = 6$, what is the integer nearest to KY?

361. In triangle ABC, let O be the circumcentre. The incircle touches BC, CA and AB at D, E, F. Let G be the centroid of $\triangle DEF$. Suppose the inradius and circumradius of ABC are 3 and 8. Over all such triangles, pick one maximising the area of $\triangle AGO$. Writing $AG^2 = \frac{m}{n}$ in lowest terms, find the sum of the digits of m.

362. Let ABC be a triangle with $AB = 5$, $BC = 6$ and $CA = 7$. Let Γ be the incircle with centre I. The circumcircle of BIC meets Γ at X_1, X_2; the circumcircle of CIA meets Γ at Y_1, Y_2; the circumcircle of AIB meets Γ at Z_1, Z_2. The area of the triangle determined by lines X_1X_2, Y_1Y_2, Z_1Z_2 equals $\frac{m\sqrt{p}}{n}$ with $\gcd(m, n) = 1$ and p squarefree. Compute $m + n + p$.

363. In $\triangle ABC$ with incentre I, the incircle meets BC, CA, AB at D, E, F. Given $EF \cap BC = J$, $BJ = 100$ and $CJ = 60$. Let the A-excircle meet BC at K; line IK passes through L, a point dividing CA internally in ratio $5 : 3$. If the inradius is r, calculate $\dfrac{r^2}{5}$.

364. Let I be the incentre of $\triangle ABC$ and ω its incircle. Let D be the intersection of ω with BC. Suppose X, Y are points on ω such that lines IX, IY and AD are tangent to the circumcircle of $\triangle AXY$. Prove that line XY bisects segment BC.

365. Let M be the midpoint of BC in triangle ABC. The circle with diameter BC, call it ω, meets AB, AC at D, E. A point P lies inside $\triangle ABC$ with $\angle PBA = \angle PAC$, $\angle PCA = \angle PAB$, and $2PM \cdot DE = BC^2$. A point X lies outside ω with $XM \parallel AP$ and $\dfrac{XB}{XC} = \dfrac{AB}{AC}$. Determine $\dfrac{\angle BXC + \angle BAC}{2}$.

3.3. Circles, Tangency and Power of a Point

366. A circle with centre O and radius 1 contains a chord AB of length 1, and M is the midpoint of AB. If the perpendicular to AO through M meets AO at P, compute $[MAP]$. If the answer is $\frac{\sqrt{a}}{b}$ with a, b coprime, find $a + b$.

367. Define a common chord of two intersecting circles to be the segment joining their intersection points. Let $\omega_1, \omega_2, \omega_3$ have radii $3, 5, 7$, arranged so that the common chord of ω_1, ω_2 is a diameter of ω_1, the common chord of ω_1, ω_3 is a diameter of ω_1, and the common chord of ω_2, ω_3 is a diameter of ω_2. Compute the square of the area of the triangle formed by the three centres.

368. Let λ_1 and λ_2 be circles centred at O and P, externally tangent at Q. Draw D on λ_1 and E on λ_2 so that line DE is tangent to both circles. If $OQ = 1$ and the area of quadrilateral $ODEP$ is 520, what is PQ?

369. Let AD be the diameter of a circle, B a point on the circle and C on AD such that A, B, C form a right triangle with the right angle at C. The hypotenuse is four times the square root of the triangle's area. If $BC = 30$, what is half the radius of the circle?

370. Circle J_1 of radius 77 is centred at X and circle J_2 of radius 39 is centred at Y. A point A lies on J_1 and on line XY, with A and Y on opposite sides of X. Let Ω be the unique circle simultaneously tangent to the tangent segments from A to J_2 and internally tangent to J_1. If $XY = 157$, what is the radius of Ω?

371. Let O be a circle with chord AB. The perpendicular bisector of AB meets O at C and D and meets AB at its midpoint E. A circle O' with diameter ED meets chord AD at F. Given $EC = 12$ and $EF = 7$, compute the radius of O; if the answer is $\frac{m}{n}$ in lowest terms, find $m + n$.

372. Two non-intersecting circles ω and Ω have centres C_ω and C_Ω, with the radius of Ω strictly larger. The two common external tangents meet at P, and an internal tangent meets the external tangents at X and Y. Suppose the radius of ω is 4, the circumradius of $\triangle PXY$ is 9, and XY bisects PC_Ω. Compute XY to the nearest integer.

373. Two non-intersecting circles ω and Ω have centres C_ω and C_Ω, with the radius of Ω strictly larger. The two common external tangents meet at P, and an internal tangent meets the external tangents at X and Y. Suppose the radius of ω is 4, the circumradius of $\triangle PXY$ is 9, and XY bisects PC_Ω. Compute $\dfrac{XY^2}{16}$.

374. Let circles ω_1 and ω_2 meet at P and Q. The external common tangent closer to Q touches ω_1 at A and ω_2 at B. A point T lies on segment PQ with $\angle ATB = 90^\circ$. Given $AT = 6$, $BT = 8$ and $PT = 4$, compute $9PQ$.

375. Circle ω has radius 10, centre O, and diameter AB. A point C on ω satisfies $AC = 12$. Let P be the circumcentre of $\triangle AOC$. Line AP meets ω again at Q. Then $\dfrac{AP}{AQ} = \frac{m}{n}$ in lowest terms; compute $m + n$.

376. Let ω be a circle with centre O and a chord AB of length 6. Let M be a point outside ω with $AM = AB$. Let X be the foot of the perpendicular from O to AB. Line XO extended meets the bisector of $\angle BAM$ at E. Let Y be a point on AM with $EX = EY$. Given $XY = \sqrt{162/5}$, find BE^2.

377. AB is the diameter of a semicircle with centre O. C is the midpoint of arc AB, M is the midpoint of chord AC, and $CH \perp BM$ at H. Prove that $CH^2 = AH \cdot OH$.

378. Let Γ be a circumcircle. A circle with centre O touches segment BC at P and touches the arc BC of Γ not containing A at Q. If $\angle BAO = \angle CAO$, prove that $\angle PAO = \angle QAO$.

379. It is given that the circle with centre O touches the sides AB and AC of $\triangle ABC$ at P and Q respectively, and is internally tangent to the circumcircle of $\triangle ABC$ at D. Prove that $\angle POQ = 2\angle MDC$, where M is the midpoint of PQ.

380. Let ω_1 and ω_2 be intersecting circles with radii 12 and 15. Suppose Γ is a circle such that ω_1 and ω_2 are internally tangent to Γ at X_1 and X_2. Similarly ℓ is a line tangent to ω_1 and ω_2 at Y_1 and Y_2. If $X_1X_2 = 18$ and $Y_1Y_2 = 9$, what is the closest integer to the radius of Γ?

381. A, B are two fixed points on circle W_1. Circle W_2 is tangent to AB at B and its centre lies on W_1. Line ADE meets W_2 at D, E, and line BD meets W_1 again at F. Prove that BE is tangent to W_1 if and only if $DF = DB$.

382. Points A and B are fixed on a circle c_1. A circle c_2, whose centre lies on c_1, touches line AB at B. Another line through A meets c_2 at D and E, with D between A and E. Line BD meets c_1 again at F. Prove that EB is tangent to c_1 if and only if D is the midpoint of segment BF.

383. Circles Ω and Γ meet at A and B. The line through their centres meets Ω and Γ at P and Q respectively, on the same side of line AB, with Q closer to AB. A circle δ on the same side of AB as P and Q touches segment AB at D and touches Γ at T. Line PD meets δ and Ω again at K and L. Prove that $\angle QTK = \angle DTL$.

384. Let ω be a circle. Points A, B, C, X, D, Y lie on ω in this order, with BD a diameter and $DX = DY = DP$, where P is the intersection of AC and BD. Let E, F be the intersections of line XP with lines AB, BC. Prove that B, E, F, Y lie on a circle.

385. The tangent at C to Ω, the circumcircle of scalene triangle ABC, meets AB at D. Through D a line is drawn meeting segments AC and BC at K and L. On segment AB, points M and N are marked with $AC \parallel NL$ and $BC \parallel KM$. Lines NL and KM meet at P inside $\triangle ABC$. Let ω be the circumcircle of MNP, and suppose CP meets ω again at Q. Show that DQ is tangent to ω.

386. Let ω_1, ω_2 be two non-intersecting circles with centres O_1, O_2 and radii $r_1 < r_2$. Let AB and XY be the two internal common tangents, where A, X lie on ω_1 and B, Y on ω_2. The circle with diameter AB meets ω_1 and ω_2 at P and Q. If $\angle AO_1P + \angle BO_2Q = 180^\circ$, find $\left(\dfrac{PX}{QY}\right)^4$ in terms of r_1, r_2.

387. A triangle ABC is given with $|AB| > |AC|$. A line ℓ is tangent at A to the circumcircle of ABC. A circle centred at A with radius $|AC|$ cuts AB at D and cuts ℓ at E, F, where C and E are in the same half-plane with respect to AB. Prove that line DE passes through the incentre of ABC.

388. Quadrilateral $ABCD$ is inscribed in circle ω. Segments AC and BD meet at E. Circle γ passes through E and is tangent to ω at A. Suppose the circumcircle of $\triangle BCE$ is tangent to γ at E and tangent to line CD at C. Suppose ω has radius 3 and γ has radius 2. Find the nearest integer to BD.

389. Two circles O_1 and O_2 meet at M and N. The common tangent nearer M touches O_1 and O_2 at A and B. Let C, D be the reflections of A, B in M. The circumcircle of $\triangle DCM$ meets O_1 and O_2 again at E and F. Prove that triangles MEF and NEF have equal circumradii.

390. Given circles ω_1 and ω_2 intersecting at X and Y, let ℓ_1 be a line through the centre of ω_1 meeting ω_2 at P and Q, and ℓ_2 a line through the centre of ω_2 meeting ω_1 at R and S. Prove that if P, Q, R, S lie on a circle, then the centre of that circle lies on line XY.

391. Let three non-overlapping, mutually external circles $\Gamma_1, \Gamma_2, \Gamma_3$ be given in the plane. For each point P outside all three, construct A_i, B_i on Γ_i so that PA_i and PB_i are both tangent to Γ_i. Call P *exceptional* if the three lines A_1B_1, A_2B_2, A_3B_3 are concurrent. Show that every exceptional point lies on one circle.

392. Let ABC be a triangle with sides $AB = 34$, $BC = 15$, $AC = 35$ and let Γ be the circle of smallest possible radius passing through A and tangent to BC. Let the second intersections of Γ with sides AB, AC be X, Y. Let ray XY meet the circumcircle of $\triangle ABC$ at Z. If $AZ = \frac{p}{q}$ for coprime integers p, q, find $p - 15q$.

393. Let A, B, C, D be fixed points in this order on a line. Let ω be a variable circle through C and D meeting the perpendicular bisector of CD at X and Y. Let Z and T be the second intersections of AX and BY with ω. Prove that ZT passes through a fixed point independent of ω.

3.4. Lengths, Areas and Computation

394. In $\triangle ABC$, $AB = 4$, $BC = 6$ and $AC = 8$. Squares $ABQR$ and $BCST$ are drawn external to and in the same plane as $\triangle ABC$. If $QT = a\sqrt{b}$ with a, b coprime integers, find ab.

395. Polygon $A_1A_2 \ldots A_n$ is a regular n-gon. For some integer $k < n$, quadrilateral $A_1A_2A_kA_{k+1}$ is a rectangle of area 6. If the area of $A_1A_2 \ldots A_n$ is 60, find n.

396. Given $\triangle BMT$ with $BM = 4$, $BT = 6$ and $MT = 8$. A point A lies on line BM and Y on line BT so that $AY \parallel MT$ and the incentre of $\triangle BAY$ lies on MT. Compute $5AY$.

397. Let $ABCD$ be a square of side 5, and E a point on BC with $BE = 3$, $EC = 2$. Let P be a variable point on the diagonal BD. Determine the nearest integer to PB when $PE + PC$ is smallest.

398. Soham constructs a square $BERK$ of side length 10. Points T, O, W, N are the midpoints of BE, ER, RK, KB. A square $CALI$ is constructed whose edges contain the vertices of $BERK$, with CA parallel to BO. Compute half the area of $CALI$.

399. The convex quadrilateral $ABCD$ has area 441. Let E be the intersection of the diagonals, with $AE = 12$, $BE = 16$, $CE = 30$ and $DE = 5$. The perimeter of $ABCD$ is $m + n\sqrt{p}$ with m, n positive integers and p prime. Find $m + n - p$.

400. In rectangle $ABCD$, $AB = 20$ and $AD = 19$. A point E lies on AD with $AE = 4$. Let the incircle of $\triangle CDE$ be tangent to CE at F. A circle tangent to AB and BC is tangent to CE at G. The distance FG can be written $\frac{m}{n}$ with m, n coprime; find $m + n$.

401. An isosceles triangle R has side lengths $17, 17, 16$. Region S consists of the points inside R at distance at least 2 from the sides of R. The ratio of the area of S to the area of R is $\frac{m}{n}$ in lowest terms. Find $n - m$.

402. Points A, B, C lie on a line in that order with $AB = 7$ and $BC = 4$. Points D and E lie on the same side of line AC so that $\triangle ABD$ and $\triangle BCE$ are equilateral. The line through the centroids of $\triangle ABD$ and $\triangle BCE$ meets BE and CE at F and G. The ratio of the area of $\triangle EFG$ to the area of $\triangle BCE$ is $\frac{m}{n}$ in lowest terms. Find $n - 2m$.

403. If the area of the smallest triangle which can contain six congruent, non-overlapping unit circles is $a + b\sqrt{c}$ with $a, b, c \in \mathbb{Z}$ and c squarefree, find $a + b + c$.

404. Let A, B, M, C, D be distinct points on a line with $AB = BM = MC = CD = 6$. Circles ω_1 and ω_2 with centres O_1, O_2 of radii 4 and 9 are tangent to line AD at A and D, with O_1, O_2 on the same side. Let P satisfy $PB \perp O_1M$ and $PC \perp O_2M$. Determine the product of the non-zero digits of $PO_2^2 - PO_1^2$.

405. In triangle $\triangle ABC$, E and F are the feet of the altitudes from B to AC and from C to AB. Line BC and the tangent to the circumcircle at A meet at X. Let Y be the intersection of line EF with the line through A parallel to BC. If $XB = 4$, $BC = 8$ and $EF = 4\sqrt{3}$, compute XY^2.

406. Let $ABCDEF$ be a regular hexagon with side length 1, and let G be the midpoint of CD. Define H to be the unique point on side DE such that $AGHF$ is a trapezoid. Find the length of the altitude from H to AG. If the answer is $\frac{a\sqrt{b}}{c}$, find b/c.

407. In square $ABCD$ with side length 2, let P and Q both lie on side AB with $AP = BQ = \frac{1}{2}$. Let E be a point on the edge of the square maximising $\angle PEQ$. Find the area of triangle PEQ.

408. In quadrilateral $ABCD$, $AB = 2$, $AD = 3$, $BC = CD = \sqrt{7}$ and $\angle DAB = 60°$. Semicircles γ_1, γ_2 are erected exterior to the quadrilateral on diameters AB and AD; points $E \neq B$ on γ_1 and $F \neq D$ on γ_2 are chosen so that $\triangle CEF$ is equilateral. If the area of $\triangle CEF$ is $\frac{a\sqrt{b}}{c}$, find $a - bc$.

409. Suppose $T = A_0A_1A_2A_3$ is a tetrahedron with $\angle A_1A_0A_3 = \angle A_2A_0A_1 = \angle A_3A_0A_2 = 90°$, $A_0A_1 = 5$, $A_0A_2 = 12$, $A_0A_3 = 9$. A cube of side s is inscribed in T with one vertex at A_0, three edges along A_0A_1, A_0A_2, A_0A_3, and the opposite vertex on $\triangle A_1A_2A_3$. If $s = \frac{a}{b}$ in lowest terms, find $a - 2b$.

410. A rhombus R has short diagonal of length 1 and long diagonal 2023. Let R' be the rotation of R by 90° about its centre, U their union and I their intersection. Compute the sum of the digits of the reciprocal of the ratio of the area of I to the area of U.

411. George draws an isosceles triangle AB_0C_0 with $AB_0 = AC_0 = 41$ and $B_0C_0 = 18$. He then draws points B_i and C_i on sides AB_0 and AC_0 with $B_iB_{i+1} = 1$ and $C_iC_{i+1} = 1$ (with $B_{41} = C_{41} = A$), and colours in the triangles $B_iC_iC_{i+1}$ for i from 0 to 40. What is the closest integer to half the total area he colours in?

412. Given the pentagon $ABCDE$ of area $\frac{\sqrt{3}}{4}AB \cdot BC + 4$ with $\angle ABC = 120^\circ$, let BF, AH and CG be the angle bisectors of $\triangle ABC$ with $DE \parallel AC$ and $\angle DCA = \angle GFH$. Given $\frac{AC}{DE} = 3$, $HF = \frac{3}{2}$ and $GF = \sqrt{3}$, and that a circle can be inscribed in $ACDE$, find $R^2 - r^2$, where R is the circumradius of $\triangle ABC$ and r the circumradius of $\triangle GBH$.

413. In an isosceles triangle ABC with $AC = BC$, AD and CF are altitudes on BC and AB. Let ℓ be the line through B parallel to AD, meeting CF at G. Assume $GD \parallel AC$; let GD cut AB at Y, and let ω be the circumcircle of $\triangle GFA$ with radius 10. Given $FY = 3$ and $GY = 5$, the perimeter of $\triangle ABC$ has the form $\frac{a\sqrt{b}}{c}$ with a, b, c coprime. Find $a + b + c$.

414. Jiraa draws $\triangle ABC$ with $AB = \sqrt{3}$, $BC = 1$ and $AC = 2$, and cuts it out. The bisector of $\angle ABC$ meets AC at D. He folds $\triangle ABD$ over BD, sending A to A'. After folding $\triangle A'CD$ over CD, what area of the resulting figure is covered by three layers of paper? Find the integer closest to 100 times your answer.

415. On regular hexagon $GOBEAR$ with side length 2, bears are placed at G, B, A, forming an equilateral triangle. At $t = 0$ all move clockwise along the sides at the same pace, stopping once each has travelled 1 unit. What is the total area swept out by the triangle formed by the three bears? If the answer in simplest form is $\frac{a\sqrt{b}}{c}$, find $\frac{ac}{b}$.

416. In the quadrilateral $MARE$ inscribed in a unit circle ω with diameter AM, the point E lies on the bisector of $\angle RAM$. Given that triangles RAM and REM have the same area, find the area of quadrilateral $MARE$. If the answer in simplest form is $\frac{a\sqrt{b}}{c}$, find $a + b + c$.

417. Three circles with radii 23, 46, 69 are mutually externally tangent. Find the area of the largest circle that can be drawn in the region enclosed by these three circles.

418. Let O be the centre of the circumcircle of right triangle ABC with $\angle ACB = 90^\circ$. Let M be the midpoint of minor arc AC and let N be a point on line BC with $MN \parallel BC$. Let P be the intersection of line AN with the circle and Q the intersection of line BP with MN. If $QN = 2$ and $BN = 8$, compute the radius of the circle.

419. Let $ABCD$ be a trapezoid with $AB \parallel CD$, $AD = 13$, $BC = 15$, $AB = 20$ and $CD = 34$. A point X lies inside with $\angle XAB = 2\angle XBA$ and $\angle XDC = 2\angle XCD$. Compute $XD - XA$.

420. $ABCD$ is a trapezium with CD and AB the upper and lower bases. Given $\angle ADC = 90^\circ$ and $AC \perp BD$, and $DE \perp BC$ at E, prove that $\dfrac{AE}{BE} = \dfrac{AC \cdot CD}{AC^2 - CD^2}$.

421. Let $\triangle XYZ$ have $YZ = 7$, $ZX = 6$, $XY = 5$, incentre I, and incircle touching YZ, ZX, XY at D, E, F. Let the circumcircle of $\triangle XEF$ meet the circumcircle of $\triangle XYZ$ again at $M \neq X$. Let P be the intersection of MI and EF. If $MP \cdot IP = \frac{a}{b}$ for coprime positive integers a, b, find the sum of the digits of $a + b$.

422. The trisectors of $\angle A$ in $\triangle ABC$ meet the circumcircle at D and E so that A, B, D, E, C lie on the circumcircle in this order. Given $AB = 4$, $AD = 16$ and $AC = 14$, compute BC. If the answer is $a\sqrt{b}$ with a, b coprime positive integers, find $a + b$.

423. Let $\triangle ABC$ satisfy $AB = 17$, $AC = \frac{70}{3}$ and $BC = 19$. Let I be the incentre and E the excentre opposite A. Suppose the circle with diameter IE meets AB beyond B at D. If $BD = \frac{a}{b}$ with a, b coprime, find $a + b$.

424. Let $\triangle ABC$ be a right triangle with $\angle A = 90^\circ$ and $BC = 38$. There exist points K and L inside the triangle with

$$AK = AL = BK = CL = KL = 14.$$

The area of quadrilateral $BKLC$ can be expressed as $n\sqrt{3}$ for a positive integer n. Find the product of the non-zero digits of n.

425. Let M be the midpoint of AC in $\triangle ABC$. Line $MD \parallel AB$ meets the tangent at A to the circumcircle at D. A point E lies on line AD with A the midpoint of DE. Let P be the second intersection of the circumcircle of $\triangle ABE$ with AC, and Q the second intersection of the circumcircle of $\triangle ADP$ with DM. Find $\dfrac{\angle QCB}{\angle BAC}$.

426. Let $\triangle ABC$ be acute with $AB < AC$ and circumcentre O. On sides AB and AC select points T and P with $OT = OP$. Let M, K and Λ be the midpoints of PT, PB and CT. Find the nearest integer value of $\dfrac{\angle TMK}{\angle M\Lambda K}$.

427. Let $ABCDEFGH$ be a regular octagon with side length $\sqrt{60}$. Let K denote the locus of all points K such that the circumcircles (possibly degenerate) of triangles HAK and DCK are tangent. Find the nearest integer to the hundredth part of the area of the region that K encloses.

428. Let $ABCDEFGH$ be a regular octagon with side length $\sqrt{60}$, and let K be the locus above. If the area enclosed by K is $(a + b\sqrt{c})\pi$, find $a - \dfrac{30c}{b}$.

429. Let ABC be a triangle with circumcircle ω, $AB = 11$, $AC = 13$ and $\angle A = 30^\circ$. Points D and E lie on segments AB and AC with $AD = 7$ and $AE = 8$. There is a unique point $F \neq A$ on minor arc AB of ω with $\angle FDA = \angle FEA$. Compute FA^2; if the answer is $\frac{a+b\sqrt{c}}{d}$, find $a + b + c - d$.

430. Triangle ABC has $AB = 15$, $BC = 22$, $AC = 20$. Points D, E, F lie on BC, AC, AB so that AD, BE, CF all pass through a point K. Let L be the second intersection of the circumcircles of BFK and CEK. Suppose $\frac{AK}{KD} = \frac{11}{7}$ and $BD = 6$. If $KL^2 = \frac{a}{b}$ with a, b coprime, find the sum of the digits of $a + b$.

431. Triangle ABC has $AB = 13$, $BC = 14$, $CA = 15$, circumcircle Γ and orthocentre H. Let AH meet Γ again at D, let BH meet AC at F and meet Γ again at G, and suppose DG meets AC at X. Compute the greatest integer at most the area of quadrilateral $HDXF$.

432. Points A and B lie on circle ω with centre O. Let X be a point inside ω with $XO = 2\sqrt{2}$, $XA = 1$, $XB = 3$ and $\angle AXB = 90^\circ$. Points Y and Z on ω satisfy $Y \neq A$ and $\triangle AXB \sim \triangle YXZ$ with the same orientation. Compute XY; if the answer is $\frac{a}{b}$ with a, b integers, find the minimum value of $a + b$.

433. Let $\triangle ABC$ have circumcentre O and circumradius r. Let E be on line AB and F, G points such that $EF \perp AE$ and O lies appropriately, as in the original configuration. Given $AC = \frac{65}{63}$, $BC = \frac{24}{13}r$ and $AB = \frac{126}{65}r$, compute $AF \cdot EG$.

434. Let O be the circumcentre of $\triangle ABC$, M the midpoint of BC, and E, F the feet of the altitudes from B and C. Line EF meets BC at P. The line through O perpendicular to BC meets the circumcircle at L (on major arc BC) and N, and meets BC at M. A point Q lies on line LA with $OQ \perp AP$. Given $\angle BAC = 60^\circ$ and $\angle AMC = 60^\circ$, compute the closest integer to $\dfrac{10\,OQ}{AP}$.

435. Let M be the centre of equilateral triangle $A_1A_2A_3$, and let N be an arbitrary point in the plane. The circle with MN as diameter meets line MA_i at B_i for $i = 1, 2, 3$. Prove that

$$MB_1^2 + MB_2^2 + MB_3^2 = NB_1^2 + NB_2^2 + NB_3^2.$$

436. Let ABC be a triangle. In a triangle ABC let D be the intersection of the bisector of $\angle C$ with side AB, and let F be the area of $\triangle ABC$. Prove that

$$2F\left|\frac{1}{AD} - \frac{1}{BD}\right| \leq AB.$$

437. Let ABC be a triangle and let the points $D \in (BC)$, $E \in (CA)$, $F \in (AB)$ satisfy $\dfrac{BD}{DC} = \dfrac{CE}{EA} = \dfrac{AF}{FB}$. Prove that if the circumcentres of triangles DEF and ABC coincide, then ABC is equilateral.

438. Quadrilateral $ABCD$ has an inscribed circle centred at O with radius r. AB meets CD at P, AD meets BC at Q, and the diagonals AC and BD meet at K. If the distance from O to line PQ is k, prove that $OK \cdot k = r^2$.

3.5. Collinearity, Concurrency and Configurations

439. In triangle ABC with $AB > AC$, the tangent at B to the circumcircle meets line AC at P. Let D be the reflection of B in P, and E the reflection of C in line BP. Prove that quadrilateral $ABED$ is cyclic.

440. In an acute triangle ABC, AM is the median on BC, and BK, KL are altitudes with M, K, L on BC, CA, AB. The line perpendicular to AM at A meets lines CL and BK at E and F. Prove that A is the midpoint of EF.

441. The circle taking side BC of $\triangle ABC$ as diameter meets lines AB and AC at E and F. The tangent lines at E and F meet at P. Line AP meets line BF at D. Prove that D, C, E are collinear.

442. In an acute triangle ABC, D, E, F are the midpoints of BC, AC, AB. On the extensions of EF, FD, DE take points P, Q, R with $AP = BQ = CR$. Prove that the circumcentre of $\triangle PQR$ is the orthocentre of $\triangle ABC$.

443. Given that line segments AB, CD, EF are three non-intersecting chords of a circle. Three quadrilaterals are formed by taking any two chords as a pair of opposite sides, and M, N, P are the intersections of the diagonals of these quadrilaterals. Prove that M, N, P are collinear.

444. Let ABC be a right-angled triangle with $\angle A = 90°$, let D be the foot of the altitude from A to BC, and let E be the midpoint of DC. The circumcircle of ABD meets AE again at F. Let X be the intersection of lines AB and DF. Prove that $XD = XC$.

445. In triangle ABC with $AB < AC$, D is a point on segment AC with $BD = CD$. A line parallel to BD meets segment BC at E and line AB at F. Let G be the intersection of AE and BD. Show that $\angle BCG = \angle BCF$.

446. In triangle ABC, the points D, E are on AB, AC so that $DE \parallel BC$. The incircle of $\triangle ADE$ touches DE at M, and the excircle on side BC of $\triangle ABC$ touches BC at N. BE and CD meet at P. Prove that M, N, P are collinear.

447. The points M, N are on sides AB, AC of $\triangle ABC$. Given $MC = AC$, $NB = AB$ and P the reflection of A in line BC, prove that PA bisects $\angle MPN$.

448. Let X, X_1, X_2 be points on sides AB, AC, BC of $\triangle ABC$ with $XX_1 \perp AC$, $X_1X_2 \perp BC$, $X_2X \perp AB$. Let Y, Y_1, Y_2 be points on BC, AC, AB with $YY_1 \perp AC$ and $Y_1Y_2 \perp AB$. Prove that $YY_2 \perp BC$ if $XY \parallel AC$.

449. Let $ABCD$ be a parallelogram whose diagonals meet at P. Let M be the midpoint of AB. Let Q be a point such that QA is tangent to the circumcircle of MAD and QB is tangent to the circumcircle of MBC. Prove that Q, M, P are collinear.

450. Let ABC be an acute triangle with orthocentre H, and let K be the foot of the altitude from A. A circle c through A and K meets sides AB and AC at M and N. The line through A parallel to BC meets the circumcircles of triangles AHM and AHN again at X and Y. Prove that $|XY| = |BC|$.

451. $ABCD$ is a convex quadrilateral. The circumcircle of $\triangle ABC$ meets CD, DA at P, Q, and the circumcircle of $\triangle ACD$ meets AB, BC at R, S. Lines BP, BQ meet line RS at M, N. Prove that M, N, P, Q are concyclic.

452. In an acute triangle ABC, D is on AB, and the circumcircles of triangles BCD and ACD meet AC and BC at E and F. Let O be the circumcentre of $\triangle CEF$. Prove that the circumcentres of triangles ADE, ADC, DBF, DBC and the points D and O are concyclic, and that $OD \perp AB$.

453. Let $ABCD$ be a convex quadrilateral with AC bisecting $\angle BAD$. Points E and F are on sides BC and CD with $EF \parallel BD$. Extend FA and EA to points P and Q so that the circle ω_1 through A, B, P and the circle ω_2 through A, D, Q are both tangent to line AC. Prove that B, P, Q, D are concyclic.

454. Let ABC be an acute scalene triangle. Let D and E be points on sides AB and AC with $BD = CE$. Denote by O_1 and O_2 the circumcentres of triangles ABE and ACD. Prove that the circumcircles of triangles ABC, ADE and AO_1O_2 have a common point different from A.

455. Let ABC be a scalene triangle with smallest side BC, and let D be a point on BC with $\angle CAD = \angle DAB$. Let ω_1 and ω_2 be the circumcircles of $\triangle ABD$ and $\triangle ACD$. Line AC meets ω_1 again at F, and line AB meets ω_2 again at E. Line DE meets ω_1 again at G, and line DF meets ω_2 again at H. Prove that the circumcircles of $\triangle ABC$, $\triangle AEF$ and $\triangle AGH$ have a common point other than A.

456. Let ABC be an acute-angled scalene triangle and D a point on side BC. Points E and F are on AD with $EB \perp AB$ and $FC \perp AC$. Points S and T are on BC with $SE \parallel AC$ and $TF \parallel AB$. The circumcircle of $\triangle BSE$ meets AB again at M, and the circumcircle of $\triangle CTF$ meets AC again at N. Prove that lines MS, NT and AD are concurrent.

457. Let ABC be a triangle not isosceles, with incircle w touching BC, CA, AB at D, E, F. If lines FD, DE, EF meet lines CA, AB, BC at U, V, W, and the midpoints of DW, EU, FV are L, M, N, prove that L, M, N are collinear.

458. Let $AXYZB$ be a convex pentagon inscribed in a semicircle of diameter AB. Denote by P, Q, R, S the feet of the perpendiculars from Y onto lines AX, BX, AZ, BZ. Prove that the acute angle formed by lines PQ and RS is half the size of $\angle XOZ$, where O is the midpoint of AB.

459. Let ABC be a triangle with circumcentre O. Points P and Q are interior points of sides CA and AB. Let K, L and M be the midpoints of BP, CQ and PQ, and let Γ be the circle through K, L and M. Suppose that line PQ is tangent to Γ. Prove that $OP = OQ$.

460. Let MN be a line parallel to side BC of triangle ABC, with M on AB and N on AC. Lines BN and CM meet at P. The circumcircles of triangles BMP and CNP meet at two distinct points P and Q. Prove that $\angle BAQ = \angle CAP$.

461. Let O be the circumcentre of acute triangle ABC, K a point on BC other than the midpoint, and D on the extension of AK. Lines BD and AC meet at N, and lines AB and CD meet at M. Prove that if $OK \perp MN$ then A, B, C, D are concyclic.

462. The three excircles of triangle ABC touch the segments AB, BC, CA at M, N, P. Let I and O be the incentre and circumcentre. Prove that if quadrilateral $AMNP$ is cyclic then

(a)M, N, P are collinear;

(b)I, O, N are collinear.

463. Let ABC be a triangle and BD its internal bisector with $D \in BC$. The line BD meets the circumcircle Ω of $\triangle ABC$ at B and E. A circle ω with diameter DE cuts Ω again at F. Prove that BF is the symmedian line of triangle ABC.

464. Let ABC be an acute-angled scalene triangle with circumcircle Γ. Let K be the foot of the internal bisector of $\angle BAC$ on BC, and let M be the midpoint of the arc BC containing A. Line MK meets Γ again at A'. Let T be the intersection of the tangents at A and A', and let R be the intersection of the perpendicular to AK at A and the perpendicular to $A'K$ at A'. Show that T, R and K are collinear.

465. Let (O) be a circle, and let ABP be a line such that A, B lie on (O) and P is outside (O). Let C be a point on (O) such that PC is tangent to (O), and let D be the point on (O) such that CD is a diameter of (O) and CD intersects AB inside (O). Suppose lines DB and OP meet at E. Prove that $AC \perp CE$.

466. Let $ABCD$ be a quadrilateral with $AD \nparallel BC$ inscribed in a circle (O) with larger edge BC. Let P be a point on the extension of segment BC. Line PA cuts (O) at N (so PA is not tangent to (O)), the circle with diameter PD meets (O) at E, and DE meets BC at M. Prove that MN always passes through a fixed point.

467. Let ABC be a triangle with $\angle C = 60^\circ$. Let B_1 and A_1 be points on sides AC and BC. The circumcircles of $\triangle BCB_1$ and $\triangle ACA_1$ meet at C and D. Prove that D lies on side AB if and only if $\dfrac{CB_1}{CB} + \dfrac{CA_1}{CA} = 1$.

468. Let A, B be two fixed points and C a variable point with $\angle ACB = \alpha$ constant, $0^\circ \leq \alpha \leq 180^\circ$. Let D, E, F be the projections of the incentre I of $\triangle ABC$ onto BC, CA, AB. Let M, N be the intersections of AI, BI with EF. Prove that MN has constant length and that the circumcircle of $\triangle DMN$ always passes through a fixed point.

469. Construct the semicircle h with diameter AB and midpoint M. Construct the semicircle k with diameter MB on the same side as h. Let X and Y be points on k such that arc BX is 1.5 times arc BY. Line MY meets line BX at D and meets h at C. Show that Y is the midpoint of CD.

470. The inscribed circle w of triangle ABC touches BC, CA, AB at X, Y, Z. Points K, L on w satisfy

$$\angle AKY + \angle BKX = \angle ALY + \angle BLX = 180^\circ.$$

Prove that the distances from X, Y, Z to line KL are equal.

471. Two circles w_1 and w_2 meet at A and B. A line through B meets w_1 and w_2 at C and D, and another line through B meets w_1 and w_2 at E and F. Line CF meets w_1 and w_2 at P and Q. Let M, N be the midpoints of arcs PB and QB. Prove that C, F, M, N are concyclic if $CD = EF$.

472. Given that the circle with centre O touches segment AB at M and touches the semicircle of diameter AB at E. A point C on the semicircle satisfies $CD \perp AB$ at D and CD is tangent to the circle centred at O at F. Connect CA, CM. Prove that

(a) A, F, E are collinear;

(b) $AC = AM$;

(c) $MC^2 = 2\,MD \cdot MA$.

473. Given a point P outside a circle with centre O, with PA, PB tangent at A and B. Let M, N be the midpoints of AP and AB. The extension of MN meets the circle at C, with N between M and C. Line PC meets the circle again at D, and the extension of ND meets PB at Q. Prove that quadrilateral $MNPQ$ is a rhombus.

474. A circle Γ and a line ℓ are given in a plane with ℓ not cutting Γ. Determine the intersection set of the circles having $[AB]$ as diameter for all pairs $\{A, B\}$ lying on ℓ satisfying $P, Q, R, S \in \Gamma$ with $PQ \cap RS = \{A\}$ and $PS \cap QR = \{B\}$.

475. Given a trapezoid $ABCD$ with parallel sides AB and CD, assume there exist points E on line BC outside segment BC, and F inside segment AD, such that $\angle DAE = \angle CBF$. Denote by I the intersection of CD and EF, and by J the intersection of AB and EF. Let K be the midpoint of segment EF, and assume it does not lie on line AB. Prove that I belongs to the circumcircle of ABK if and only if K belongs to the circumcircle of CDJ.

476. In a convex cyclic hexagon $ABCDEF$, $BC = EF$ and $CD = AF$. Diagonals AC and BF meet at Q, and diagonals EC and DF meet at P. Points R and S are on segments DF and BF with $FR = PD$ and $BQ = FS$. Segments RQ and PS meet at T. Prove that line TC bisects diagonal DB.

477. In parallelogram $ABCD$ with acute angle at A, a point N is chosen on segment AD and a point M on segment CN so that $AB = BM = CM$. Point K is the reflection of N in line MD, and line MK meets segment AD at L. Let P be the common point of the circumcircles of AMD and CNK on the same side of line MK as A. Prove that $\angle CPM = \angle DPL$.

478. Let $ABCD$ be a circumscribed quadrilateral and let X be the intersection point of its diagonals AC and BD. Let I_1, I_2, I_3, I_4 be the incentres of $\triangle DXC$, $\triangle BXC$, $\triangle AXB$, $\triangle DXA$. The circumcircle of $\triangle CI_1I_2$ meets sides CB and CD at P and Q; the circumcircle of $\triangle AI_3I_4$ meets sides AB and AD at M and N. Prove that $AM + CQ = AN + CP$.

479. Let $ABCD$ be a cyclic quadrilateral with circumcentre O in the interior. Let E and F be the midpoints of segments BC and AD. Let X be the point on the same side of line EF as C such that $\triangle EXF$ and $\triangle BOA$ are similar. Prove that $XC = XD$.

480. Let $ABCD$ be a convex quadrilateral with non-perpendicular diagonals and with sides AB and CD non-parallel. Denote by O the intersection of the diagonals, H_1 the orthocentre of $\triangle AOB$ and H_2 the orthocentre of $\triangle COD$. Let M be the midpoint of AB and N the midpoint of CD. Prove that H_1H_2 and MN are parallel if and only if $AC = BD$.

481. Inside triangle ABC, a point P is marked. Point Q is on segment AB and R on segment AC such that the circumcircles of triangles BPQ and CPR are tangent to line AP. Lines are drawn through B and C passing through the circumcentre of $\triangle BPC$, and through Q and R passing through the circumcentre of $\triangle PQR$. Prove that there exists a circle tangent to all four drawn lines.

482. Let ABC be an isosceles triangle with $AB = BC$. Let D be on segment AB, E on segment BC, and P on segment DE with $AD = DP$ and $CE = PE$. Let M be the midpoint of DE. The line through M parallel to AB meets AC at X, and the line through M parallel to BC meets AC at Y. Lines DX and EY meet at F. Prove that $FP \perp DE$.

483. In acute triangle ABC with $AB > AC$, M is the midpoint of minor arc BC, O is the circumcentre of (ABC) and AK is its diameter. The line through O parallel to AM meets segment AB at D and meets CA extended at E. Lines BM and CK meet at P; lines BK and CM meet at Q. Prove that $\angle OPB + \angle OEB = \angle OQC + \angle ODC$.

484. Point P is inside acute triangle $\triangle ABC$ with $\angle BPC = 90^\circ$ and $\angle BAP = \angle PAC$. Let D be the projection of P onto BC, and let M and N be the incentres of $\triangle ABD$ and $\triangle ADC$. Prove that quadrilateral $BMNC$ is cyclic.

485. In convex cyclic quadrilateral $ABCD$, lines AC and BD meet at E, lines AB and CD meet at F, and lines BC and DA meet at G. Suppose that the circumcircle of $\triangle ABE$ meets line CB at B and P, and the circumcircle of $\triangle ADE$ meets line CD at D and Q, where C, B, P, G and C, Q, D, F are collinear in that order. Prove that if lines FP and GQ meet at M, then $\angle MAC = 90^\circ$.

486. Let ABC be an acute-angled triangle and let D, E, F be the feet of the altitudes from A, B, C to sides BC, CA, AB. Denote by ω_B and ω_C the incircles of triangles BDF and CDE, tangent to segments DF and DE at M and N. Let line MN meet ω_B and ω_C again at $P \neq M$ and $Q \neq N$. Prove that $MP = NQ$.

487. Chords AB and CD of a circle meet at a point E inside the circle. Let M be an interior point of segment EB. The tangent line at E to the circle through D, E, M meets lines BC and AC at F and G. If $\dfrac{AM}{AB} = t$, find $\dfrac{EG}{EF}$ in terms of t.

488. Let ABC be a triangle with incentre I. Let U, V, W be the intersections of the angle bisectors from A, B, C with the incircle, with V between B and I, and similarly for U and W. Let X, Y, Z be the points of tangency of the incircle with BC, AC, AB. Prove that triangles UVW and XYZ are congruent if and only if ABC is equilateral.

489. Triangle ABC is inscribed in circle ω. A circle with chord BC meets segments AB and AC again at S and R. Segments BR and CS meet at L, and rays LR and LS meet ω at D and E. The internal bisector of $\angle BDE$ meets line ER at K. Prove that if $BE = BR$ then $\angle ELK = \frac{1}{2}\angle BCD$.

490. Let ABC be a triangle. Let θ be a fixed angle with $\theta < \frac{1}{2}\min(\angle A, \angle B, \angle C)$. Points S_A and T_A lie on segment BC with $\angle BAS_A = \angle T_A AC = \theta$. Let P_A and Q_A be the feet from B and C to AS_A and AT_A. Then ℓ_A is the perpendicular bisector of $P_A Q_A$. Define ℓ_B and ℓ_C analogously. Prove that ℓ_A, ℓ_B, ℓ_C are concurrent or all parallel.

491. Triangle ABC is inscribed in circle (O). A point P lies on arc BC not containing A such that AP is the symmedian of triangle ABC. Points E, F are the reflections of P in CA, AB. K is the reflection of A in EF, and L is the projection of K onto the line through A parallel to BC. Find $\dfrac{PA}{PL}$.

3.6. Constructions

Straightedge and compass, with occasional variations in the toolkit — a splitter, a cyclos, a T-square, a parallel-only straightedge. In each case the first task is to work out exactly what the given instrument can and cannot produce.

492. Let $ABCD$ be a square and let a line ℓ pass through C without meeting the square at another point. Using a ruler, find a point T on ℓ such that $AT \perp \ell$.

493. Construct, using only ruler and compass, a trapezoid $ABCD$ with bases AB and CD such that if E is the midpoint of side AD, then $EC = BC = 4$, $CD = 2$ and $\angle ECB = 120^\circ$. Indicate the steps and calculate the area of $ABCD$.

494. What should the angle at the vertex of an isosceles triangle be so that it is possible to construct a triangle with sides equal to the height, base, and one of the other sides of the isosceles triangle?

495. Given a point A, a line p, and a circle k, construct a triangle ABC with $\angle A = 60^\circ$, $\angle B = 90^\circ$, vertex B on line p, and vertex C on circle k.

496. Construct a triangle ABC given t_a, h_b and $b + c$.

497. Two different points A, M are given in a plane with $AM = d > 0$. Let a number $v > 0$ be given. Construct a rhombus $ABCD$ with height v and M the midpoint of BC. Discuss the conditions of solvability and determine the number of solutions. Can the resulting quadrilateral be a square?

498. Construct a right triangle with a given hypotenuse c such that the median to the hypotenuse is the geometric mean of the two legs.

499. Construct triangle ABC, given h_a, h_b (the altitudes from A and B), and m_a, the median from A.

500. Construct a triangle ABC if $AC = b$, $AB = c$ and $\angle AMB = \omega$ ($\omega < 90^\circ$), where M is the midpoint of BC. Prove that the construction has a solution if and only if $b \tan \frac{\omega}{2} \leq c < b$. In what case does equality hold?

501. On a circle K three distinct points A, B, C are given. Construct, using only straightedge and compass, a fourth point D on K such that a circle can be inscribed in the resulting quadrilateral.

502. Let AOB be a given angle less than 180° and let P be an interior point of the angular region determined by $\angle AOB$. Show, with proof, how to construct using only ruler and compass a segment CD through P with C on ray OA and D on ray OB and $CP : PD = 1 : 2$.

503. An ancient civilisation had a single geometric instrument with exactly two functions: draw a line through two points, and draw the perpendicular to a line through a given point. Give a procedure for bisecting a given 60° angle using only this instrument.

504. Let ABC be a scalene triangle. Given the incentre I and the points K_1, K_2, K_3 where the incircle touches BC, AC, AB, construct the circumcentre of $\triangle ABC$ using only a ruler.

505. Let I be the incentre of triangle ABC, and let K_1, K_2 be the points on BC and AC where the incircle is tangent. Using a ruler and compass, find the incentre of triangle CK_1K_2 in the minimal possible number of steps (each step being one circle or one line).

506. With an unmarked ruler only, reconstruct the trapezoid $ABCD$ with $AD \parallel BC$ given the vertices A and B, the intersection point O of the diagonals, and the midpoint M of the base AD.

507. Given an ABC triangle and a program allowing only these operations — mark points on segments, on lines, or outside them; draw the line joining two points; find the intersection of two lines; and reflect a point in a line — construct the circumcentre of the triangle.

508. Construct with ruler and compass a triangle ABC right-angled at A with perimeter 10 and altitude from A equal to 2. Indicate the construction steps and justify why the conditions are satisfied.

509. Given three points L, M, N, construct three points A, B, C so that L, M, N are the vertices of the equilateral triangles constructed on segments AB, BC, CA outside triangle ABC.

510. Let ABC be a given acute-angled triangle. Show a way to construct with ruler and compass the equilateral triangle DEF, with D on BC, E on AC and F on AB, such that the perpendiculars to BC at D, to AC at E and to AB at F are concurrent.

511.

(a) Construct a trapezoid $ABCD$ with $AB \parallel DC$ and $|AB| = 2 \cdot |DC|$, and construct the midpoint M of side DC.

(b) Describe how, given $ABCD$ and M, one can construct all straight lines through M dividing the trapezoid into two pieces whose areas are in ratio $1 : 2$.

(c) Prove that all lines so constructed satisfy the condition, and that no others do.

512. A pentagon $ABCDE$ is given. Find all points P inside it for which quadrilateral $ABCP$ has the same area as pentagon $CDEAP$. Describe a construction, and prove that every point resulting from your construction satisfies the condition.

513. Euclid has a tool called a *splitter* which can do exactly two things: given three non-collinear marked points X, Y, Z, draw the interior bisector of $\angle XYZ$; and mark the intersection of two previously drawn non-parallel lines. Given only three non-collinear marked points A, B, C, prove that Euclid can use the splitter several times to draw the centre of the circle through A, B and C.

514. Euclid has a tool called a *cyclos* which allows him to draw the circle through three given non-collinear marked points, or the circle with two marked points as endpoints of a diameter, and to mark intersection points of drawn circles or new points on a drawn circle. Show that given two marked points, Euclid can draw a circle centred at one of them and passing through the other, using only the cyclos.

515. Two distinct circles K_1 and K_2 intersect at A and B, where AB is a diameter of K_1. A point P on K_2 and inside K_1 is given. Using only a T-square (an instrument producing the line through two points, and the perpendicular to a line through a point on or off the line), construct two points C and D on K_1 such that $CD \perp AB$ and $\angle CPD$ is a right angle.

516. Six segments $S_1, \ldots, S_6$ are given in a plane, congruent to the edges AB, AC, AD, BC, BD, CD of a tetrahedron $ABCD$. Show how to construct with straightedge and compass a segment congruent to the altitude of the tetrahedron from vertex A.

517. Prove that if n is not a multiple of 3, then the angle $\frac{\pi}{n}$ can be trisected with ruler and compasses.

518. $ABCD$ and $A'B'C'D'$ are square maps of the same region, drawn to different scales and superimposed. Prove that there is exactly one point O on the small map lying directly over the point O' of the large map representing the same place, and give a Euclidean construction for O.

519. Two planes P and Q intersect along the line p. The point A is in plane P and the point C is in plane Q; neither lies on p. Construct an isosceles trapezoid $ABCD$ with $AB \parallel DC$ in which a circle can be inscribed, with vertices B and D lying in planes P and Q respectively.

520. Given three pairwise skew lines a, b, c, is it possible to construct a parallelepiped whose edges lie on the lines a, b, c?

521. Let A be a point in the plane and ℓ a line not through A. Evan has no straightedge, but has a special compass which draws the circle through three given distinct non-collinear points (the centre is not marked). He can mark intersections of drawn objects, and mark an arbitrary point on a given object or in the plane.

(i) Can Evan construct the reflection of A over ℓ?

(ii) Can Evan construct the foot of the altitude from A to ℓ?

522. Carl is given three distinct non-parallel lines ℓ_1, ℓ_2, ℓ_3 and a circle ω. In addition to a normal straightedge, Carl has a special straightedge which, given a line ℓ and a point P, constructs the line through P parallel to ℓ. (He has no compass.) Show that Carl can construct a triangle with circumcircle ω whose sides are parallel to ℓ_1, ℓ_2, ℓ_3 in some order.

523. A regular pyramid $SABC$ with base ABC, together with the height AH of face SAB, is drawn on the plane by orthogonal projection. Using only a compass and a ruler, construct the image of the circumcentre of the pyramid on the paper.

524. Given a point M, a circle k around M with radius r, and two different points P and Q with $PQ = x < 2r$ lying outside the circle. On k, four points A, B, C, D in this order form quadrilateral $ABCD$ where AB, CD and PQ are parallel and $|AB| = |CD| = |PQ|$.

(a) Calculate the area of $ABCD$ for $r = 13$ and $|PQ| = 10$.

(b) Give a general formula for the area in terms of r and x.

(c) Describe a compass-and-ruler construction producing A, B, C, D.

525. Construct a convex hexagon $ABCDEF$ satisfying: opposite sides are always parallel and at distance $d = 8$; the angles BAF and EDC are right angles; and $\angle CBA = 150°$. The diagonals BE and CF meet at S. Describe your construction and determine $\angle ESF$.

526. Let $A_0B_0C_0$ and $A_1B_1C_1$ be any two acute-angled triangles. Consider all triangles ABC similar to $\triangle A_1B_1C_1$ (with A_1, B_1, C_1 corresponding to A, B, C) and circumscribed about triangle $A_0B_0C_0$ (with A_0 on BC, B_0 on CA, C_0 on AB). Of all such triangles, determine the one with maximum area and construct it.

527. You are given a circle (without its centre) on which points B, C are marked with $BC < \sqrt{3}\,R$, where R is the radius. Construct a point A on this circle with $AB < AC$ using at most four dollars, such that the distance from the incentre of $\triangle ABC$ to the tangent at the midpoint of arc BC not containing A equals the distance from the incentre to A. Marking intersections is free; each of line, circle, midpoint, perpendicular bisector, parallel line, perpendicular line and tangent costs one dollar.

Chapter 4

Combinatorics

Combinatorics probably has the shortest yet hardest arguments! The IOQM papers want a number, and the technique is almost always a careful bijection or a well-set-up recursion. The RMO and INMO material wants a strategy, an invariant, or an extremal argument, and the answer is a proof. Both are here.

The sections are ordered by method. A problem about placing coins on a chessboard is filed under counting if the work is a binomial coefficient and under grids and boards if the work is a colouring argument. Games are separated out because the technique — backward induction, pairing strategies, strategy stealing — is genuinely its own thing.

4.1. Counting and Enumeration

528. How many ways are there to place 3 identical coins on a 4×4 chessboard such that no two coins are in the same row or the same column?

529. A drawer contains shirts which may be red, blue or yellow. If we choose shirts at random, the minimum number we must take to be guaranteed a red one is 8, the minimum to guarantee a blue one is 7, and the minimum to guarantee a yellow one is 12. How many shirts are in the drawer?

530. Find the number of ordered triples of positive integers (a, b, c) such that abc divides $(ab+1)(bc+1)(ca+1)$.

531. Determine the number of sequences of sets $S_1, S_2, \ldots, S_{999}$ such that

$$S_1 \subseteq S_2 \subseteq \cdots \subseteq S_{999} \subseteq \{1, 2, \ldots, 999\}.$$

Report the remainder when your answer is divided by 1000.

532. Let $S = \{1, 2, 3, \ldots, 12\}$. How many subsets of S, excluding the empty set, have an even sum but not an even product?

533. Find the number of ways to choose two nonempty subsets X and Y of $\{1, 2, \ldots, 2001\}$ such that $|Y| = 1001$ and the smallest element of Y equals the largest element of X. Report the answer modulo 100.

534. For a set $S \subset \mathbb{Z}$, write $\pi(S)$ for the product of all its elements. Compute the sum of the digits of

$$\sum_{\emptyset \neq S \subseteq \{1,2,\ldots,2024\}} \frac{1}{\pi(S)},$$

the sum being over all nonempty subsets. (For a one-element set the product is the sole element.)

535. Let $a \diamond b = ab - 4(a+b) + 20$. Evaluate

$$1 \diamond (2 \diamond (3 \diamond (\cdots (99 \diamond 100) \cdots))).$$

536. How many integers in the range $[2, 100]$ can be represented in the form

$$3^{m_1}2^{n_1} + 3^{m_2}2^{n_2} + \cdots + 3^{m_k}2^{n_k}$$

where $m_1 > m_2 > \cdots > m_k \geq 0$ and $0 \leq n_1 < n_2 < \cdots < n_k$ are integers?

537. A *weak binary representation* of a non-negative integer n is a representation $n = a_0 + 2a_1 + 2^2a_2 + \cdots$ with $a_i \in \{0, 1, 2, 3, 4, 5\}$. Determine the last two digits of the number of such representations of 513.

538. A binary string is a word containing only 0s and 1s. A 1-run is a non-extendable substring containing only 1s. For a positive integer n, let $B(n)$ be the number of 1-runs in the binary representation of n; for example $B(107) = 3$ since 107 is 1101011 in binary. Compute

$$B(1) + B(2) + \cdots + B(255) \pmod{99},$$

expressing the answer in the reduced residue class $R_n = \{a \in \{1, \ldots, n-1\} : \gcd(a, n) = 1\}$.

539. Find the product of the digits of the total number of natural numbers $n < 2015$ that are divisible by $\lfloor \sqrt[3]{n} \rfloor$.

540. Find the number of ordered pairs of positive integers (a, b) with $a + b$ prime, $1 \leq a, b \leq 100$, and $\dfrac{ab+1}{a+b}$ an integer.

541. How many ordered triples (a, b, c) of positive integers are there such that a, b, c are all divisors of $2^{20} \cdot 3^{16} \cdot 5^{12}$ and $\operatorname{lcm}(a, b, c) = 2^{20} \cdot 3^{16} \cdot 5^{12}$? Report the answer modulo 100.

542. There are $2n$ people at a party. Each person has an even number of friends at the party (friendship is mutual and no one is their own friend). Prove that there exist two people with an even number of common friends.

543. Consider an equilateral triangle of side length n divided into unit triangles. Let $f(n)$ be the number of paths from the triangle in the top row to the middle triangle in the bottom row, such that adjacent triangles in the path share a common edge, and the path never travels up or revisits a triangle. Find the value of $f(2005)$.

544. Suppose that n people each know exactly one piece of information, and all n pieces are different. Every time person A phones person B, A tells B everything A knows, while B tells A nothing. What is the minimum number of phone calls needed for everyone to know everything? Prove that your answer is a minimum.

545. Ten distinct numbers from $\{0, 1, 2, \ldots, 13, 14\}$ are chosen to fill the ten circles in a diagram so that the integers in every straight line of three circles add up to t. Find the sum of the maximum and minimum possible values of t.

546. A mathematical contest had three problems, each scored between 0 and 7 inclusive. For any two contestants there is at most one problem on which they obtained the same score. (For example there are no two contestants with ordered scores $7, 1, 2$ and $7, 1, 5$, but there might be two with $7, 1, 2$ and $7, 2, 1$.) Find the maximum number of contestants.

547. For each positive integer n, find the number of n-digit positive integers satisfying both: no two consecutive digits are equal, and the last digit is a prime.

548. Let S be a set of size 11. A collection F of subsets of S has the property that any two subsets in F have non-empty intersection. Prove that some element of S belongs to more than half of the subsets in F.

549. Suppose $a_1, a_2, \ldots, a_{100}$ are positive integers. Find the number of distinct values that $\sum_{i=1}^{100} \gcd(a_i, a_{i+1})$ can take, where $a_{101} = a_1$.

550. In an exhibition there are 100 paintings, each made with exactly k colours. Find the minimum possible value of k if any 20 paintings have a common colour but there is no colour used in all paintings.

551. There are dwarves in a forest, each owning exactly three hats numbered with numbers from $1, 2, \ldots, 28$. The three hats of a dwarf have different numbers, and there are three festivals in a day. At the first festival each dwarf wears the hat with the smallest value, at the second the second smallest, and at the third the largest. It is then realised that there is no pair of dwarves wearing the same value in at least two festivals. Find the maximum number of dwarves.

552. Martu wants to build a set of cards with these properties: each card has a positive integer on it; the number on each card is one of five possible numbers; and if any two cards are taken and added together, it is always possible to find two other cards in the set with the same sum. Determine the fewest number of cards Martu's set can have, and give an example.

553. We want to choose telephone numbers for a city. The numbers have 10 digits and 0 is not used. Our aim is that we do not choose numbers such that every two telephone numbers are different in more than one digit, or every two telephone numbers are different in a digit by more than 1. What is the maximum number of telephone numbers which can be chosen? In how many ways can we choose the numbers in this maximum situation?

554. A 5×5 table is called *regular* if each of its cells contains one of four pairwise distinct real numbers, such that each of them occurs exactly once in every 2×2 subtable. The sum of all numbers of a regular table is called its total sum. With any four numbers, one constructs all possible regular tables, computes their total sums, and counts the distinct outcomes. Determine the maximum possible count.

555. We call a 5-tuple of integers *arrangeable* if its elements can be labelled a, b, c, d, e in some order so that $a - b + c - d + e = 29$. Determine all 2017-tuples of integers $n_1, n_2, \ldots, n_{2017}$ such that if we place them in a circle in clockwise order, then any 5-tuple of numbers in consecutive positions on the circle is arrangeable.

556. Let n be a positive integer. On a blackboard, Bobo writes a list of n non-negative integers, then performs a sequence of moves, each as follows: for each $i = 1, \ldots, n$ he computes the number a_i of integers currently on the board that are at most i; he erases all integers; he writes the numbers $a_1, \ldots, a_n$.

(a) Show that whatever n and whatever the initial configuration, the numbers on the board eventually stop changing.

(b) As a function of n, determine the minimum integer k such that, whatever the initial configuration, moves from the k-th onwards do not change the numbers on the board.

557. A music streaming service offers songs classified into 10 genres, each song belonging to exactly one genre. The songs are played one after another: the first 17 are chosen by the user, but from the eighteenth onwards the service determines which song to play. Elisabetta notices that, ranking the genres by how often they appear among the last 17 songs played, the new song always belongs to the genre at the top of the ranking, or in case of a tie to one of the first genres. Prove that however the first 17 tracks are chosen, from a certain point onwards the songs proposed are all of the same kind.

4.2. Probability and Expectation

558. Three distinct vertices are chosen at random from the vertices of a given regular polygon of $2n + 1$ sides. If all such choices are equally likely, what is the probability that the centre of the polygon lies in the interior of the triangle determined by the three chosen points?

559. A bag has 19 blue and 98 red balls. Kevin removes balls uniformly at random without replacement until all balls remaining are the same colour. The expected number of balls remaining is $\frac{a}{b}$ in lowest terms; find the sum of the digits of $a + b$.

560. Andy the ant starts at a vertex of a regular octahedron. Each minute he walks along a random edge to an adjacent vertex. The probability he is back at his starting vertex after 6 minutes is $\frac{m}{n}$ in lowest terms. Find the sum of the digits of $m + n$.

561. Two positive integers m and n are chosen uniformly at random from $\{1, 2, \ldots, 2024\}$. The probability that $2^m + 2^n$ is a perfect square is $\frac{a}{b}$ with a, b coprime. Find $b - a$ modulo 100.

562. A fair coin is flipped repeatedly. What is the expected number of flips until the sequence HTTH first appears? Report the answer modulo 100.

563. Alice and Bob play a game with a stack of 2024 tokens, alternating turns and each removing 1, 2 or 3 tokens, Alice going first. Whoever removes the last token wins. Suppose instead each player on their turn removes a uniformly random legal number of tokens. Let p be the probability Alice wins. Find $\lfloor 1000p \rfloor$.

4.3. Graphs and Networks

564. In a country of n cities, an express train runs both ways between any two cities. For any train, ticket prices in either direction are equal, but for any different routes these prices are different. Prove that a traveller can select a starting city, leave it and go on, successively, $n - 1$ trains, such that each fare is smaller than the previous fare. (A traveller may enter the same city several times.)

565. In a soccer league with 2020 teams every two teams have played exactly once and no game has led to a draw. The teams are ordered first by points (3 for a win, 1 for a draw, 0 for a loss) and then by goal difference. Is it possible for the goal difference in such a table to be strictly increasing from top to bottom?

566. There are $n \geq 3$ islands in a city. Initially, the ferry company offers some routes between the islands so that it is impossible to divide the islands into two groups such that no two islands in different groups are connected by a ferry route. After each year, the ferry company will close a ferry route between some two islands X and Y, and open new routes from X and Y to each island Z that is not currently connected to both. Prove that after some years, there will be routes between any two islands.

567. Consider a graph G on n vertices in which every vertex has degree at least $\frac{n}{2}$. Prove that G contains a Hamiltonian cycle.

568. On a circle there are 1000 marked points, each coloured in one of k colours. It is known that among any 5 pairwise intersecting segments whose endpoints are 10 distinct marked points, there are at least 3 segments each of which has its endpoints coloured in different colours. Determine the smallest possible value of k.

569. There are 5 points on a plane. Prove that you can choose some of them and shift them so that distances between shifted points do not change, and as a result the set of 5 points becomes symmetric about some line.

570. Let $n \geq 5$ be an integer. Consider n squares with side lengths $1, 2, \ldots, n$. The squares are arranged in the plane with their sides parallel to the x and y axes. Suppose that no two squares touch, except possibly at their vertices. Show that it is possible to arrange these squares so that every square touches exactly two other squares.

571. Given a positive integer n, Susann fills a square of $n \times n$ boxes, inscribing an integer in each so that each row and each column contains distinct numbers. An imp then destroys some of the boxes. Show that Susann can choose some of the remaining boxes and colour them red so that: no two red boxes are in the same row or column; and for each box which is neither destroyed nor coloured, there is a red box with a larger number in the same row, or a red box with a smaller number in the same column.

4.4. Games and Strategy

572. A and B play a game. Given a positive integer N, A writes down 1 first, then every player sees the last number written and if it is n then on their turn they write $n+1$ or $2n$, but the number cannot exceed N. The player who writes N wins. For how many values of N less than 100 does B win?

573. A black pawn and a white pawn are placed on the first and last squares of a $1 \times n$ chessboard. Wiwit and Siti move alternately; Wiwit has the white pawn and moves first. In every move a player moves her pawn one or two squares to the right or left, without passing the opponent's pawn. The first player who cannot move loses. Which player has a winning strategy? Explain the strategy.

574. Let n be a positive integer. Two persons play a game in which they call an integer m with $1 \leq m \leq n$ alternately, and may not call a number already said. The game is over when no one can call numbers. If the sum of the numbers the leader has said is divisible by 3, the leader wins; otherwise the second player wins. Find n for which the second player has a winning strategy.

575. Player A places an odd number of boxes around a circle and distributes 2013 balls into some of these boxes. Then player B chooses one of these boxes and takes the balls in it. After that player A chooses half of the remaining boxes such that no two are consecutive and takes the balls in them. If player A guarantees to take k balls, find the maximum possible value of k.

576. Shubham has n coins, among which at most one is counterfeit. The counterfeit coin, if it exists, is either heavier or lighter than a real coin. He has an old weighing balance on which he can place any number of coins on either side, and it shows which side is heavier. The balance can be used only 4 more times. What is the largest n for which it is possible for Shubham to find the counterfeit coin, if it exists?

577. William is thinking of an integer between 1 and 50 inclusive. Victor can choose a positive integer m and ask William "does m divide your number?", to which William must answer truthfully. Victor continues asking until he determines William's number. What is the minimum number of questions Victor needs to guarantee this?

578. Alice and Bob play the following game: starting with the number 2 written on a blackboard, each player in turn changes the current number n to a number $n+p$, where p is a prime divisor of n. Alice goes first and the players alternate. The game

is lost by the one forced to write a number greater than $\underbrace{22\ldots2}_{2020}$. Assuming perfect play, who wins?

579. Alice and Bob play the following game on a 100×100 grid, taking turns with Alice first. Initially the grid is empty. On their turn they choose an integer from 1 to 100^2 not yet written in any cell, choose an empty cell, and place it there. When no empty cell is left, Alice computes the sum of the numbers in each row and her score is the maximum of these 100 numbers; Bob computes the sum in each column and his score is the maximum of those. Alice wins if her score exceeds Bob's, Bob wins if his exceeds hers, otherwise no one wins. Determine whether one of the players has a winning strategy, and if so which.

580. Let $k \leq 2022$ be a positive integer. Alice and Bob play a game on a 2022×2022 board; initially all cells are white. Alice starts and the players alternate. In her turn, Alice can either colour one white cell red or pass. In his turn, Bob can either colour a $k \times k$ square of white cells blue or pass. Once both players pass, the game ends and the person who coloured more cells wins (a draw can occur). For each $1 \leq k \leq 2022$, determine which player, if any, has a winning strategy.

581. Let k be a fixed positive integer. A finite sequence of integers $x_1, x_2, \ldots, x_n$ is written on a blackboard. Pepa and Geoff play a game in rounds. In each round, Pepa first partitions the sequence currently on the blackboard into two or more contiguous subsequences; however, if the number of these subsequences is larger than 2, then the sum of numbers in each of them has to be divisible by k. Then Geoff selects one of the subsequences Pepa has formed and wipes all the others from the blackboard. The game finishes once there is only one number left. Prove that Pepa may choose his moves so that, independently of Geoff's moves, the game finishes after at most $3k$ rounds.

582. Anna and Bob, with Anna starting first, alternately colour the integers of the set $S = \{1, 2, \ldots, 2022\}$ red or blue. At their turn each may colour any uncoloured number of S with any colour. The game ends when all numbers are coloured. Let N be the number of pairs (a, b) with $a, b \in S$ such that a and b have the same colour and $b - a = 3$. Anna wishes to maximise N. What is the maximum value of N she can achieve regardless of how Bob plays?

583. Nina and Tadashi play the following game. Initially, a triple (a, b, c) of non-negative integers with $a + b + c = 2021$ is written on a blackboard. Nina goes first, and thereafter the players alternate. A player, on their turn, may replace the current triple (x, y, z) on the blackboard with one of

$$(x - 1, y, z + 1), \qquad (x - 1, y + 1, z), \qquad (x + 1, y - 1, z)$$

provided the resulting triple consists of non-negative integers. A player loses if they are unable to make a move. Find the number of initial triples (a, b, c) for which Tadashi has a winning strategy.

584. On her blackboard, Alice has written n integers strictly greater than 1. She may then, as often as she likes, erase two numbers a and b with $a \neq b$ and replace them with q and q^2, where q is the product of the prime factors of ab (each prime factor counted only once). For instance, if Alice erases 4 and 6, then $ab = 2^3 \times 3$ has prime factors 2 and 3, so she writes $q = 6$ and $q^2 = 36$. Prove that after some time, and whatever Alice's strategy is, the list of numbers written on the blackboard will never change anymore.

4.5. Invariants, Colourings and Monovariants

585. A cube of side 3 is divided into 27 unit cubes, arbitrarily labelled 1 to 27 (each cube receiving a different number). A move consists of swapping the cube labelled 27 with one of its 6 neighbours. Is it possible to find a finite sequence of moves at the end of which cube 27 is in its original position, but cube n has moved to the position originally occupied by $27 - n$ for each $n = 1, 2, \ldots, 26$?

586. A basket is called a *stuff basket* if it includes 10 kilograms of rice and 30 eggs. A market is to distribute 100 stuff baskets. There are in total 1000 kilograms of rice and 3000 eggs in the baskets, but some baskets include more or less rice or eggs. In each step, market workers can select two baskets and move an arbitrary amount of rice or eggs between them. Starting from an arbitrary situation, what is the minimum number of steps in which the workers can provide 100 stuff baskets?

587. Consider the following operation on positive real numbers written on a blackboard: choose a number r written on the blackboard, erase it, and write a pair of positive reals a and b satisfying $2r^2 = ab$. Assume you start with just one positive real number r on the blackboard, and apply this operation $k^2 - 1$ times to end up with k^2 positive reals, not necessarily distinct. Show that there exists a number on the board which does not exceed kr.

588. The integers $1, 2, 3, \ldots, 2016$ are written on a board. You can choose any two numbers on the board and replace them with their average. For example, you can replace 1 and 2 with 1.5, or replace 1 and 3 with a second copy of 2. After 2015 replacements of this kind, the board will have only one number left.

(a) Prove that there is a sequence of replacements making the final number equal to 2.

(b) Prove that there is a sequence of replacements making the final number equal to 1000.

589. A rectangle of size 10×14 is divided into 140 unit squares and painted red and white like a chessboard. We put 0 or 1 in each square so that each row and each column has an odd number of 1s. Prove that the number of 1s contained in red-painted squares is even.

590. Initially, a word of 250 letters with 125 letters A and 125 letters B is written on a blackboard. In each operation, we may choose a contiguous string of any length with an equal number of letters A and an equal number of letters B, reverse those letters, and then swap each B with A and each A with B. (For example $ABABBA$ becomes $BAABAB$.) Decide whether it is possible to choose an initial word so that after some operations it becomes the same as the first word, but in reverse order.

591. In the lake, there are 23 stones arranged along a circle. There are 22 frogs numbered $1, 2, \ldots, 22$, each number appearing once. Initially, each frog randomly sits on a stone (several frogs might sit on the same stone). Every minute, all frogs jump at the same time as follows: frog number i jumps i stones forward in the clockwise direction. (In particular, frog number 22 jumps 1 stone in the counter-clockwise direction.) Prove that at some point, at least 6 stones will be empty.

592. In each square of a 2012×2012 board there is a person. People are either honest, who always tell the truth, or liars, who always lie. At a given moment, each person makes the same statement: "In my row there are the same number of liars as in my column." Determine the minimum number of honest people that can be on the board.

593. In a class of 26 students, everyone is graded on five subjects with one of three possible marks. Prove that if 25 of these students have received their marks, then we can grade the last one in such a way that their marks differ from those of any other student in at least two subjects.

594. Let n be a positive integer. We are given a $3n \times 3n$ board whose unit squares are coloured black and white in such a way that starting with the top left square, every third diagonal is coloured black and the rest of the board is white. In one move, one can take a 2×2 square and change the colour of all its squares in such a way that white squares become orange, orange ones become black, and black ones become white. Find all n for which, using a finite number of moves, we can make all squares which were initially black white, and all squares which were initially white black.

595. Let $n \geq 2$ be an integer. Alex writes the numbers $1, 2, \ldots, n$ in some order on a circle such that any two neighbours are coprime. Then, for any two numbers that are not coprime, Alex draws a line segment between them. For each such segment s we denote by d_s the difference of the numbers written at its extremities, and by p_s the number of all other drawn segments which intersect s in its interior. Find the greatest n for which Alex can write the numbers on the circle such that $p_s \leq |d_s|$ for each drawn segment s.

4.6. Extremal Combinatorics and Pigeonhole

596. Prove that from a set of ten distinct two-digit numbers (in the decimal system), it is possible to select two disjoint subsets whose members have the same sum.

597. Let S be a subset of $\{1, 2, \ldots, 2017\}$ such that no two elements of S differ by 3 or 5. What is the largest possible size of S?

598. Let A be a set of n positive integers, none exceeding $2n$. Prove that some element of A divides another, unless A consists of n pairwise coprime numbers.

599. Prove that among any $2^n + 1$ distinct positive integers whose prime divisors all lie in a fixed set of n primes, there are two whose product is a perfect square.

600. Consider a 5×100 table divided into 500 unit square cells, where n of them are coloured black and the rest white. Two unit square cells are called adjacent if they share a common side. Each of the unit square cells has at most two adjacent black unit square cells. Find the largest possible value of n.

601. Given an integer $n \geq 3$, determine the smallest positive number k such that any two points in any n-gon (or on its boundary) in the plane can be connected by a polygonal path consisting of k line segments contained in the n-gon (including its boundary).

602. Let $n \geq 2$ be a positive integer and let A be a set of $2n$ points in the plane, no three collinear. Suppose exactly n of the points are coloured red and the remaining n are coloured blue. Prove that there exist n pairwise disjoint segments, each joining one red and one blue point, so that every point of A is an endpoint of exactly one segment.

603. Determine the largest N for which there exists a table T of integers with N rows and 100 columns satisfying: every row contains the numbers $1, 2, \ldots, 100$ in some order; and for any two distinct rows r and s, there is a column c such that $|T(r, c) - T(s, c)| \geq 2$.

4.7. Grids, Tilings and Boards

604. You have an infinite stack of T-shaped tetrominoes (composed of four squares of side length 1), and an $n \times n$ board. You are allowed to place some tetrominoes on the board, possibly rotated, as long as no two overlap and none extends off the board. For which values of n can you cover the entire board?

605. Each cell of an $m \times n$ board is filled with some non-negative integer. Two numbers in the filling are said to be *adjacent* if their cells share a common side. (Two numbers in cells sharing only a corner are not adjacent.) The filling is called a *garden* if it satisfies: the difference between any two adjacent numbers is either 0 or 1; and if a number is less than or equal to all of its adjacent numbers, then it is equal to 0. Determine the number of distinct gardens in terms of m and n.

606. On a $(4n + 2) \times (4n + 2)$ square grid, a turtle can move between squares sharing a side. The turtle begins in a corner square and enters each square exactly once, ending in the square where she started. In terms of n, what is the largest positive integer k such that there must be a row or column that the turtle has entered at least k distinct times?

607. Let $n \geq 3$ be an odd number. We play a game using an $n \times n$ grid. The game comprises n^2 turns; in every turn we perform the following operations in sequence. We choose a square with an unwritten integer, and write down an integer among 1 through n^2; each integer can be written only once during the game. Then, for each row and column including that square, if the sum of integers is a multiple of n, we get 1 point (if both sums are multiples of n, we get 2 points). Determine the maximum possible total number of points.

608. Suppose that each square of a 4×7 chessboard is coloured either black or white.

(a) Prove that with any such colouring, the board must contain a rectangle formed by the horizontal and vertical lines of the board whose four distinct unit corner squares are all of the same colour.

(b) Exhibit a black-white colouring of a 4×6 board in which the four corner squares of every such rectangle are not all of the same colour.

609. A square is drawn in the plane whose vertices are $(0, 0)$, $(1, 0)$, $(0, 1)$ and $(1, 1)$. The line $y = kx$ divides the square into two regions, the area of the upper region being 27 times the area of the lower region. What is the inverse of k?

610. Let n be a positive integer. Consider an $n \times n$ grid of unit squares. How many ways are there to place non-attacking rooks on the grid such that every empty square is attacked by at least one rook? Give your answer as a closed form and evaluate it at $n = 5$.

611. Determine all positive integers n for which there exists an infinite subset A of the set $\mathbb{N}$ of positive integers such that for all pairwise distinct $a_1, \ldots, a_n \in A$, the numbers $a_1 + \cdots + a_n$ and $a_1 a_2 \cdots a_n$ are coprime.

Appendix A

Answer Key and Sources

Every problem in this book has a row below, and each row carries up to three pieces of information.

The **answer** column gives a numerical answer wherever the problem has one — essentially everything drawn from the IOQM mock papers, where answers are integers in $[00, 99]$. A dash means the problem asks for a proof, a construction.

The **Olympiad Arena material** column says which of our documents the problem appeared in, and where in that document. This is how you reassemble an original paper if you want to sit it under time conditions, and how you find the rest of a handout if one problem from it interests you.

The **original source** column says which competition the problem came to us from. A dash here means either that the problem is ours, or that the handout it came from recorded no attribution, solely because of the reason that several of our early handouts listed their source competitions only in bulk, at the top, with no per-problem mapping.

A.1. Algebra

Systems of Equations

#	Ans.	Olympiad Arena material	Original source
1	04	IOQM Mock 2 (2025), Q7	—
2	570	IOQM Mock 2 (2024), Q13	—
3	28	IOQM Mock 2 (2024), Q3	—
4	—	RMO Handout Series	Czech–Polish–Slovak 2004
5	—	RMO Handout Series	Vietnam 2007
6	—	RMO Handout Series	Croatia TST 2007
7	—	RMO Handout Series	Austria 2005
8	—	RMO Handout Series	Moldova TST 2008
9	—	RMO Handout Series	Bulgaria 2003
10	—	RMO Handout Series	USAMO TST 2001
11	—	RMO Handout Series	Czech–Polish–Slovak 2005
12	—	RMO Handout Series	USAMO TST 2001
13	47	IOQM Mock 2 (2024), Q23	—
14	39	IOQM Mock 1 (2025), Q11	—
15	10	IOQM Mock 2 (2025), Q12	—

#	Ans.	Olympiad Arena material	Original source
16	39	IOQM Mock 2 (2024), Q15	—
17	—	RMO Handout Series	Czech–Polish–Slovak 2008
18	—	RMO Handout Series	Australia 2008
19	—	RMO Handout Series	Australia 2012
20	—	RMO Handout Series	Belarus 2005
21	—	RMO Handout Series	Greece TST 2009
22	—	RMO Handout Series	Vietnam 2009
23	04	IOQM Mock 2 (2025), Q3	—
24	33	IOQM Mock 4 (2024), Q22	—
25	98	IOQM Mock 6 (2024), Q30	—
26	17	IOQM Mock 2 (2024), Q25	—
27	—	RMO Mock 2025, Paper 3, P3	—

Polynomials

#	Ans.	Olympiad Arena material	Original source
28	15	IOQM Mock 4 (2024), Q1	—
29	23	IOQM Mock 5 (2024), Q1	—
30	—	RMO Handout Series	USAMO 1975
31	12	IOQM Mock 1 (2024), Q1	—
32	03	IOQM Mock 1 (2025), Q6	—
33	91	IOQM Mock 5 (2024), Q5	—
34	—	RMO Handout Series	USAMO 1984
35	16	IOQM Mock 3 (2024), Q10	—
36	—	RMO Handout Series	Korea 2000
37	—	RMO Handout Series	Austria 2017
38	06	IOQM Mock 4 (2024), Q15	—
39	26	IOQM Mock 6 (2024), Q2	—
40	—	RMO Handout Series	China 2008
41	41	IOQM Mock 6 (2024), Q12	—
42	05	IOQM Mock 3 (2024), Q12	—
43	06	IOQM Mock 2 (2026), Q20	—
44	03	IOQM Mock 1 (2025), Q10	—
45	—	INMO Mock 1 (2024), P2	—
46	01	IOQM Mock 1 (2026), Q13	—
47	—	RMO Handout Series	Russia (RMO) 2021
48	—	RMO Handout Series	Russia (RMO) 2024

#	Ans.	Olympiad Arena material	Original source
49	12	IOQM Mock 1 (2024), Q28	—
50	32	IOQM Mock 3 (2024), Q24	—
51	21	IOQM Mock 2 (2025), Q23	—
52	12	IOQM Mock 6 (2024), Q23	—
53	—	RMO Mock 2025, Paper 2, P4	—

Inequalities

#	Ans.	Olympiad Arena material	Original source
54	—	RMO Mock 5 (2024), P1	—
55	—	RMO Handout Series	Bosnia 2008
56	—	RMO Handout Series	Germany 2010
57	—	RMO Handout Series	China (CHNMO) TST 2008
58	—	RMO Handout Series	Greece 2009
59	—	RMO Handout Series	Czech–Polish–Slovak 2011
60	—	RMO Handout Series	IMO 2000, Problem 2
61	—	RMO Mock 6 (2024), P6	—
62	—	RMO Mock 2025, Paper 6, P4	—
63	—	RMO Handout Series	Iran TST 2008
64	—	RMO Handout Series	China (CMO) 2008
65	—	RMO Handout Series	Ukraine 2008
66	—	RMO Handout Series	Balkan MO 2010
67	—	RMO Handout Series	Turkey 2009
68	27	IOQM Mock 5 (2024), Q17	—
69	—	RMO Mock 4 (2024), P5	—
70	—	RMO Handout Series	USAMO 2009
71	—	RMO Handout Series	Turkey 2007
72	—	RMO Handout Series	Iran 2008
73	—	RMO Handout Series	Mediterranean MO 2010
74	—	RMO Handout Series	APMO 2007
75	—	RMO Handout Series	IMO 2005, Problem 3
76	—	RMO Handout Series	IMO Shortlist 2006
77	—	RMO Handout Series	APMO 2005
78	—	RMO Handout Series	JBMO 2007
79	—	RMO Handout Series	China 2008
80	—	RMO Handout Series	Japan TST 2016

#	Ans.	Olympiad Arena material	Original source
81	—	RMO Handout Series	China (CMC) 2009
82	—	RMO Handout Series	China (CHNMO) 2010
83	—	RMO Mock 3 (2024), P3	—
84	—	RMO Mock 2025, Paper 1, P2	—
85	—	RMO Mock 2025, Paper 5, P4	—
86	—	RMO Mock 7.3 (2024), P5	—
87	—	RMO Mock 7.2 (2024), P4	—
88	—	RMO Mock 1 (2024), P5	—

Functional Equations

#	Ans.	Olympiad Arena material	Original source
89	06	IOQM Mock 2 (2026), Q7	—
90	47	IOQM Mock 1 (2025), Q22	—
91	—	RMO Mock 5 (2024), P5	—
92	00	IOQM Mock 1 (2026), Q15	—
93	23	IOQM Mock 5 (2024), Q11	—

Sequences and Recurrences

#	Ans.	Olympiad Arena material	Original source
94	61	IOQM Mock 6 (2024), Q1	—
95	87	IOQM Mock 2 (2025), Q11	—
96	01	IOQM Mock 1 (2026), Q2	—
97	77	IOQM Mock 1 (2026), Q14	—
98	12	IOQM Mock 2 (2025), Q24	—
99	25	IOQM Mock 6 (2024), Q22	—
100	01	IOQM Mock 1 (2024), Q30	—
101	20	IOQM Mock 2 (2026), Q19	—
102	—	RMO Handout Series	APMO 1995
103	02	IOQM Mock 1 (2026), Q24	—
104	—	RMO Handout Series	IMO Shortlist 2006
105	—	Combinatorics Handout, More Sequences P8	—
106	—	RMO Handout Series	USAJMO 2015
107	13	IOQM Mock 1 (2025), Q19	—
108	—	RMO Mock 7.1 (2024), P3	APMO 1995

Miscellaneous Algebra

#	Ans.	Olympiad Arena material	Original source
109	00	IOQM Mock 2 (2025), Q5	—
110	40	IOQM Mock 1 (2024), Q2	—
111	16	IOQM Mock 2 (2026), Q12	—
112	19	IOQM Mock 2 (2025), Q1	—
113	13	IOQM Mock 1 (2026), Q1	—
114	02	IOQM Mock 6 (2024), Q17	—
115	—	RMO Mock 2 (2024), P6	—
116	—	RMO Handout Series	APMO 2005
117	—	Proficiency Test (2025), P3	—
118	—	Proficiency Test (2025), P8	—
119	—	Proficiency Test (2025), P4	—

A.2. Number Theory

Divisibility, GCD and LCM

#	Ans.	Olympiad Arena material	Original source
120	15	IOQM Mock 4 (2024), Q10	—
121	07	IOQM Mock 6 (2024), Q10	—
122	94	IOQM Mock 5 (2024), Q3	—
123	—	RMO Handout Series	Singapore Junior 2019
124	03	IOQM Mock 2 (2026), Q15	—
125	—	NT Pset (INMO), Divisibility P3	—
126	—	RMO Handout Series	Balkan MO 2007
127	—	Diophantine Handout, P1	—
128	—	NT Handout (RMO), P4	IMO 1988, Problem 6
129	—	NT Handout (RMO), P4	—
130	—	NT Handout (RMO), P29	Czech–Austrian–Polish–Slovak 2021
131	—	NT Pset (INMO), Divisibility P5	—
132	—	NT Handout (RMO), P16	—
133	—	RMO Mock 7.1 (2024), P5	—
134	—	NT Handout (RMO), P29	—
135	—	RMO Mock 2025, Paper 3, P1	—
136	—	RMO Handout Series	Kosovo TST 2019
137	—	RMO Handout Series	Serbia 2016
138	—	RMO Handout Series	Argentina 2022

#	Ans.	Olympiad Arena material	Original source
139	—	RMO Handout Series	IMO Shortlist 2005
140	—	RMO Handout Series	Russia (RMO) 2018
141	—	RMO Handout Series	Spain 2022
142	—	RMO Mock 2025, Paper 2, P3	—
143	—	RMO Mock 2025, Paper 3, P5	—
144	—	NT Pset (INMO), Divisibility P7	USA TST 2010
145	—	Combinatorics Handout, Recurrences P12	—
146	—	NT Handout (RMO), P2	—

Congruences and Orders

#	Ans.	Olympiad Arena material	Original source
147	02	IOQM Mock 2 (2026), Q1	—
148	03	IOQM Mock 2 (2025), Q10	—
149	—	NT Handout (RMO), P11	—
150	—	NT Handout (RMO), P15	—
151	—	NT Handout (RMO), P25	—
152	—	NT Handout (RMO), P20	—
153	—	NT Handout (RMO), P28	—
154	—	RMO Handout Series	Iran 2007
155	61	IOQM Mock 1 (2025), Q14	—
156	16	IOQM Mock 3 (2024), Q17	—
157	14	IOQM Mock 3 (2024), Q7	—
158	10	IOQM Mock 2 (2025), Q27	—
159	27	IOQM Mock 3 (2024), Q28	—
160	04	IOQM Mock 3 (2024), Q27	—
161	—	NT Handout (RMO), P26	—
162	—	RMO Handout Series	IMO 2003, Problem 6
163	—	Post-INMO Pset, P8	ELMO Shortlist 2010
164	—	Diophantine Handout, P11	—
165	—	Post-INMO Pset, P6	—
166	—	Diophantine Handout, P13	—
167	—	RMO Mock 2025, Paper 5, P2	—
168	—	RMO Mock 2025, Paper 1, P1	—

Primes

#	Ans.	Olympiad Arena material	Original source
169	23	IOQM Mock 4 (2024), Q2	—
170	34	IOQM Mock 2 (2026), Q8	—
171	—	RMO Handout Series	China (CHNMO) 2009
172	—	RMO Handout Series	Korea 2007
173	—	NT Handout (RMO), P27	—
174	—	RMO Mock 3 (2024), P1	—
175	—	RMO Mock 7.2 (2024), P2	—
176	—	RMO Mock 2025, Paper 4, P3	—
177	—	RMO Mock 6 (2024), P4	—
178	—	RMO Mock 6 (2024), P3	—
179	—	NT Handout (RMO), P12	—
180	—	NT Handout (RMO), P13	—
181	—	RMO Handout Series	JBMO 2019
182	—	RMO Handout Series	Korea 2003
183	—	RMO Handout Series	Turkey 2007
184	—	RMO Handout Series	JBMO Shortlist 2023
185	—	RMO Handout Series	JBMO Shortlist 2020
186	40	IOQM Mock 5 (2024), Q27	—
187	—	RMO Handout Series	USAMO 2017
188	—	RMO Mock 5 (2024), P6	—
189	—	RMO Handout Series	Spain 2024
190	98	IOQM Mock 6 (2024), Q20	—

Diophantine Equations

#	Ans.	Olympiad Arena material	Original source
191	06	IOQM Mock 1 (2024), Q7	—
192	01	IOQM Mock 1 (2024), Q11	—
193	—	Diophantine Handout, P7	—
194	—	Diophantine Handout, P27	—
195	—	Diophantine Handout, P26	—
196	—	NT Handout (RMO), P3	—
197	—	Diophantine Handout, P30	—
198	—	Diophantine Handout, P25	—
199	—	Diophantine Handout, P24	—
200	—	RMO Handout Series	Korea Junior 2022
201	01	IOQM Mock 4 (2024), Q7	—
202	—	Diophantine Handout, P28	—

#	Ans.	Olympiad Arena material	Original source
203	—	RMO Handout Series	Canada 2009
204	—	NT Handout (RMO), P1	—
205	—	RMO Handout Series	Philippines 2018
206	—	RMO Handout Series	Italy TST 2009
207	—	RMO Mock 4 (2024), P3	—
208	—	RMO Mock 4 (2024), P6	—
209	—	RMO Handout Series	Germany 2008
210	—	RMO Handout Series	Japan 2019
211	—	RMO Handout Series	JBMO 2018
212	—	RMO Handout Series	JBMO 2015
213	—	RMO Handout Series	Balkan MO 2013
214	—	Diophantine Handout, P4	—
215	—	RMO Mock 7.3 (2024), P4	—
216	—	Diophantine Handout, P12	—
217	04	IOQM Mock 1 (2026), Q3	—
218	02	IOQM Mock 1 (2026), Q4	—
219	72	IOQM Mock 3 (2024), Q19	—
220	07	IOQM Mock 1 (2025), Q27	—
221	—	Diophantine Handout, P23	—
222	—	Diophantine Handout, P21	—
223	—	NT Handout (RMO), P18	—
224	—	NT Handout (RMO), P19	—
225	—	Diophantine Handout, P8	—
226	—	Diophantine Handout, P22	—
227	—	Diophantine Handout, P20	—
228	—	Diophantine Handout, P15	—
229	—	Diophantine Handout, P10	—
230	—	RMO Handout Series	Iran 2010
231	—	Diophantine Handout, P19	—
232	—	RMO Handout Series	IMO 1992, Problem 1
233	—	RMO Handout Series	Macedonia 2009
234	—	Diophantine Handout, P29	—
235	—	Diophantine Handout, P9	—
236	—	Diophantine Handout, P16	—
237	—	RMO Handout Series	Slovenia TST 2008–09
238	—	Diophantine Handout, P14	—
239	—	Diophantine Handout, P17	—
240	—	Diophantine Handout, P18	—

#	Ans.	Olympiad Arena material	Original source
241	—	Diophantine Handout, XYZ section P1	—

Arithmetic Functions

#	Ans.	Olympiad Arena material	Original source
242	28	IOQM Mock 2 (2024), Q2	—
243	—	RMO Mock 1 (2024), P1	—
244	—	NT Handout (RMO), P5	—
245	—	NT Pset (INMO), Arithmetic Functions P6	—
246	—	RMO Handout Series	Canada Junior 2022
247	—	RMO Handout Series	Canada 2017
248	—	NT Pset (INMO), Arithmetic Functions P1	—
249	—	NT Pset (INMO), Arithmetic Functions P2	—
250	—	NT Pset (INMO), Arithmetic Functions P8	—
251	—	NT Pset (INMO), Arithmetic Functions P5	—
252	—	NT Pset (INMO), Arithmetic Functions P7	—
253	—	NT Handout (RMO), P10	—
254	—	NT Handout (RMO), P8	—
255	—	NT Handout (RMO), P22	—
256	—	NT Pset (INMO), Arithmetic Functions P3	—
257	—	RMO Handout Series	IMO Shortlist 2004
258	—	Post-INMO Pset, P3	USA TSTST 2016
259	—	Post-INMO Pset, Advanced P1	USA TSTST 2020
260	—	Post-INMO Pset, Advanced P2	—
261	—	Post-INMO Pset, Advanced P3	ELMO Shortlist 2010
262	—	Post-INMO Pset, P7	—
263	—	Post-INMO Pset, P4	—
264	—	Post-INMO Pset, P1	—
265	—	Post-INMO Pset, P5	—
266	—	Post-INMO Pset, Advanced P4	—
267	—	Post-INMO Pset, P2	IMO Shortlist 2015
268	—	Post-INMO Pset, Advanced P6	China TST 2019

Digits and Base Representation

#	Ans.	Olympiad Arena material	Original source
269	20	IOQM Mock 1 (2024), Q9	—

#	Ans.	Olympiad Arena material	Original source
270	90	IOQM Mock 5 (2024), Q9	—
271	45	IOQM Mock 1 (2024), Q29	—
272	18	IOQM Mock 3 (2024), Q16	—
273	03	IOQM Mock 2 (2025), Q15	—
274	32	IOQM Mock 3 (2024), Q1	—
275	29	IOQM Mock 2 (2026), Q9	—
276	33	IOQM Mock 2 (2026), Q10	—
277	61	IOQM Mock 1 (2025), Q13	—
278	68	IOQM Mock 1 (2024), Q18	—
279	—	RMO Handout Series	Singapore 2009
280	—	RMO Handout Series	Mediterranean MO 2022
281	—	RMO Handout Series	APMO 2001
282	—	NT Handout (RMO), P14	—
283	01	IOQM Mock 1 (2026), Q6	—
284	—	Post-INMO Pset, P11	USA TSTST 2022
285	—	RMO Mock 2025, Paper 4, P1	—
286	—	NT Handout (RMO), P21	—
287	—	RMO Handout Series	Australia 2024
288	—	RMO Handout Series	France TST 2007
289	47	IOQM Mock 2 (2025), Q21	—
290	—	Combinatorics Handout, CNT P3	—

Advanced and Miscellaneous

#	Ans.	Olympiad Arena material	Original source
291	—	NT Handout (RMO), P9	—
292	—	NT Handout (RMO), P17	—
293	—	Combinatorics Handout, Fibonacci P2	—
294	—	Combinatorics Handout, Recurrences P1	—
295	—	Combinatorics Handout, Recurrences P2	—
296	—	Combinatorics Handout, Recurrences P3	—
297	—	Combinatorics Handout, Recurrences P11	—
298	—	RMO Handout Series	Korea Junior 2023
299	—	Combinatorics Handout, More Sequences P3	—
300	—	RMO Handout Series	Philippines 2018
301	—	NT Handout (RMO), P7	—
302	—	NT Handout (RMO), P6	—

#	Ans.	Olympiad Arena material	Original source
303	—	NT Pset (INMO), Divisibility P4	—
304	—	Diophantine Handout, P2	—
305	—	Diophantine Handout, P3	—
306	—	RMO Handout Series	IMO 2023, Problem 1
307	—	RMO Handout Series	Mediterranean MO 2020
308	—	RMO Handout Series	JBMO 2017
309	—	RMO Handout Series	JBMO Shortlist 2023
310	—	Combinatorics Handout, Recurrences P4	—
311	—	Combinatorics Handout, Recurrences P5	—
312	—	Combinatorics Handout, Recurrences P6	—
313	—	Combinatorics Handout, Recurrences P9	—
314	—	NT Handout (RMO), P23	—
315	—	NT Handout (RMO), P24	—
316	—	RMO Mock 2025, Paper 6, P1	—
317	—	RMO Handout Series	JBMO Shortlist 2019
318	—	Post-INMO Pset, P9	—
319	—	Post-INMO Pset, Advanced P5	—
320	—	Post-INMO Pset, P10	Simon Marais 2021

A.3. Geometry

Angle Chasing and Cyclic Quadrilaterals

#	Ans.	Olympiad Arena material	Original source
321	20	IOQM Mock 1 (2024), Q3	—
322	39	IOQM Mock 1 (2024), Q14	—
323	20	IOQM Mock 6 (2024), Q13	—
324	30	IOQM Mock 5 (2024), Q14	—
325	39	IOQM Mock 6 (2024), Q4	—
326	20	IOQM Mock 3 (2024), Q3	—
327	95	IOQM Mock 1 (2025), Q9	—
328	03	IOQM Mock 1 (2026), Q9	—
329	20	IOQM Mock 5 (2024), Q2	—
330	72	IOQM Mock 2 (2025), Q5	—
331	39	IOQM Mock 3 (2024), Q11	—
332	—	Geometry Handout (IOQM), P17	—
333	—	Geometry Handout (IOQM), P14	—
334	—	Geometry Handout (IOQM), P19	—
335	—	RMO Handout Series	China (CMC) 2010

#	Ans.	Olympiad Arena material	Original source
336	04	IOQM Mock 2 (2026), Q6	—
337	69	IOQM Mock 5 (2024), Q25	—
338	20	IOQM Mock 4 (2024), Q21	—
339	00	IOQM Mock 2 (2026), Q27	—
340	—	Geometry Handout (IOQM), P18	—
341	90	IOQM Mock 2 (2026), Q29	—
342	—	RMO Handout Series	APMO 2007
343	—	RMO Handout Series	Hong Kong 2006
344	—	Geometry Handout (IOQM), P16	—
345	—	RMO Mock 2025, Paper 2, P2	—

Triangle Centres

#	Ans.	Olympiad Arena material	Original source
346	22	IOQM Mock 3 (2024), Q4	—
347	28	IOQM Mock 1 (2025), Q4	—
348	11	IOQM Mock 2 (2026), Q14	—
349	26	IOQM Mock 5 (2024), Q26	—
350	13	IOQM Mock 1 (2025), Q17	—
351	24	IOQM Mock 4 (2024), Q9	—
352	12	IOQM Mock 6 (2024), Q21	—
353	—	RMO Handout Series	China (CMC) 2010
354	—	RMO Handout Series	Croatia 2009
355	—	RMO Handout Series	China (CMC) 2009
356	—	Geometry Handout (IOQM), P15	—
357	—	RMO Mock 2025, Paper 5, P3	—
358	—	RMO Handout Series	Turkey 2008
359	—	RMO Handout Series	Iran TST 2009
360	16	IOQM Mock 5 (2024), Q16	—
361	17	IOQM Mock 6 (2024), Q25	—
362	53	IOQM Mock 3 (2024), Q21	—
363	20	IOQM Mock 1 (2025), Q26	—
364	—	RMO Mock 2025, Paper 6, P3	—
365	90	IOQM Mock 1 (2025), Q29	—

Circles, Tangency and Power of a Point

#	Ans.	Olympiad Arena material	Original source
366	20	IOQM Mock 1 (2024), Q4	—
367	96	IOQM Mock 6 (2024), Q3	—
368	39	IOQM Mock 1 (2024), Q15	—
369	20	IOQM Mock 3 (2024), Q9	—
370	39	IOQM Mock 4 (2024), Q29	—
371	61	IOQM Mock 2 (2025), Q19	—
372	24	IOQM Mock 5 (2024), Q23	—
373	14	IOQM Mock 2 (2026), Q16	—
374	90	IOQM Mock 6 (2024), Q14	—
375	09	IOQM Mock 6 (2024), Q27	—
376	16	IOQM Mock 4 (2024), Q30	—
377	—	RMO Handout Series	China (CMC) 2008
378	—	RMO Handout Series	Japan 2009
379	—	RMO Handout Series	China (CMC) 2008
380	20	IOQM Mock 4 (2024), Q21	—
381	—	RMO Handout Series	Estonia TST 2008
382	—	Geometry Handout (IOQM), P12	—
383	—	Geometry Handout (IOQM), P13	—
384	—	Geometry Handout (IOQM), P11	—
385	—	Geometry Handout (IOQM), P10	—
386	20	IOQM Mock 1 (2026), Q20	—
387	—	Geometry Handout (IOQM), P9	—
388	06	IOQM Mock 1 (2026), Q21	—
389	—	RMO Handout Series	Italy TST 2009
390	—	RMO Handout Series	USAMO 2009
391	—	RMO Handout Series	APMO 2009
392	07	IOQM Mock 3 (2024), Q29	—
393	—	RMO Mock 2025, Paper 1, P4	—

Lengths, Areas and Computation

#	Ans.	Olympiad Arena material	Original source
394	20	IOQM Mock 1 (2024), Q5	—
395	20	IOQM Mock 1 (2024), Q6	—
396	20	IOQM Mock 2 (2024), Q5	—
397	04	IOQM Mock 2 (2025), Q6	—
398	56	IOQM Mock 6 (2024), Q5	—
399	86	IOQM Mock 6 (2024), Q6	—

#	Ans.	Olympiad Arena material	Original source
400	22	IOQM Mock 3 (2024), Q4	—
401	35	IOQM Mock 1 (2024), Q5	—
402	46	IOQM Mock 6 (2024), Q7	—
403	22	IOQM Mock 1 (2024), Q8	—
404	72	IOQM Mock 2 (2024), Q5	—
405	64	IOQM Mock 5 (2024), Q7	—
406	04	IOQM Mock 2 (2024), Q8	—
407	01	IOQM Mock 1 (2026), Q8	—
408	60	IOQM Mock 2 (2024), Q4	—
409	15	IOQM Mock 5 (2024), Q8	—
410	08	IOQM Mock 1 (2026), Q16	—
411	60	IOQM Mock 6 (2024), Q8	—
412	10	IOQM Mock 1 (2026), Q20	—
413	15	IOQM Mock 1 (2026), Q17	—
414	27	IOQM Mock 1 (2026), Q18	—
415	19	IOQM Mock 1 (2026), Q19	—
416	98	IOQM Mock 6 (2024), Q9	—
417	06	IOQM Mock 1 (2026), Q10	—
418	09	IOQM Mock 3 (2024), Q9	—
419	16	IOQM Mock 6 (2024), Q11	—
420	—	RMO Handout Series	Slovenia TST 2008–09
421	11	IOQM Mock 4 (2024), Q11	—
422	17	IOQM Mock 4 (2024), Q11	—
423	12	IOQM Mock 4 (2024), Q12	—
424	20	IOQM Mock 6 (2024), Q13	—
425	01	IOQM Mock 2 (2026), Q17	—
426	01	IOQM Mock 1 (2026), Q13	—
427	19	IOQM Mock 5 (2024), Q19	—
428	14	IOQM Mock 5 (2024), Q20	—
429	09	IOQM Mock 3 (2024), Q20	—
430	16	IOQM Mock 5 (2024), Q24	—
431	42	IOQM Mock 4 (2024), Q23	—
432	21	IOQM Mock 5 (2024), Q24	—
433	66	IOQM Mock 4 (2024), Q25	—
434	28	IOQM Mock 6 (2024), Q21	—
435	—	RMO Handout Series	China (CMC) 2010
436	—	Geometry Handout (IOQM), P8	—
437	—	RMO Handout Series	Romania 2008

#	Ans.	Olympiad Arena material	Original source
438	—	RMO Handout Series	Turkey TST 2009

Collinearity, Concurrency and Configurations

#	Ans.	Olympiad Arena material	Original source
439	—	RMO Handout Series	Russia (RMO) 2008
440	—	RMO Handout Series	Italy TST 2008
441	—	RMO Handout Series	China (CMC) 2008
442	—	RMO Handout Series	China (CMC) 2008
443	—	RMO Handout Series	China (CMC) 2009
444	—	Geometry Handout (IOQM), P7	—
445	—	Geometry Handout (IOQM), P6	—
446	—	RMO Handout Series	China (CMC) 2009
447	—	RMO Handout Series	Russia (RMO) 2008
448	—	RMO Handout Series	Belarus 2009
449	—	Geometry Handout (IOQM), P5	—
450	—	Geometry Handout (IOQM), P4	—
451	—	RMO Handout Series	Croatia 2009
452	—	RMO Handout Series	Slovenia TST 2008–09
453	—	RMO Mock 2025, Paper 3, P4	—
454	—	RMO Mock 6 (2024), P2	—
455	—	RMO Mock 5 (2024), P4	—
456	—	RMO Mock 4 (2024), P4	—
457	—	RMO Handout Series	India TST 2008
458	—	RMO Handout Series	USAMO 2010
459	—	RMO Handout Series	IMO 2009, Problem 4
460	—	RMO Handout Series	Balkan MO 2009
461	—	RMO Handout Series	China (CMC) 2010
462	—	RMO Handout Series	Bulgaria TST 2009
463	—	RMO Handout Series	Russia (RMO) 2009
464	—	RMO Handout Series	Italy 2009
465	—	RMO Handout Series	Singapore TST 2008
466	—	RMO Handout Series	Vietnam 2007
467	—	RMO Handout Series	Belarus 2009
468	—	RMO Handout Series	Vietnam 2009
469	—	RMO Handout Series	Austria 2005
470	—	RMO Handout Series	Russia (RMO) 2008–09
471	—	RMO Handout Series	China (CMC) 2010

#	Ans.	Olympiad Arena material	Original source
472	—	RMO Handout Series	China (CMC) 2009
473	—	RMO Handout Series	Thailand 2007
474	—	RMO Handout Series	Turkey 2008
475	—	RMO Handout Series	IMO Shortlist 2008
476	—	RMO Mock 3 (2024), P4	—
477	—	RMO Mock 2 (2024), P4	—
478	—	RMO Mock 1 (2024), P4	—
479	—	RMO Mock 7.3 (2024), P2	—
480	—	RMO Mock 7.2 (2024), P3	—
481	—	RMO Mock 7.1 (2024), P4	—
482	—	RMO Mock 2025, Paper 4, P4	—
483	—	RMO Mock 2025, Paper 4, P2	—
484	—	RMO Mock 2025, Paper 6, P2	—
485	—	INMO Mock 1 (2024), P4	—
486	—	INMO Mock 1 (2024), P6	—
487	—	Proficiency Test (2025), P5	—
488	—	Proficiency Test (2025), P6	—
489	—	Proficiency Test (2025), P7	—
490	—	Proficiency Test (2025), P9	—
491	01	IOQM Mock 2 (2026), Q17	—

Constructions

#	Ans.	Olympiad Arena material	Original source
492	—	Constructions Handout, P1	—
493	—	Constructions Handout, P2	—
494	—	Constructions Handout, P3	—
495	—	Constructions Handout, P4	—
496	—	Constructions Handout, P5	—
497	—	Constructions Handout, P6	—
498	—	Constructions Handout, P7	—
499	—	Constructions Handout, P8	—
500	—	Constructions Handout, P9	—
501	—	Constructions Handout, P10	—
502	—	Constructions Handout, P11	—
503	—	Constructions Handout, P12	—
504	—	Constructions Handout, P13	—
505	—	Constructions Handout, P14	—

#	Ans.	Olympiad Arena material	Original source
506	—	Constructions Handout, P15	—
507	—	Constructions Handout, P16	—
508	—	Constructions Handout, P17	—
509	—	Constructions Handout, P18	—
510	—	Constructions Handout, P19	—
511	—	Constructions Handout, P20	—
512	—	Constructions Handout, P21	—
513	—	Constructions Handout, P22	—
514	—	Constructions Handout, P23	—
515	—	Constructions Handout, P24	—
516	—	Constructions Handout, P25	—
517	—	Constructions Handout, P26	—
518	—	Constructions Handout, P27	—
519	—	Constructions Handout, P28	—
520	—	Constructions Handout, P29	—
521	—	Constructions Handout, P30	—
522	—	Constructions Handout, P31	—
523	—	Constructions Handout, P32	—
524	—	Constructions Handout, P33	—
525	—	Constructions Handout, P34	—
526	—	Constructions Handout, P35	—
527	—	Constructions Handout, P36	—

A.4. Combinatorics

Counting and Enumeration

#	Ans.	Olympiad Arena material	Original source
528	96	IOQM Mock 2 (2026), Q2	—
529	12	IOQM Mock 2 (2026), Q3	—
530	20	IOQM Mock 1 (2024), Q3	—
531	00	IOQM Mock 1 (2026), Q12	—
532	64	IOQM Mock 2 (2024), Q7	—
533	84	IOQM Mock 2 (2024), Q20	—
534	08	IOQM Mock 2 (2026), Q11	—
535	04	IOQM Mock 2 (2026), Q4	—
536	99	IOQM Mock 2 (2026), Q18	—
537	90	IOQM Mock 2 (2026), Q24	—
538	81	IOQM Mock 2 (2026), Q26	—

#	Ans.	Olympiad Arena material	Original source
539	72	IOQM Mock 2 (2026), Q21	—
540	28	IOQM Mock 2 (2024), Q22	—
541	47	IOQM Mock 2 (2024), Q30	—
542	—	RMO Mock 3 (2024), P5	—
543	—	RMO Handout Series	Canada 2005
544	—	RMO Handout Series	Canada 1971
545	20	IOQM Mock 5 (2024), Q13	—
546	—	RMO Handout Series	Italy 2016
547	—	RMO Handout Series	USAJMO 2018
548	—	RMO Mock 2 (2024), P2	—
549	91	IOQM Mock 5 (2024), Q21	—
550	—	RMO Handout Series	Turkey Junior 2015
551	—	RMO Handout Series	Turkey Junior 2020
552	—	RMO Handout Series	Argentina 2021
553	—	RMO Handout Series	Iran 2008
554	—	RMO Handout Series	JBMO 2016
555	—	RMO Handout Series	APMO 2017
556	—	RMO Handout Series	Italy 2023
557	—	RMO Handout Series	Italy 2015

Probability and Expectation

#	Ans.	Olympiad Arena material	Original source
558	—	RMO Handout Series	USAMO 1973
559	13	IOQM Mock 3 (2024), Q13	—
560	11	IOQM Mock 4 (2024), Q11	—
561	09	IOQM Mock 1 (2026), Q22	—
562	20	IOQM Mock 1 (2026), Q23	—
563	26	IOQM Mock 1 (2026), Q28	—

Graphs and Networks

#	Ans.	Olympiad Arena material	Original source
564	—	RMO Handout Series	Russia (RMO) 2014
565	—	RMO Handout Series	Iran (ICO) 2020
566	—	RMO Mock 2025, Paper 1, P3	—
567	—	Combinatorics Handout, Graphs P4	—

#	Ans.	Olympiad Arena material	Original source
568	—	RMO Handout Series	Russia (RMO) 2021
569	—	RMO Handout Series	Russia (RMO) 2019
570	—	RMO Handout Series	APMO 2023
571	—	RMO Handout Series	Germany 2018

Games and Strategy

#	Ans.	Olympiad Arena material	Original source
572	07	IOQM Mock 2 (2026), Q28	—
573	—	RMO Handout Series	Indonesia 2006
574	—	RMO Handout Series	Japan 2007
575	—	RMO Handout Series	Turkey Junior 2013
576	39	IOQM Mock 2 (2026), Q22	—
577	—	RMO Handout Series	Canada 2023
578	—	RMO Handout Series	JBMO Shortlist 2020
579	—	RMO Handout Series	JBMO 2023
580	—	RMO Handout Series	Czech–Polish–Slovak 2022
581	—	RMO Handout Series	Czech–Polish–Slovak 2017
582	—	RMO Handout Series	JBMO Shortlist 2022
583	—	RMO Mock 2025, Paper 4, P5	—
584	—	RMO Handout Series	France 2023

Invariants, Colourings and Monovariants

#	Ans.	Olympiad Arena material	Original source
585	—	RMO Handout Series	Balkan MO 2001
586	—	RMO Handout Series	Iran 2014
587	—	RMO Handout Series	APMO 2009
588	—	RMO Handout Series	Canada 2016
589	—	RMO Handout Series	Japan 1991
590	—	RMO Handout Series	Russia (RMO) 2023
591	—	RMO Handout Series	Iran (ICO) 2021
592	—	RMO Handout Series	Argentina 2012
593	—	RMO Handout Series	Bulgaria 2023
594	—	RMO Handout Series	JBMO Shortlist 2021

#	Ans.	Olympiad Arena material	Original source
595	—	RMO Handout Series	JBMO Shortlist 2022

Extremal Combinatorics and Pigeonhole

#	Ans.	Olympiad Arena material	Original source
596	—	RMO Handout Series	IMO 1972, Problem 1
597	—	Combinatorics Handout, Extremal P2	—
598	—	Combinatorics Handout, Extremal P3	—
599	—	Combinatorics Handout, CNT P4	—
600	—	RMO Handout Series	JBMO 2019
601	—	RMO Handout Series	Czech–Polish–Slovak 2023
602	—	RMO Mock 7.3 (2024), P3	—
603	—	INMO Mock 1 (2024), P3	—

Grids, Tilings and Boards

#	Ans.	Olympiad Arena material	Original source
604	—	RMO Handout Series	Canada Junior 2022
605	—	RMO Handout Series	USAJMO 2013
606	—	RMO Handout Series	Canada 2015
607	—	RMO Handout Series	Japan 2019
608	—	RMO Handout Series	USAMO 1976
609	14	IOQM Mock 2 (2026), Q5	—
610	—	Combinatorics Handout, Grids P7	—
611	—	RMO Mock 2025, Paper 5, P1	—

Appendix B

Index of Source Material

Mock test papers

Document	Problems	Format
IOQM Mock Test 1 (July 2024)	30	Integer answer, 3 hours
IOQM Mock Test 2 (July 2024)	30	Integer answer, 3 hours
IOQM Mock Test 3 (July 2024)	30	Integer answer, 3 hours
IOQM Mock Test 4 (August 2024)	30	Integer answer, 3 hours
IOQM Mock Test 5 (August 2024)	30	Integer answer, 3 hours
IOQM Mock Test 6 (September 2024)	30	Integer answer, 3 hours
IOQM Mock Test 1 (2025)	30	Integer answer, 3 hours
IOQM Mock Test 2 (2025)	30	Integer answer, 3 hours
IOQM Mock Test 1 (June 2026)	30	Integer answer, 3 hours
IOQM Mock Test 2 (June 2026)	30	Integer answer, 3 hours
RMO Mock Test 1 (2024)	6	Proof, 3 hours
RMO Mock Test 2 (2024)	6	Proof, 3 hours
RMO Mock Test 3 (2024)	6	Proof, 3 hours
RMO Mock Test 4 (2024)	6	Proof, 3 hours
RMO Mock Test 5 (2024)	6	Proof, 3 hours
RMO Mock Test 6 (2024)	6	Proof, 3 hours
RMO Diwali Special 7.1 / 7.2 / 7.3 (2024)	18	Proof, three difficulty tiers
RMO Diwali Mocks, Papers 1–6 (2025)	36	Proof, 3 hours each
INMO Mock Test 1 (2024)	6	Proof, 4.5 hours

Handouts and problem sets

Document	Problems	Subject
RMO Handout Series	200	Algebra, NT, geometry, combinatorics; 50 each
Number Theory Handout (RMO level)	30	Mixed number theory
Diophantine Equations Handout	30	Equations in integers
Number Theory Pset (INMO level)	30	Divisibility, arithmetic functions
Post-INMO Problem Set	15	Advanced arithmetic functions
Geometry Handout (IOQM level)	19	Angle chasing, configurations
Constructions Handout	36	Ruler-and-compass constructions
Combinatorics and PnC Handout	65	Counting, recurrences, sequences
Mathematics Proficiency Test (2025)	13	Mixed

Appendix C

A Reference of Olympiad Mathematics

C.1. Algebra

C.1.1. Polynomials

Definition C.1.1. A *polynomial* over a ring R is a formal expression $P(x) = a_n x^n + \cdots + a_1 x + a_0$ with $a_i \in R$. If $a_n \neq 0$ then n is the *degree* $\deg P$, and a_n is the *leading coefficient*; P is *monic* when $a_n = 1$. The zero polynomial is conventionally assigned degree $-\infty$.

Theorem C.1.2. For the quadratic $ax^2 + bx + c$ with $a \neq 0$ and real coefficients, the roots are $x = (-b \pm \sqrt{D})/2a$ where $D = b^2 - 4ac$ is the *discriminant*. The roots are real and distinct when $D > 0$, equal when $D = 0$, and a conjugate pair of non-real complex numbers when $D < 0$.

Theorem C.1.3 (Division algorithm). For polynomials A, B over a field with $B \neq 0$ there are unique Q, R with $A = BQ + R$ and $\deg R < \deg B$. Over $\mathbb{Z}$ the same holds provided B is monic.

Theorem C.1.4 (Factor theorem). $(x - a)$ divides $P(x)$ if and only if $P(a) = 0$. More generally the remainder on dividing P by $(x - a)$ is $P(a)$.

Theorem C.1.5. A nonzero polynomial of degree n over a field has at most n roots. Consequently two polynomials of degree at most n agreeing at $n + 1$ points are identical — the workhorse behind most identity-proving arguments.

Theorem C.1.6 (Fundamental theorem of algebra). Every non-constant polynomial with complex coefficients has a complex root, hence splits into linear factors over $\mathbb{C}$. A real polynomial factors over $\mathbb{R}$ into linear and irreducible quadratic factors, and its non-real roots occur in conjugate pairs.

Theorem C.1.7 (Vieta's formulas). If $P(x) = x^n + c_1 x^{n-1} + \cdots + c_n$ has roots $\alpha_1, \ldots, \alpha_n$ counted with multiplicity, then $c_k = (-1)^k e_k(\alpha_1, \ldots, \alpha_n)$, where e_k is the k-th elementary symmetric polynomial. In particular $\sum \alpha_i = -c_1$ and $\prod \alpha_i = (-1)^n c_n$.

Definition C.1.8. The *elementary symmetric polynomials* in $x_1, \ldots, x_n$ are $e_k = \sum x_{i_1} \cdots x_{i_k}$, the sum running over all k-element subsets of indices. The *power sums* are $p_k = x_1^k + \cdots + x_n^k$.

Theorem C.1.9 (Fundamental theorem of symmetric polynomials). Every symmetric polynomial in $x_1, \ldots, x_n$ is a polynomial in $e_1, \ldots, e_n$, uniquely so.

Theorem C.1.10 (Newton's identities). For $k \geq 1$,

$$p_k - e_1 p_{k-1} + e_2 p_{k-2} - \cdots + (-1)^{k-1} e_{k-1} p_1 + (-1)^k k e_k = 0,$$

with $e_j = 0$ for $j > n$. These convert between power sums and elementary symmetric functions in either direction.

Theorem C.1.11 (Binomial theorem). For x, y in a commutative ring and $n \in \mathbb{N}$, $(x+y)^n = \sum_{i=0}^{n} \binom{n}{i} x^{n-i} y^i$.

Theorem C.1.12. Useful binomial identities: $\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$ (Pascal); $\sum_k \binom{n}{k} = 2^n$; $\sum_k (-1)^k \binom{n}{k} = 0$ for $n \geq 1$; $\sum_i \binom{m}{i}\binom{n}{k-i} = \binom{m+n}{k}$ (Vandermonde); and the hockey stick $\sum_{j=r}^{n} \binom{j}{r} = \binom{n+1}{r+1}$.

Theorem C.1.13 (Lagrange interpolation). Given $n+1$ distinct nodes $x_0, \ldots, x_n$ and values $y_0, \ldots, y_n$ there is exactly one polynomial of degree at most n with $P(x_i) = y_i$, namely

$$P(x) = \sum_{i=0}^{n} y_i \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}.$$

Definition C.1.14. The *finite difference* operator is $(\Delta P)(x) = P(x+1) - P(x)$. If $\deg P = n$ then $\deg \Delta P = n-1$, and $\Delta^{n+1} P = 0$. Hence $\sum_k (-1)^k \binom{n+1}{k} P(x+k) = 0$ for any polynomial of degree at most n — the standard tool for problems specifying P at consecutive integers.

Theorem C.1.15 (Rational root theorem). If $P(x) = a_n x^n + \cdots + a_0$ has integer coefficients and p/q is a rational root in lowest terms, then $p \mid a_0$ and $q \mid a_n$.

Theorem C.1.16 (Gauss's lemma). A polynomial with integer coefficients that factors over $\mathbb{Q}$ factors over $\mathbb{Z}$ into factors of the same degrees. A primitive polynomial is irreducible over $\mathbb{Z}$ exactly when it is irreducible over $\mathbb{Q}$.

Theorem C.1.17 (Eisenstein's criterion). Let $P(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$. If a prime p divides $a_0, \ldots, a_{n-1}$ but not a_n, and $p^2 \nmid a_0$, then P is irreducible over $\mathbb{Q}$.

Remark. In the extended form, if $p \mid a_0, \ldots, a_k$, $p \nmid a_{k+1}$ and $p^2 \nmid a_0$, then P has an irreducible factor of degree exceeding k. Applying the criterion after a substitution $x \mapsto x + c$ often rescues an apparently unpromising polynomial; $\Phi_p(x) = 1 + x + \cdots + x^{p-1}$ is the standard example, via $x \mapsto x + 1$.

Theorem C.1.18. For $P \in \mathbb{Z}[x]$ and integers $a \neq b$, $(a-b) \mid P(a) - P(b)$. This single observation settles a surprising proportion of polynomial problems posed over the integers.

Definition C.1.19. A root α of P has *multiplicity* m if $(x-\alpha)^m \mid P$ but $(x-\alpha)^{m+1} \nmid P$; equivalently $P(\alpha) = P'(\alpha) = \cdots = P^{(m-1)}(\alpha) = 0 \neq P^{(m)}(\alpha)$. A polynomial has a repeated root exactly when $\gcd(P, P') \neq 1$.

C.1.2. Complex Numbers and Roots of Unity

Definition C.1.20. A complex number is $z = a + bi$ with $i^2 = -1$; its *conjugate* is $\bar{z} = a - bi$, its *modulus* $|z| = \sqrt{a^2 + b^2}$, and $z\bar{z} = |z|^2$. In polar form $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$.

Theorem C.1.21 (De Moivre). $(\cos\theta + i\sin\theta)^n = \cos n\theta + i \sin n\theta$ for $n \in \mathbb{Z}$. Expanding the left side and comparing parts yields the multiple-angle formulas.

Definition C.1.22. The n-th *roots of unity* are $\zeta^k = e^{2\pi i k/n}$, $k = 0, \ldots, n-1$. They form a cyclic group of order n under multiplication, sum to 0 for $n \geq 2$, and are exactly the roots of $x^n - 1$.

Theorem C.1.23 (Roots of unity filter). If ζ is a primitive n-th root of unity and $f(x) = \sum a_k x^k$, then

$$\sum_{k \equiv r \pmod{n}} a_k = \frac{1}{n} \sum_{j=0}^{n-1} \zeta^{-jr} f(\zeta^j).$$

This extracts every n-th coefficient of a generating function.

Definition C.1.24. $\Phi_n(x)$, the n-th *cyclotomic polynomial*, is the monic polynomial whose roots are the primitive n-th roots of unity. It has integer coefficients, is irreducible over $\mathbb{Q}$, has degree $\varphi(n)$, and $x^n - 1 = \prod_{d|n} \Phi_d(x)$.

Theorem C.1.25. Any prime divisor p of $\Phi_n(a)$ for an integer a either divides n or satisfies $p \equiv 1 \pmod{n}$. This gives a short proof that there are infinitely many primes congruent to 1 modulo n.

C.1.3. Recurrences and Generating Functions

Definition C.1.26. A *linear homogeneous recurrence with constant coefficients* of order k is $x_n = c_1 x_{n-1} + \cdots + c_k x_{n-k}$. Its *characteristic polynomial* is $t^k - c_1 t^{k-1} - \cdots - c_k$.

Theorem C.1.27. If the characteristic polynomial factors as $\prod_{i=1}^{r}(t-\alpha_i)^{m_i}$ with distinct α_i, the general solution is $x_n = \sum_i q_i(n)\alpha_i^n$ where $\deg q_i < m_i$. Distinct roots give constant q_i. The initial values $x_0, \ldots, x_{k-1}$ determine the q_i uniquely.

Definition C.1.28. The *ordinary generating function* of (a_n) is $A(x) = \sum_{n\ge 0} a_n x^n$, treated as a formal power series; the *exponential* generating function is $\sum a_n x^n/n!$. Multiplication of ordinary generating functions corresponds to convolution, $[x^n]A(x)B(x) = \sum_k a_k b_{n-k}$.

Theorem C.1.29. Standard expansions: $\dfrac{1}{1-x} = \sum x^n$, $\dfrac{1}{(1-x)^k} = \sum \binom{n+k-1}{k-1} x^n$, and $(1+x)^\alpha = \sum \binom{\alpha}{n} x^n$ for arbitrary α, where $\binom{\alpha}{n} = \alpha(\alpha-1)\cdots(\alpha-n+1)/n!$.

C.1.4. Inequalities

Theorem C.1.30 (Trivial inequality). $x^2 \ge 0$ for real x, with equality only at $x = 0$. Nearly every inequality below is this statement in disguise, and a surprising number of competition inequalities yield to nothing more after the right substitution.

Theorem C.1.31 (AM–GM). For non-negative reals, $\dfrac{x_1 + \cdots + x_n}{n} \ge \sqrt[n]{x_1 \cdots x_n}$, with equality exactly when all x_i are equal.

Theorem C.1.32 (Power mean inequality). For positive reals and $p \in \mathbb{R}$ set $M_p = \left(\frac{1}{n}\sum x_i^p\right)^{1/p}$ for $p \neq 0$, with M_0, $M_{\pm\infty}$ defined as limits. Then $M_p \le M_q$ whenever $p \le q$. The special values $M_{-\infty}, M_{-1}, M_0, M_1, M_2, M_\infty$ are $\min x_i$, the harmonic, geometric, arithmetic and quadratic means, and $\max x_i$.

Theorem C.1.33 (Cauchy–Schwarz). For reals, $\left(\sum a_i b_i\right)^2 \le \left(\sum a_i^2\right)\left(\sum b_i^2\right)$, with equality when the vectors are proportional.

Theorem C.1.34 (Engel form / Titu's lemma). For reals a_i and positive b_i,

$$\frac{a_1^2}{b_1} + \cdots + \frac{a_n^2}{b_n} \ge \frac{(a_1 + \cdots + a_n)^2}{b_1 + \cdots + b_n}.$$

This is Cauchy–Schwarz rewritten, and is the single most useful form of it in competition.

Theorem C.1.35 (Hölder). For non-negative reals and $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$, $\sum a_i b_i \le \left(\sum a_i^p\right)^{1/p}\left(\sum b_i^q\right)^{1/q}$. Cauchy–Schwarz is the case $p = q = 2$. The three-sequence form $\left(\sum a_i\right)\left(\sum b_i\right)\left(\sum c_i\right) \ge \left(\sum \sqrt[3]{a_i b_i c_i}\right)^3$ is what usually appears in practice.

Theorem C.1.36 (Minkowski). For non-negative reals and $p \ge 1$, $\left(\sum (a_i + b_i)^p\right)^{1/p} \le \left(\sum a_i^p\right)^{1/p} + \left(\sum b_i^p\right)^{1/p}$.

Theorem C.1.37 (Chebyshev's sum inequality). If $a_1 \geq \cdots \geq a_n$ and $b_1 \geq \cdots \geq b_n$ then

$$n\sum a_i b_i \;\geq\; \Big(\sum a_i\Big)\Big(\sum b_i\Big) \;\geq\; n\sum a_i b_{n+1-i}.$$

Theorem C.1.38 (Rearrangement). For $a_1 \leq \cdots \leq a_n$ and $b_1 \leq \cdots \leq b_n$, the sum $\sum a_i b_{\sigma(i)}$ over permutations σ is largest when σ is the identity and smallest when σ reverses the order.

Definition C.1.39. A function f on an interval I is *convex* if $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$ for all $x, y \in I$ and $\lambda \in [0,1]$; *concave* if $-f$ is convex. For twice differentiable f, convexity is equivalent to $f'' \geq 0$.

Theorem C.1.40 (Jensen). For f convex on I, weights $\lambda_i \geq 0$ summing to 1 and points $x_i \in I$, $f(\sum \lambda_i x_i) \leq \sum \lambda_i f(x_i)$. The inequality reverses for concave f.

Theorem C.1.41 (Tangent line trick). If f is convex and ℓ is the tangent to f at the conjectured equality point, then $f(x) \geq \ell(x)$ pointwise, and summing the linear bounds proves the inequality. This handles most symmetric inequalities with a fixed sum constraint.

Definition C.1.42. A sequence a *majorizes* b (both nonincreasing, equal total) if every partial sum of a is at least the corresponding partial sum of b.

Theorem C.1.43 (Karamata). If f is convex and a majorizes b, then $\sum f(a_i) \geq \sum f(b_i)$.

Theorem C.1.44 (Muirhead). Write $T_a(x) = \sum_\sigma x_{\sigma(1)}^{a_1} \cdots x_{\sigma(n)}^{a_n}$, summed over all permutations. If a majorizes b then $T_a \geq T_b$ for positive reals, with equality only when all x_i coincide.

Remark. Muirhead applies only to symmetric sums of monomials, and a solution that invokes it should say why the hypotheses hold; Schur's inequality is the standard example of a true inequality that Muirhead cannot reach.

Theorem C.1.45 (Schur). For non-negative reals x, y, z and $t > 0$,

$$x^t(x-y)(x-z) + y^t(y-x)(y-z) + z^t(z-x)(z-y) \geq 0.$$

The case $t = 1$ expands to $x^3 + y^3 + z^3 + xyz \cdot 1 \geq xy(x+y) + yz(y+z) + zx(z+x)$ after rearrangement, and is the form usually needed.

Theorem C.1.46 (SOS method). Many symmetric three-variable inequalities can be written $S_a(y-z)^2 + S_b(z-x)^2 + S_c(x-y)^2 \geq 0$; if the coefficients S_a, S_b, S_c are non-negative the inequality follows, and there are standard criteria covering the case where one of them is negative.

Theorem C.1.47 (Abel summation). $\sum_{k=1}^{n} a_k b_k = A_n b_n + \sum_{k=1}^{n-1} A_k(b_k - b_{k+1})$ where $A_k = a_1 + \cdots + a_k$. The discrete analogue of integration by parts, and the usual route to inequalities with monotone weights.

Theorem C.1.48 (Bernoulli). For $x > -1$: $(1+x)^n \geq 1 + nx$ for integers $n \geq 1$; more generally $(1+x)^\alpha \geq 1 + \alpha x$ when $\alpha \geq 1$ or $\alpha \leq 0$, and $\leq$ when $0 \leq \alpha \leq 1$.

Theorem C.1.49 (Nesbitt). For positive a, b, c, $\dfrac{a}{b+c} + \dfrac{b}{c+a} + \dfrac{c}{a+b} \geq \dfrac{3}{2}$.

C.1.5. Functional Equations

Definition C.1.50. A function $f : \mathbb{R} \to \mathbb{R}$ is *injective* if $f(a) = f(b)$ forces $a = b$, *surjective* if every value is attained, *bijective* if both. An *involution* satisfies $f(f(x)) = x$.

Theorem C.1.51 (Cauchy's functional equation). The solutions of $f(x + y) = f(x) + f(y)$ on $\mathbb{R}$ that are additionally continuous at a point, or monotone on an interval, or bounded on a set of positive measure, are exactly $f(x) = cx$. Without a regularity hypothesis, pathological solutions exist.

Remark. Over $\mathbb{Q}$ no hypothesis is needed: additivity alone forces $f(x) = cx$. The related equations $f(x + y) = f(x)f(y)$, $f(xy) = f(x) + f(y)$ and $f(xy) = f(x)f(y)$ reduce to this one by taking logarithms or exponentials on the appropriate domain.

Theorem C.1.52. Standard opening moves for a functional equation: substitute $x = y = 0$ and other fixed points; look for $f(0)$; test whether f is injective or surjective; find the fixed points of f; substitute $y \mapsto$ something that makes an argument constant; and check whether the equation is symmetric in x and y, since comparing the two substitutions often yields a relation for free.

C.1.6. Algebraic Structures

Definition C.1.53. A *group* is a set with an associative binary operation, an identity, and inverses. It is *abelian* if the operation commutes. The *order* of a group is its cardinality; the order of an element g is the least $n \geq 1$ with $g^n = e$.

Theorem C.1.54 (Lagrange). In a finite group the order of any subgroup, and hence of any element, divides the order of the group.

Definition C.1.55. A *ring* has two operations, addition making it an abelian group and an associative multiplication distributing over addition. A *field* is a commutative ring with $1 \neq 0$ in which every nonzero element is invertible.

Theorem C.1.56. Common examples: the groups $(\mathbb{Z}_n, +)$ and $(\mathbb{Z}_p \setminus \{0\}, \cdot)$; the rings $\mathbb{Z}$, $\mathbb{Z}_n$, $\mathbb{Z}[x]$; the fields $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}_p$, and $\mathbb{Q}(\sqrt{d})$.

C.2. Number Theory

C.2.1. Divisibility

Definition C.2.1. For integers a, b not both zero, $\gcd(a, b)$ is the largest integer dividing both and $\operatorname{lcm}(a, b)$ the least positive integer divisible by both. They satisfy $\gcd(a, b) \operatorname{lcm}(a, b) = |ab|$. Integers with $\gcd = 1$ are *coprime*.

Theorem C.2.2 (Euclidean algorithm). $\gcd(a, b) = \gcd(b, a - b) = \gcd(b, a \bmod b)$, which terminates and computes the gcd. Running it in reverse gives Bézout's identity.

Theorem C.2.3 (Bézout). There exist integers x, y with $ax+by = \gcd(a, b)$, and $\gcd(a, b)$ is the least positive integer of the form $ax + by$.

Theorem C.2.4 (Euclid's lemma). If p is prime and $p \mid ab$ then $p \mid a$ or $p \mid b$.

Theorem C.2.5 (Fundamental theorem of arithmetic). Every integer greater than 1 factors into primes uniquely up to order.

Remark. Unique factorisation also holds in $\mathbb{Z}[i]$, $\mathbb{Z}[\omega]$ with ω a primitive cube root of unity, and $\mathbb{Z}[\sqrt{2}]$, up to units — which is why $x^2 + y^2$ and $x^2 + xy + y^2$ are tractable. It fails in $\mathbb{Z}[\sqrt{-5}]$, where $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$.

Theorem C.2.6. $\gcd(a^m-1, a^n-1) = a^{\gcd(m,n)}-1$ for $a>1$. The analogous statement for Fibonacci numbers, $\gcd(F_m, F_n) = F_{\gcd(m,n)}$, is proved the same way.

Theorem C.2.7 (Infinitude of primes). There are infinitely many primes. Dirichlet's theorem strengthens this: every arithmetic progression $an+b$ with $\gcd(a,b)=1$ contains infinitely many primes.

Theorem C.2.8 (Bertrand's postulate). For every $n \geq 1$ there is a prime p with $n < p \leq 2n$.

C.2.2. Congruences

Definition C.2.9. $a \equiv b \pmod{n}$ means $n \mid a-b$. Congruences may be added, subtracted and multiplied, but divided only by integers coprime to the modulus.

Theorem C.2.10 (Chinese remainder theorem). If $m_1, \ldots, m_k$ are pairwise coprime, the system $x \equiv c_i \pmod{m_i}$ has a unique solution modulo $m_1 \cdots m_k$. Equivalently $\mathbb{Z}_{m_1 \cdots m_k} \cong \mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_k}$ as rings.

Theorem C.2.11 (Fermat's little theorem). For a prime p and $p \nmid a$, $a^{p-1} \equiv 1 \pmod{p}$; without the coprimality hypothesis, $a^p \equiv a \pmod{p}$.

Definition C.2.12. *Euler's totient* $\varphi(n)$ counts the integers in $[1,n]$ coprime to n. It is multiplicative, and for $n = \prod p_i^{\alpha_i}$, $\varphi(n) = n\prod(1-1/p_i)$. Also $\sum_{d \mid n} \varphi(d) = n$.

Theorem C.2.13 (Euler). If $\gcd(a,n)=1$ then $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Theorem C.2.14 (Wilson). p is prime if and only if $(p-1)! \equiv -1 \pmod{p}$.

Definition C.2.15. The *order* of a modulo n, written $\mathrm{ord}_n(a)$, is the least positive d with $a^d \equiv 1$. It divides every exponent e with $a^e \equiv 1$, and in particular divides $\varphi(n)$.

Theorem C.2.16 (Primitive roots). A primitive root modulo n — an element of order $\varphi(n)$ — exists precisely when $n = 1, 2, 4, p^k$ or $2p^k$ for an odd prime p. When one exists the multiplicative group modulo n is cyclic.

Theorem C.2.17 (Lifting the exponent). Let p be an odd prime with $p \mid a-b$ and $p \nmid a, b$. Then $v_p(a^n-b^n) = v_p(a-b) + v_p(n)$. For $p=2$ and odd a, b: $v_2(a^n-b^n) = v_2(a-b) + v_2(a+b) + v_2(n) - 1$ when n is even, and $v_2(a^n-b^n) = v_2(a-b)$ when n is odd.

Remark. Here $v_p(m)$ denotes the exponent of p in m. LTE is the fastest route through a large fraction of exponential Diophantine problems.

Theorem C.2.18 (Zsygmondy). For coprime $a > b \geq 1$, the number $a^n - b^n$ has a prime factor dividing no $a^k - b^k$ with $k < n$, with the exceptions $n=1$ with $a-b=1$; $n=2$ with $a+b$ a power of 2; and $(a,b,n) = (2,1,6)$.

Theorem C.2.19 (Hensel's lemma). If $f \in \mathbb{Z}[x]$, $f(a) \equiv 0 \pmod{p^k}$ and $p \nmid f'(a)$, then the root lifts uniquely to a root modulo p^{k+1}, and hence to every higher power.

Theorem C.2.20 (Frobenius / Chicken McNugget). For coprime positive a, b the largest integer not expressible as $ax+by$ with $x, y \geq 0$ is $ab-a-b$, and exactly $(a-1)(b-1)/2$ non-negative integers are inexpressible.

C.2.3. Arithmetic Functions

Definition C.2.21. $\tau(n)$ or $d(n)$ counts divisors, $\sigma(n)$ sums them, and $\sigma_k(n) = \sum_{d \mid n} d^k$. All are multiplicative, and for $n = \prod p_i^{\alpha_i}$ we have $\tau(n) = \prod(\alpha_i+1)$ and $\sigma(n) = \prod \frac{p_i^{\alpha_i+1}-1}{p_i-1}$.

Definition C.2.22. The *Möbius function* is $\mu(n) = 0$ if n is not squarefree and $(-1)^k$ if n is a product of k distinct primes. Then $\sum_{d|n} \mu(d)$ is 1 for $n = 1$ and 0 otherwise.

Theorem C.2.23 (Möbius inversion). If $g(n) = \sum_{d|n} f(d)$ for all n then $f(n) = \sum_{d|n} \mu(d) g(n/d)$, and conversely.

Theorem C.2.24 (Legendre's formula). The exponent of a prime p in $n!$ is $v_p(n!) = \sum_{i \geq 1} \lfloor n/p^i \rfloor = \dfrac{n - s_p(n)}{p-1}$, where $s_p(n)$ is the digit sum of n in base p.

Theorem C.2.25 (Kummer). The exponent of p in $\binom{m+n}{n}$ equals the number of carries when m and n are added in base p.

Theorem C.2.26 (Lucas). For a prime p, writing m, n in base p with digits m_i, n_i, $\binom{m}{n} \equiv \prod_i \binom{m_i}{n_i} \pmod{p}$.

C.2.4. Quadratic Residues

Definition C.2.27. An integer a coprime to an odd prime p is a *quadratic residue* modulo p if $x^2 \equiv a$ is solvable. Exactly half the nonzero residues are quadratic residues.

Definition C.2.28. The *Legendre symbol* $\left(\frac{a}{p}\right)$ is 1 if a is a quadratic residue and $p \nmid a$, -1 if a is a non-residue, and 0 if $p \mid a$. It is completely multiplicative in a.

Theorem C.2.29 (Euler's criterion). $\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}$.

Theorem C.2.30 (Supplementary laws). For odd primes p: $\left(\frac{-1}{p}\right) = 1$ exactly when $p \equiv 1 \pmod 4$; $\left(\frac{2}{p}\right) = 1$ exactly when $p \equiv \pm 1 \pmod 8$; and for $p > 3$, $\left(\frac{-3}{p}\right) = 1$ exactly when $p \equiv 1 \pmod 3$.

Theorem C.2.31 (Quadratic reciprocity). For distinct odd primes p, q,

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}.$$

So the two symbols agree unless both primes are $\equiv 3 \pmod 4$.

Definition C.2.32. The *Jacobi symbol* extends the Legendre symbol to odd positive $b = \prod p_i^{\alpha_i}$ by $\left(\frac{a}{b}\right) = \prod \left(\frac{a}{p_i}\right)^{\alpha_i}$. Reciprocity and the supplementary laws survive; Euler's criterion does not, and $\left(\frac{a}{b}\right) = 1$ no longer implies that a is a residue.

Theorem C.2.33 (Sums of two squares). A prime p is a sum of two squares exactly when $p = 2$ or $p \equiv 1 \pmod 4$. A positive integer is a sum of two squares exactly when every prime $\equiv 3 \pmod 4$ occurs to an even power in its factorisation.

Theorem C.2.34 (Lagrange's four-square theorem). Every non-negative integer is a sum of four squares. Legendre's three-square theorem: n is a sum of three squares exactly when n is not of the form $4^a(8b + 7)$.

C.2.5. Diophantine Equations

Theorem C.2.35 (Pythagorean triples). The primitive solutions of $a^2 + b^2 = c^2$ in positive integers are $a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$ with $m > n > 0$ coprime of opposite parity, up to swapping a and b.

Definition C.2.36. A *Pell equation* is $x^2 - Dy^2 = 1$ with D a positive non-square. If (x_1, y_1) is the least positive solution then all solutions are given by $x_n + y_n\sqrt{D} = (x_1 + y_1\sqrt{D})^n$, and a least solution always exists.

Theorem C.2.37 (Simon's favourite factoring trick). $xy + ax + by = c$ becomes $(x + b)(y + a) = c + ab$. More generally, forcing a product form and then enumerating divisors settles most two-variable Diophantine equations of low degree.

Theorem C.2.38 (Vieta jumping). If (x, y) solves a symmetric quadratic $x^2 - kxy + y^2 = c$, then fixing y and using Vieta on the quadratic in x produces a second root $x' = ky - x$, giving a new solution. Descending to a minimal solution then bounds k or characterises all solutions.

Theorem C.2.39 (Fermat's method of infinite descent). If a solution in positive integers yields a strictly smaller one, no solution exists. The classical application is $x^4 + y^4 = z^2$.

Theorem C.2.40 (Mihăilescu (Catalan's conjecture)). The only solution of $x^a - y^b = 1$ in integers greater than 1 is $3^2 - 2^3 = 1$.

Remark. Quotable but essentially never the intended route; a problem that appears to need it usually has an elementary solution by size or modular arguments.

C.2.6. Rational Approximation

Definition C.2.41. The *Farey sequence* F_n lists the reduced fractions in $[0, 1]$ with denominator at most n, in increasing order. Consecutive terms $p/q < p'/q'$ satisfy $p'q - pq' = 1$, and the *mediant* $(p + p')/(q + q')$ is the first fraction to appear between them.

Theorem C.2.42 (Dirichlet's approximation theorem). For any real α and any N there are integers p, q with $1 \le q \le N$ and $\left|\alpha - \frac{p}{q}\right| < \frac{1}{qN}$. Consequently every irrational has infinitely many rational approximations with $\left|\alpha - \frac{p}{q}\right| < \frac{1}{q^2}$.

Definition C.2.43. The *continued fraction* of α is $[a_0; a_1, a_2, \dots]$ with $a_0 = \lfloor \alpha \rfloor$ and the process repeated on $1/(\alpha - a_0)$. It terminates exactly for rationals and is eventually periodic exactly for quadratic irrationals. Its convergents are the best rational approximations, and the convergents of $\sqrt{D}$ produce the solutions of the Pell equation.

C.3. Geometry

C.3.1. The Triangle and its Centres

Definition C.3.1. In $\triangle ABC$ we write $a = BC$, $b = CA$, $c = AB$, s for the semiperimeter, S for the area, R for the circumradius and r for the inradius.

Definition C.3.2. The *circumcentre* O is the common point of the perpendicular bisectors; the *incentre* I of the internal angle bisectors; the *centroid* G of the medians; the *orthocentre* H of the altitudes. All four exist and are unique for a non-degenerate triangle.

Theorem C.3.3 (Euler line). O, G and H are collinear with $\overrightarrow{OH} = 3\overrightarrow{OG}$, so $HG = 2GO$. The incentre lies on this line only for an isosceles triangle.

Theorem C.3.4 (Nine-point circle). The feet of the three altitudes, the midpoints of the three sides, and the midpoints of AH, BH, CH all lie on one circle, of radius $R/2$, centred at the midpoint of OH.

Theorem C.3.5 (Feuerbach). The nine-point circle is tangent to the incircle and to all three excircles.

Theorem C.3.6 (Incentre–excentre lemma). If AI meets the circumcircle again at M, then M is the arc midpoint of BC and $MB = MC = MI = MI_A$, where I_A is the excentre opposite A. Equivalently B, I, C, I_A lie on a circle centred at M.

Remark. This is the single most frequently used lemma in olympiad triangle geometry.

Definition C.3.7. The *orthic triangle* has vertices at the feet of the altitudes. For an acute triangle, the orthocentre of $\triangle ABC$ is the incentre of its orthic triangle, and the sides of the orthic triangle make equal angles with the sides of $\triangle ABC$.

Definition C.3.8. Two cevians through A are *isogonal* if they make equal angles with AB and AC. The *symmedian* is the isogonal of the median; it passes through the pole of BC with respect to the circumcircle, and the three symmedians concur at the *Lemoine point*.

Definition C.3.9. The *pedal triangle* of a point P has vertices at the feet of the perpendiculars from P to the sides.

Theorem C.3.10 (Simson line). The pedal triangle of P degenerates to a line exactly when P lies on the circumcircle. That line bisects PH.

Theorem C.3.11 (Carnot). Perpendiculars to BC, CA, AB at points X, Y, Z are concurrent if and only if $BX^2 - XC^2 + CY^2 - YA^2 + AZ^2 - ZB^2 = 0$.

Theorem C.3.12 (Fermat point). If every angle of $\triangle ABC$ is less than 120°, the point minimising $PA + PB + PC$ sees each side at 120°; it is the common point of AA', BB', CC' for equilateral triangles erected outward. If some angle is at least 120°, the minimiser is that vertex.

C.3.2. Concurrency and Collinearity

Theorem C.3.13 (Ceva). For X, Y, Z on lines BC, CA, AB, the cevians AX, BY, CZ concur if and only if $\frac{\overline{BX}}{\overline{XC}} \cdot \frac{\overline{CY}}{\overline{YA}} \cdot \frac{\overline{AZ}}{\overline{ZB}} = 1$ in signed ratios. The trigonometric form replaces each ratio by a ratio of sines of the subtended angles.

Theorem C.3.14 (Menelaus). With the same notation, X, Y, Z are collinear if and only if the same signed product equals -1.

Theorem C.3.15 (Stewart). If D lies on line BC then $AB^2 \cdot \overline{DC} + AC^2 \cdot \overline{BD} - AD^2 \cdot \overline{BC} = \overline{BC} \cdot \overline{BD} \cdot \overline{DC}$. For D the midpoint this reduces to the median length formula $4AD^2 = 2AB^2 + 2AC^2 - BC^2$.

Theorem C.3.16 (Desargues). Two triangles are perspective from a point if and only if they are perspective from a line.

Theorem C.3.17 (Pappus). If A_1, A_2, A_3 lie on one line and B_1, B_2, B_3 on another, the three points $A_iB_j \cap A_jB_i$ are collinear.

Theorem C.3.18 (Pascal). For six points on a conic, the three intersections of opposite sides of the hexagon they form are collinear. Degenerate cases, where two adjacent points coincide and the side becomes a tangent, are often the useful ones.

Theorem C.3.19 (Brianchon). For a hexagon circumscribed about a conic, the three main diagonals concur.

Theorem C.3.20 (Newton–Gauss line). In a complete quadrilateral the midpoints of the three diagonals are collinear. If the quadrilateral has an incircle, its centre also lies on this line.

C.3.3. Circles

Theorem C.3.21 (Inscribed angle theorem). An inscribed angle is half the central angle subtending the same arc; angles subtending the same arc from the same side are equal. The tangent–chord angle equals the inscribed angle in the alternate segment.

Definition C.3.22. The *power* of X with respect to a circle with centre O and radius ρ is $\operatorname{pow}(X) = OX^2 - \rho^2$. For any line through X meeting the circle at A, B, $\operatorname{pow}(X) = \overline{XA} \cdot \overline{XB}$; for a tangent from X touching at T, $\operatorname{pow}(X) = XT^2$.

Definition C.3.23. The *radical axis* of two non-concentric circles is the locus of points of equal power; it is a line perpendicular to the line of centres. The three radical axes of three circles concur at the *radical centre* or are parallel.

Theorem C.3.24 (Cyclic quadrilaterals). $ABCD$ is cyclic if and only if $\angle ADC + \angle ABC = 180^\circ$, if and only if $\angle ACB = \angle ADB$, if and only if the power condition $\overline{PA} \cdot \overline{PC} = \overline{PB} \cdot \overline{PD}$ holds at the intersection P of the diagonals or of two opposite sides.

Theorem C.3.25 (Ptolemy). For a cyclic quadrilateral $ABCD$, $AC \cdot BD = AB \cdot CD + AD \cdot BC$. For four arbitrary points the inequality $AC \cdot BD \le AB \cdot CD + AD \cdot BC$ holds, with equality exactly in the cyclic case with the correct ordering.

Theorem C.3.26 (Casey). A generalisation of Ptolemy in which the four points are replaced by four circles tangent to a fifth, and each distance by the length of the common tangent between two of them.

Theorem C.3.27 (Pitot). A convex quadrilateral has an inscribed circle if and only if $AB + CD = BC + DA$.

Theorem C.3.28 (Brahmagupta). A cyclic quadrilateral with sides a, b, c, d and semiperimeter p has area $\sqrt{(p-a)(p-b)(p-c)(p-d)}$. For a general quadrilateral the area is $\sqrt{(p-a)(p-b)(p-c)(p-d) - abcd\cos^2\frac{\alpha+\gamma}{2}}$ with α, γ a pair of opposite angles.

Theorem C.3.29 (Brocard). For a cyclic quadrilateral $ABCD$ with circumcentre O, let $P = AB \cap CD$, $Q = AD \cap BC$ and $R = AC \cap BD$. Then O is the orthocentre of $\triangle PQR$.

Definition C.3.30. The *pole* of a line ℓ not through O with respect to a circle centred at O of radius ρ is the point A on the perpendicular from O with $d(O, \ell) \cdot OA = \rho^2$; the *polar* of A is ℓ. For A outside the circle the polar is the chord of contact of the two tangents from A.

Theorem C.3.31 (La Hire). A lies on the polar of B if and only if B lies on the polar of A.

C.3.4. Transformations

Definition C.3.32. A *homothety* with centre C and ratio $k \neq 0$ sends P to the point P' with $\overrightarrow{CP'} = k\overrightarrow{CP}$. It maps lines to parallel lines and circles to circles.

Theorem C.3.33 (Monge's theorem). The three external similitude centres of three circles taken pairwise are collinear; each pair of internal centres is collinear with the remaining external centre.

Definition C.3.34. A *spiral similarity* is the composition of a rotation and a homothety about the same centre.

Theorem C.3.35 (Spiral similarity lemma). If AB and CD are segments with $P = AC \cap BD$ and Q the second intersection of the circumcircles of $\triangle PAB$ and $\triangle PCD$, then Q is the centre of the unique spiral similarity taking AB to CD; it also takes AC to BD. Equivalently, the centre is the second intersection of circles (PAC) and (PBD).

Definition C.3.36. *Inversion* with centre O and radius ρ sends $P \neq O$ to P' on ray OP with $OP \cdot OP' = \rho^2$.

Theorem C.3.37 (Properties of inversion). Inversion fixes the circle of inversion pointwise, exchanges its interior and exterior, and preserves angles between curves. Lines through O map to themselves; lines not through O map to circles through O and conversely; circles not through O map to circles not through O. Distances transform by $A'B' = \dfrac{\rho^2 \cdot AB}{OA \cdot OB}$.

Remark. The choice of centre is everything: inverting at a point where many circles meet turns them all into lines. Inverting at A with radius $\sqrt{AB \cdot AC}$ followed by reflection in the bisector of $\angle A$ — the $\sqrt{bc}$ *inversion* — swaps B and C and is worth knowing by name.

Definition C.3.38. A *projective transformation* of the plane preserves collinearity. The *cross ratio* of four collinear points, $(A, B; C, D) = \dfrac{\overline{CA}}{\overline{CB}} : \dfrac{\overline{DA}}{\overline{DB}}$, is preserved by projection; a *harmonic* quadruple is one with cross ratio -1.

Theorem C.3.39 (Harmonic conjugates). If cevians AD, BE, CF of $\triangle ABC$ concur and EF meets BC at X, then $(B, C; D, X) = -1$. The polar of a point with respect to a circle is exactly the locus of its harmonic conjugates.

C.3.5. Metric Relations and Trigonometry

Theorem C.3.40 (Law of sines). $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C} = 2R.$

Theorem C.3.41 (Law of cosines). $c^2 = a^2 + b^2 - 2ab\cos C.$

Theorem C.3.42 (Heron). $S = \sqrt{s(s-a)(s-b)(s-c)}.$

Theorem C.3.43. Further area and radius identities: $S = \frac{1}{2}ab\sin C = rs = \dfrac{abc}{4R}$; $r = (s-a)\tan\frac{A}{2} = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$; and $r = R(\cos A + \cos B + \cos C - 1)$.

Theorem C.3.44 (Euler's formula for OI). $OI^2 = R(R-2r)$; consequently $R \geq 2r$, with equality only for the equilateral triangle.

Theorem C.3.45. Trigonometric identities most often needed: $\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$; $\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$; the product-to-sum formulas $2\cos\alpha\cos\beta = \cos(\alpha+\beta) + \cos(\alpha-\beta)$, $2\sin\alpha\cos\beta = \sin(\alpha+\beta) + \sin(\alpha-\beta)$, $2\sin\alpha\sin\beta = \cos(\alpha-\beta) - \cos(\alpha+\beta)$; and the Weierstrass substitution $t = \tan\frac{x}{2}$, under which $\sin x = \frac{2t}{1+t^2}$ and $\cos x = \frac{1-t^2}{1+t^2}$.

Theorem C.3.46. In any triangle, $\tan A + \tan B + \tan C = \tan A\tan B\tan C$ and $\cos^2 A + \cos^2 B + \cos^2 C + 2\cos A\cos B\cos C = 1$.

C.3.6. Analytic Methods

Definition C.3.47. In *complex coordinates* points are complex numbers, often with the circumcircle taken as the unit circle. Then $\bar{a} = 1/a$ for a on the circle; the chord through a, b has equation $z + ab\bar{z} = a + b$; the foot of the perpendicular from z to that chord is

$\frac{1}{2}(z + a + b - ab\bar{z})$; and a, b, c, d are concyclic or collinear exactly when the cross ratio $\frac{(a-c)(b-d)}{(a-d)(b-c)}$ is real.

Definition C.3.48. In *barycentric coordinates* relative to $\triangle ABC$, a point is written $(x : y : z)$ meaning $\frac{xA+yB+zC}{x+y+z}$. Then $A = (1 : 0 : 0)$, $G = (1 : 1 : 1)$, $I = (a : b : c)$, and the circumcircle is $a^2yz + b^2zx + c^2xy = 0$. Lines are linear equations and collinearity is a determinant.

Remark. Barycentrics handle cevians, ratios along lines and the standard triangle centres extremely well, and handle circles and perpendicularity clumsily. Complex coordinates are the reverse. Choosing the wrong one is the usual way a coordinate solution dies.

Theorem C.3.49 (Leibniz / Lagrange identity). For the centroid G and any point X, $XA^2 + XB^2 + XC^2 = GA^2 + GB^2 + GC^2 + 3XG^2$. More generally, for masses m_i at A_i with barycentre T, $\sum m_i XA_i^2 = \sum m_i TA_i^2 + (\sum m_i)XT^2$.

Theorem C.3.50. Vector formulas: the area of $\triangle ABC$ is $\frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}|$, and the volume of tetrahedron $ABCD$ is $\frac{1}{6}|[\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}]|$ where $[\cdot, \cdot, \cdot]$ is the scalar triple product.

C.3.7. Geometric Inequalities

Theorem C.3.51 (Triangle inequality). $AB + BC \geq AC$, with equality exactly when B lies on segment AC.

Theorem C.3.52 (Erdős–Mordell). For P inside $\triangle ABC$ with feet X, Y, Z on the sides, $PA + PB + PC \geq 2(PX + PY + PZ)$, with equality only for the equilateral triangle with P at the centre.

Theorem C.3.53 (Isoperimetric inequality). Among plane figures of given perimeter the circle has the largest area; among polygons with a fixed number of sides and given perimeter, the regular one does.

Theorem C.3.54. For a triangle with the usual notation: $a^2 + b^2 + c^2 \geq 4\sqrt{3}\,S$ (Weitzenböck), with equality only when equilateral; and the parallelogram inequality $AB^2 + BC^2 + CD^2 + DA^2 \geq AC^2 + BD^2$ for any four points, with equality exactly for a parallelogram.

C.4. Combinatorics

C.4.1. Counting

Definition C.4.1. Basic counts on an n-set: there are $n!$ permutations; $\frac{n!}{(n-k)!}$ ordered k-tuples of distinct elements; $\binom{n}{k}$ k-element subsets; and k^n functions from a k-set to an n-set. The number of multisets of size k from n types — equivalently the number of non-negative solutions of $x_1 + \cdots + x_n = k$ — is $\binom{n+k-1}{n-1}$, the *stars and bars* count.

Theorem C.4.2. The permutations of a multiset in which element i appears α_i times number $\dfrac{n!}{\alpha_1! \cdots \alpha_s!}$.

Theorem C.4.3 (Pigeonhole principle). If $nk + 1$ objects are placed into n classes, some class contains at least $k + 1$ objects. In the infinite version, if infinitely many objects are placed into finitely many classes, some class is infinite.

Theorem C.4.4 (Inclusion–exclusion). For subsets $S_1, \ldots, S_n$ of a finite set S,

$$\left|S \setminus \bigcup S_i\right| = |S| - \sum |S_i| + \sum |S_i \cap S_j| - \cdots .$$

The count of derangements, $D_n = n!\sum_{k=0}^{n}\frac{(-1)^k}{k!}$, is the standard application.

Theorem C.4.5 (Double counting). Counting one set in two ways yields an identity; counting incidences of a bipartite relation in two ways yields $\sum_x d(x) = \sum_y d(y)$, the basis of most extremal counting arguments.

Theorem C.4.6 (Bijective principle). A bijection between two sets proves their counts equal. Constructing one is almost always more informative than computing both counts.

C.4.2. Special Numbers

Definition C.4.7. The *Catalan numbers* $C_n = \frac{1}{n+1}\binom{2n}{n}$ satisfy $C_{n+1} = \sum_k C_k C_{n-k}$ and count, among many other things, balanced bracketings of length $2n$, triangulations of a convex $(n+2)$-gon, binary trees on n nodes, and lattice paths from $(0,0)$ to (n,n) staying weakly below the diagonal.

Theorem C.4.8 (Cycle lemma / reflection). The count of lattice paths from $(0,0)$ to (n,n) crossing the diagonal equals the count of all paths to the reflected endpoint, giving $C_n = \binom{2n}{n} - \binom{2n}{n+1}$.

Definition C.4.9. The *Stirling numbers of the second kind* $S(n,k)$ count partitions of an n-set into k non-empty blocks, and satisfy $S(n,k) = k\,S(n-1,k) + S(n-1,k-1)$. The *Bell number* $B_n = \sum_k S(n,k)$ counts all set partitions. The Stirling numbers of the first kind count permutations by number of cycles.

Definition C.4.10. The *Fibonacci numbers* satisfy $F_1 = F_2 = 1$, $F_{n+1} = F_n + F_{n-1}$, with Binet's formula $F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$ where φ, ψ are the roots of $x^2 = x + 1$. They satisfy $F_{m+n} = F_m F_{n+1} + F_{m-1}F_n$ and $\gcd(F_m, F_n) = F_{\gcd(m,n)}$.

C.4.3. Graph Theory

Definition C.4.11. A *graph* $G = (V, E)$ consists of vertices and edges joining pairs of them; it is *simple* if there are no loops or repeated edges. The *degree* $d(v)$ counts edges at v.

Theorem C.4.12 (Handshake lemma). $\sum_v d(v) = 2|E|$; consequently the number of vertices of odd degree is even.

Definition C.4.13. A *walk* is a sequence of adjacent vertices; a *path* repeats no vertex; a *cycle* is a closed path. A graph is *connected* if every pair of vertices is joined by a path. A *tree* is a connected acyclic graph.

Theorem C.4.14. A tree on n vertices has exactly $n-1$ edges and, for $n \geq 2$, at least two leaves. Any two of "connected", "acyclic", "$n-1$ edges" imply the third.

Theorem C.4.15 (Cayley). There are n^{n-2} labelled trees on n vertices.

Definition C.4.16. A graph is *bipartite* if its vertices split into two classes with all edges between them; equivalently, if it has no odd cycle.

Theorem C.4.17 (Euler circuits). A connected graph has an Eulerian circuit exactly when every vertex has even degree, and an Eulerian path exactly when the number of odd-degree vertices is 0 or 2.

Definition C.4.18. A *Hamiltonian* cycle visits every vertex exactly once. No simple criterion is known, but sufficient conditions exist.

Theorem C.4.19 (Dirac and Ore). A simple graph on $n \geq 3$ vertices with minimum degree at least $n/2$ (Dirac), or with $d(u) + d(v) \geq n$ for every non-adjacent pair (Ore), has a Hamiltonian cycle.

Theorem C.4.20 (Hall's marriage theorem). A bipartite graph with parts W and M has a matching saturating W if and only if every subset $U \subseteq W$ has at least $|U|$ neighbours.

Theorem C.4.21 (Kőnig). In a bipartite graph the largest matching and the smallest vertex cover have the same size.

Theorem C.4.22 (Turán). A graph on n vertices with no K_p subgraph has at most $\left(1 - \frac{1}{p-1}\right) \frac{n^2}{2}$ edges, with the extremal example the complete balanced $(p-1)$-partite graph. The case $p = 3$ is Mantel's theorem: a triangle-free graph has at most $n^2/4$ edges.

Theorem C.4.23 (Ramsey). For any r and any $q_1, \ldots, q_s$ there is a least N such that any s-colouring of the r-subsets of an N-set contains a q_i-set all of whose r-subsets have colour i. The two-colour graph case $R(3,3) = 6$, $R(3,4) = 9$, $R(3,5) = 14$, $R(4,4) = 18$ are the values worth knowing.

Definition C.4.24. A graph is *planar* if it can be drawn in the plane with no crossing edges.

Theorem C.4.25 (Euler's polyhedron formula). For a connected planar graph, $V - E + F = 2$. Consequently a simple planar graph has $E \leq 3V - 6$, and $E \leq 2V - 4$ if it is triangle-free.

Theorem C.4.26 (Kuratowski). A graph is planar exactly when it contains no subdivision of K_5 or $K_{3,3}$.

Theorem C.4.27 (Four colour theorem). Every planar graph is 4-colourable. The five colour theorem has a short proof and is the one to use in a competition.

C.4.4. Extremal and Algebraic Methods

Theorem C.4.28 (Extremal principle). In a finite configuration, consider the object that is largest, smallest, first or closest in some well-chosen measure. Most existence and impossibility proofs in combinatorics start here.

Theorem C.4.29 (Sperner). The largest antichain in the subset lattice of an n-set has size $\binom{n}{\lfloor n/2 \rfloor}$.

Theorem C.4.30 (Dilworth). In a finite partially ordered set, the minimum number of chains covering it equals the size of the largest antichain. Dually, Mirsky's theorem covers by antichains using the longest chain.

Theorem C.4.31 (Erdős–Szekeres). Any sequence of more than $(r-1)(s-1)$ distinct reals contains an increasing subsequence of length r or a decreasing subsequence of length s.

Theorem C.4.32 (Erdős–Ko–Rado). For $n \geq 2k$, a family of pairwise intersecting k-subsets of an n-set has size at most $\binom{n-1}{k-1}$.

Theorem C.4.33 (Linear algebra bound). A family of sets whose characteristic vectors are linearly independent over a field has size at most the dimension of the ambient space. This is the standard route to bounds such as the oddtown and eventown theorems.

Theorem C.4.34 (Probabilistic method). If a random object has a property with positive probability, an object with that property exists. Combined with linearity of

expectation — which needs no independence — this proves the existence of configurations no explicit construction is known for.

Theorem C.4.35 (Burnside's lemma). The number of orbits of a finite group G acting on a set X is $\frac{1}{|G|}\sum_{g\in G}|\mathrm{Fix}(g)|$. It is the standard tool for counting colourings up to symmetry; Pólya enumeration is its generating-function refinement.

C.4.5. Invariants and Games

Theorem C.4.36 (Invariant principle). A quantity unchanged by every legal move cannot distinguish reachable from unreachable states; exhibiting one that differs between the initial and target position proves impossibility. Common invariants are parity, a sum or product modulo n, a colouring class count, and the sign of a permutation.

Theorem C.4.37 (Monovariant principle). A quantity that changes monotonically under every move, and is bounded, proves that a process terminates. If it is a non-negative integer that strictly decreases, termination is immediate.

Theorem C.4.38 (Colouring arguments). Assigning weights or colours to the cells of a board so that every placeable piece covers a fixed total is the standard proof that a tiling does not exist. Checkerboard, stripe and modular colourings cover most cases.

Definition C.4.39. In a finite *impartial game* with normal play, positions are P-positions (previous player wins) or N-positions (next player wins). A position is a P-position exactly when every move leads to an N-position.

Theorem C.4.40 (Sprague–Grundy). Every position in an impartial game has a Grundy value g, computed as the least non-negative integer not among the Grundy values of its options; the position is losing exactly when $g = 0$, and the value of a disjoint sum of games is the bitwise XOR of the values. For Nim this gives the classical criterion that a position is losing exactly when the XOR of the pile sizes is zero.

Theorem C.4.41 (Strategy stealing). If in a symmetric game an extra move can never hurt, the second player cannot have a winning strategy — otherwise the first player would steal it. The argument proves who does *not* win without exhibiting a strategy.

Theorem C.4.42 (Pairing strategy). Partitioning the board into pairs and always responding within the pair the opponent just played in is the standard way to force a draw or a win in blocking games.

C.4.6. Combinatorial Geometry

Theorem C.4.43 (Helly). If finitely many convex sets in $\mathbb{R}^d$ have the property that every $d+1$ of them share a point, then all of them share a point.

Theorem C.4.44 (Radon). Any $d+2$ points in $\mathbb{R}^d$ can be split into two sets whose convex hulls intersect.

Theorem C.4.45 (Sylvester–Gallai). Given finitely many points, not all collinear, some line passes through exactly two of them.

Theorem C.4.46 (Pick). A simple lattice polygon with I interior and B boundary lattice points has area $I + \frac{B}{2} - 1$.

Theorem C.4.47 (Convex position). Among any five points in general position, four form a convex quadrilateral; the Erdős–Szekeres conjecture generalises this to $2^{n-2}+1$ points forcing a convex n-gon.

C.5. Methods of Proof

The techniques below are not results and cannot be cited, but they are what the results above are assembled with, and a solution is usually found by choosing among them before any theorem is recalled.

Method C.5.1 (Induction). To prove $P(n)$ for all $n \geq n_0$, prove $P(n_0)$ and that $P(n)$ implies $P(n+1)$. In *strong* induction the inductive step may assume all previous cases. Some statements are easier to prove in a strengthened form, because the stronger hypothesis is available in the step — if an induction stalls, try proving more.

Method C.5.2 (Well-ordering and descent). Every non-empty set of positive integers has a least element. Applied to a hypothetical counterexample, this gives proof by minimal counterexample; applied to a solution of a Diophantine equation that generates a smaller solution, it gives infinite descent.

Method C.5.3 (Contradiction and contraposition). To prove $P \Rightarrow Q$, assume P and $\neg Q$ and derive a falsehood, or prove $\neg Q \Rightarrow \neg P$ directly. The second is usually cleaner when it is available.

Method C.5.4 (Without loss of generality). A symmetry of the problem lets one case stand for several. The reduction must be genuine: state the symmetry being used, and check that it really does map every case to the chosen one.

Method C.5.5 (Working backwards). Assume the conclusion and look for a statement that would imply it, repeating until something known is reached, then reverse the chain. Every step reversed must be an implication in the required direction.

Method C.5.6 (Considering extreme and degenerate cases). Small cases reveal the pattern, extreme cases reveal the mechanism, and degenerate cases reveal the hypotheses that are actually doing work.

Method C.5.7 (Constructions). An answer of the form "the maximum is N" needs two arguments: a configuration attaining N, and a proof that $N+1$ is impossible. Losing half the marks on such a problem almost always means one of the two was omitted.